

Applied Partial Differential Equations
with Fourier Series and Boundary Value Problems
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All errors, typographical and substantive, and other offenses, are entirely my own.

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1 - Heat Equation

1.2 - Derivation of the Conduction of Heat in a One-Dimensional Rod

Question: 1.2.1

Briefly explain the minus sign:

- (a) in conservation law (1.2.3) or (1.2.5) if $Q = 0$
- (b) in Fourier's law (1.2.8)
- (c) in conservation law (1.2.12), $\frac{\partial u}{\partial t} = -\frac{\partial \phi}{\partial x}$
- (d) in Fick's law (1.2.13)

- (a) Assuming zero internal thermal energy, we have

$$\frac{\partial e}{\partial t} = -\frac{\partial \phi}{\partial x}$$

The temporal change in thermal energy density is opposite to the spatial change in heat flux. If more heat leaves a unit volume than had entered, the thermal energy density decreases, and vice-versa.

- (b) Fourier's law of heat conduction states

$$\phi = -K_0 \frac{\partial u}{\partial x}$$

Heat moves in the direction of the negative of the temperature gradient, or from hot to cold regions.

- (c) The spatial change in heat flux is opposite to the temporal change in temperature. If more heat leaves a unit volume than had entered, the temperature decreases, and vice-versa.

- (d) Fick's law states

$$\phi = -k \frac{\partial u}{\partial x}$$

The chemical moves in the direction of the negative of the gradient of the chemical concentration, or from more to less concentrated regions.

Question: 1.2.2

Derive the heat equation for a rod assuming constant thermal properties and no sources.

- (a) Consider the total thermal energy between x and $x + \Delta x$.
- (b) Consider the total thermal energy between $x = a$ and $x = b$.

- (a) For a thin slice of width Δx , the rate of change of heat energy is approximately equal to the flow of thermal energy at x and $x + \Delta x$.

$$\frac{\partial}{\partial t}[e(x, t) A \Delta x] \approx \phi(x, t) A - \phi(x + \Delta x, t) A$$

Divide through by $A \Delta x$ and in the limit as $\Delta x \rightarrow 0$, we find

$$\frac{\partial e}{\partial t} = \lim_{\Delta x \rightarrow 0} \frac{\phi(x, t) - \phi(x + \Delta x, t)}{\Delta x} = -\frac{\partial \phi}{\partial x}$$

Replacing $\partial e / \partial t$ with $c \rho \partial u / \partial t$, where c is the specific heat and ρ is the mass density, we have

$$c \rho \frac{\partial u}{\partial t} = -\frac{\partial \phi}{\partial x}$$

Lastly, we take the spatial derivative of Fourier's law, where K_0 is the thermal conductivity

$$\phi = -K_0 \frac{\partial u}{\partial x} \quad \implies \quad \frac{\partial \phi}{\partial x} = -K_0 \frac{\partial^2 u}{\partial x^2}$$

and substitute into the conservation equation to derive the heat equation.

$$\frac{\partial u}{\partial t} = \frac{K_0}{c \rho} \frac{\partial^2 u}{\partial x^2}$$

- (b) In lieu of the approximation in the previous set-up, consider the integral conservation

$$\frac{d}{dt} \int_a^b e(x, t) dx = \phi(a, t) - \phi(b, t)$$

By the Fundamental Theorem of Calculus and the constancy of x , the left side becomes

$$\int_a^b \frac{\partial e}{\partial t} dx$$

And the right side

$$-\int_a^b \frac{\partial \phi}{\partial x} dx$$

Setting equal and combining terms, we find

$$\int_a^b \left[\frac{\partial e}{\partial t} + \frac{\partial \phi}{\partial x} \right] dx = 0$$

from which we can immediately deduce the conservation law. Proceed as in part (a).

Question: 1.2.3

Derive the heat equation for a rod assuming constant thermal properties with variable cross-sectional area $A(x)$ assuming no sources by considering the total thermal energy between $x = a$ and $x = b$.

By the integral conservation law, we have

$$\frac{d}{dt} \int_a^b e A(x) dx = \int_a^b \frac{\partial e}{\partial t} A(x) dx = - \int_a^b \frac{\partial}{\partial x} [\phi(x, t) A(x)] dx$$

From which we can derive

$$\frac{\partial e}{\partial t} A(x) + \frac{\partial}{\partial x} [\phi(x, t) A(x)] = 0$$

Replacing $e(x, t)$ with $c\rho u(x, t)$ and combining with the Fourier law, conclude

$$\frac{\partial u}{\partial t} = \frac{K_0}{c\rho A(x)} \frac{\partial}{\partial x} \left[A(x) \frac{\partial u}{\partial x} \right] = \frac{K_0}{c\rho} \left[\frac{\partial^2 u}{\partial x^2} + \frac{1}{A(x)} \frac{\partial u}{\partial x} \frac{\partial A}{\partial x} \right]$$

Question: 1.2.4

Derive the diffusion equation for a chemical pollutant.

- (a) Consider the total amount of the chemical in a thin region between x and $x + \Delta x$.
 - (b) Consider the total amount of the chemical between $x = a$ and $x = b$.
- (a) The derivation is analogous to that of the heat equation for a thin slice, with chemical diffusivity k in lieu of the thermal diffusivity $K_0/c\rho$.
 - (b) See 1.2.2(b) with the same comment about the chemical diffusivity as above.

Question: 1.2.5

Derive an equation for the concentration $u(x, t)$ of a chemical pollutant if the chemical is produced due to chemical reaction at the rate of $\alpha u(\beta - u)$ per unit volume.

One conceptual understanding here is the production of the chemical originates from some *source* Q , which from the conservation law gives us

$$\frac{\partial u}{\partial t} + \frac{\partial \phi}{\partial x} = \alpha u(\beta - u)$$

From Fick's law, substitute ϕ to derive

$$\frac{\partial u}{\partial t} - k \frac{\partial^2 u}{\partial x^2} = \alpha u(\beta - u)$$

Observe the logistic form on the right. This equation is called the **Fisher-KPP equation**, named after Ronald Fisher, Andrey Kolmogorov, Ivan Petrovsky, and Nikolai Piskunov.

Question: 1.2.6

Suppose that the specific heat is a function of position and temperature, $c(x, u)$.

- (a) Show that the heat energy per unit mass necessary to raise the temperature of a thin slice of thickness Δx from 0° to $u(x, t)$ is not $c(x) u(x, t)$, but instead $\int_0^u c(x, \bar{u}) d\bar{u}$.
- (b) Rederive the heat equation in this case. Show that (1.2.3) remains unchanged.

- (a) Since specific heat is, in this instance, a function of temperature and not independent of it, we must integrate with respect to a dummy variable $\bar{u}$:

$$\int_0^u c(x, \bar{u}) d\bar{u}$$

Conceptually, the specific heat varies as a function of the temperature for any given (x, t) . For infinitesimal subdivisions of the total change in temperature from 0 to u , some amount of energy per unit mass $c(x, \bar{u}) d\bar{u}$ is needed. Because this energy per unit mass varies depending on how far we are along the temperature change to u , we integrate to sum those infinitesimal amounts.

- (b) The beginning of the derivation proceeds as in the original case: conservation of heat energy mandates that the rate of change of heat energy equals the thermal flux through the boundaries and internal sources.

$$\frac{\partial}{\partial t} [e(x, t) A \Delta x] \approx \phi(x, t) A - \phi(x + \Delta x, t) A + Q(x, t) A \Delta x$$

which implies equation (1.2.3)

$$\frac{\partial e}{\partial t} + \frac{\partial \phi}{\partial x} = Q$$

However, what changes is in the calculation of thermal energy density. Instead of the product of specific heat, mass density, and temperature, we have

$$e(x, t) = \rho(x) \int_0^u c(x, \bar{u}) d\bar{u}$$

Taking the time derivative and applying Leibniz's rule, derive

$$\begin{aligned} \frac{\partial e}{\partial t} &= \rho(x) \frac{\partial}{\partial t} \int_0^u c(x, \bar{u}) d\bar{u} \\ &= \rho(x) \left[c(x, u) \frac{\partial u}{\partial t} + \underbrace{\int_0^u \frac{\partial}{\partial t} c(x, \bar{u}) d\bar{u}}_{=0} \right] \end{aligned}$$

which is analogous to our derivation of the heat equation in the original case. Combine with Fourier's law to get

$$c(x, u) \rho(x) \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0 \frac{\partial u}{\partial x} \right) + Q$$

Question: 1.2.7

Consider conservation of thermal energy (1.2.4) for any segment of a one-dimensional rod $a \leq x \leq b$. By using the fundamental theorem of calculus,

$$\frac{\partial}{\partial b} \int_a^b f(x) dx = f(b)$$

derive the heat equation (1.2.9).

The formula implies that a is constant and b is varying. Take our integral conservation law, and differentiate both sides by b :

$$\begin{aligned} \frac{\partial}{\partial b} \int_a^b \frac{\partial e(x, t)}{\partial t} dx &= \frac{\partial}{\partial b} \left[\phi(a, t) - \phi(b, t) + \int_a^b Q dx \right] \\ \frac{\partial e(b, t)}{\partial t} &= -\frac{\partial \phi(b, t)}{\partial b} + Q \end{aligned}$$

which restores the conservation law! From here we can continue as normal to re-derive

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

Since b is the stand-in for our variable, we can revert to the more customary x .

Question: 1.2.8

If $u(x, t)$ is known, give an expression for the total thermal energy contained in a rod ($0 < x < L$).

For an arbitrary thin slice, the total thermal energy is

$$e(x, t) A \Delta x = c(x) u(x, t) \rho(x) A \Delta x$$

Integrating for over the length of the rod, we derive

$$E(t) = A \int_0^L c(x) u(x, t) \rho(x) dx$$

Question: 1.2.9

Consider a thin one-dimensional rod without sources of thermal energy whose lateral surface area is not insulated.

- (a) Assume that the heat energy flowing out of the lateral sides per unit surface area per unit time is $w(x, t)$. Derive the partial differential equation for the temperature $u(x, t)$.
- (b) Assume that $w(x, t)$ is proportional to the temperature difference between the rod $u(x, t)$ and a known outside temperature $\gamma(x, t)$. Derive

$$c\rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0 \frac{\partial u}{\partial x} \right) - \frac{P}{A} [u(x, t) - \gamma(x, t)] h(x)$$

where $h(x)$ is a positive x -dependent proportionality, P is the lateral perimeter, and A is the cross-sectional area.

- (c) Compare (1.2.15) with the equation for a one-dimensional rod whose lateral surfaces are insulated, but with heat sources.
- (d) Specialize (1.2.15) to a rod of circular cross section with constant thermal properties and 0° outside temperature.
- (e) Consider the assumptions in part (d). Suppose that the temperature in the rod is uniform [i.e., $u(x, t) = u(t)$]. Determine $u(t)$ if initially $u(0) = u_0$.

- (a) Suppose that A is the *cross-sectional area* of the rod and P is its perimeter. Since $w(x, t)$ is given as energy per unit surface area per unit time, one intuition we can pursue is to sum over all infinitesimal subdivisions of the total *lateral surface area*. **Note that it is critical that we distinguish between these two separate areas: cross-sectional (over which the heat fluxes ϕ transpire) and lateral (over which heat escapes in the form of $w(x, t)$).** Very simply, the lateral surface area is $P dx$ (for a cylinder, P would be $2\pi r$, and lateral surface area would be that times the length).

$$P \int_a^b w(x, t) dx$$

Observe that this integral is in units of energy per unit time. Taking our integral conservation equation, the temporal change in total energy must then be equal to the difference of the product of the cross-sectional area with the spatial derivative of the fluxes and the outflow of heat energy per unit time.

$$\frac{d}{dt} \int_a^b e(x, t) A dx = [\phi(a, t) - \phi(b, t)] A - P \int_a^b w(x, t) dx$$

Which simplifies to, after dividing through by cross-sectional area A

$$\int_a^b \frac{\partial e}{\partial t} dx = - \int_a^b \frac{\partial \phi}{\partial x} dx - \frac{P}{A} \int_a^b w(x, t) dx$$

and from the arbitrariness of the limits, must produce

$$\frac{\partial e}{\partial t} = -\frac{\partial \phi}{\partial x} - \frac{P}{A}w(x, t)$$

Switching out $e(x, t)$ for $c\rho u(x, t)$ and incorporating Fourier's law in place of ϕ gives us

$$c\rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0 \frac{\partial u}{\partial x} \right) - \frac{P}{A}w(x, t)$$

- (b) Suppose that the proportionality between $w(x, t)$ and $u(x, t) - \gamma(x, t)$ varies as a positive function of x . Call that function $h(x)$. Now we can write

$$w(x, t) = [u(x, t) - \gamma(x, t)] h(x)$$

Then substitute the $w(x, t)$ from the last term of the equation from part (a) with

$$c\rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0 \frac{\partial u}{\partial x} \right) - \frac{P}{A}[u(x, t) - \gamma(x, t)] h(x)$$

Since $h(x)$ is positive, we need not worry about dealing with sign alterations on that last term. Physically, suppose that $u > \gamma$. Then the outside temperature is cooler, so heat transfer is in the opposite direction of the temperature gradient (from the rod to the outside), hence the final term is negative. In the converse direction when $u < \gamma$, heat flows from the outside to the rod, and the final term is positive. At equality, the final term vanishes, and whether or not the lateral surface area is insulated becomes irrelevant in this special case.

- (c) In lieu of non-insulation and in the presence of heat sources, we have

$$c\rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0 \frac{\partial u}{\partial x} \right) + Q$$

One physical takeaway here is whether Q serves as a **source** or a **sink**, in which the sign is positive and negative, respectively. This is analogous to whether heat flows inward or outward, depending on the lateral temperature change. The chief difference is in whether the contribution or subtraction from the temporal change in energy originates from an external lateral temperature gradient or an internal source or sink.

- (d) Let $\gamma = 0$. Moreover, $h(x) = h$, $P = 2\pi r$, and $A = \pi r^2$ by the assumption of constant thermal properties, where r is the radius of the rod. We have implicitly taken c , ρ , and K_0 to be constants throughout, but let us explicitly assert that they are constants. Then

$$c\rho \frac{\partial u}{\partial t} = K_0 \frac{\partial^2 u}{\partial x^2} - \frac{2h}{r}u(x, t)$$

Now might be a good time to note the scale invariance of the equation: irrespective of the choice of units (e.g. Kelvin, Celsius, or Fahrenheit) the equation holds.

- (e) Assume spatial uniformity in u . This is a truly delightful assumption, for now the second spatial derivative is annihilated, and our partial differential equation reduces to a first-order ordinary differential equation.

$$u' = -\frac{2h}{c\rho r}u$$

By separable variables, integrate

$$\begin{aligned}\int_{u_0}^{u(t)} \frac{dU}{U} &= -\frac{2h}{c\rho r} \int_0^t dT \\ \ln\left(\frac{u(t)}{u_0}\right) &= -\frac{2h}{c\rho r}t \\ u(t) &= u_0 e^{-\frac{2h}{c\rho r}t}\end{aligned}$$

The solution under this regime of assumptions is that the temperature decreases exponentially with time from the initial temperature of $u(0) = u_0$. In the long run as $t \rightarrow \infty$, $u(t)$ tends towards 0, and the rod equilibrates with the outside temperature.

1.3 - Boundary Conditions

Question: 1.3.1

Consider a one-dimensional rod, $0 \leq x \leq L$. Assume that the heat energy flowing out of the rod at $x = L$ is proportional to the temperature difference between the end temperature of the bar and the known external temperature. Derive (1.3.5); briefly, physically explain why $H > 0$.

Recall that our sign convention for $\phi(x, t)$ is **positive** for rightward thermal energy flow per time per unit surface area. Let $u(L, t)$ and $u_B(t)$ be the temperature of the bar at the right boundary and the external temperature, respectively. Then if H , the heat transfer coefficient, is the proportionality constant, we have

$$\phi(L, t) = H[u(L, t) - u_B(t)]$$

We need to be clear about the sign convention. Suppose that $u(L, t) > u_B(t)$. Then heat energy will flow from the rod to the external bath. In order to maintain consistency with our sign convention for ϕ , this forces $H > 0$. Conversely, if $u(L, t) < u_B(t)$, then heat energy propagates from the outside to the rod (right to left), so we require $\phi < 0$. Only if $H > 0$ does that condition work. Therefore combining with the Fourier heat law, we have equation (1.3.5)

$$-K_0(L) \frac{\partial u}{\partial x}(L, t) = H[u(L, t) - u_B(t)]$$

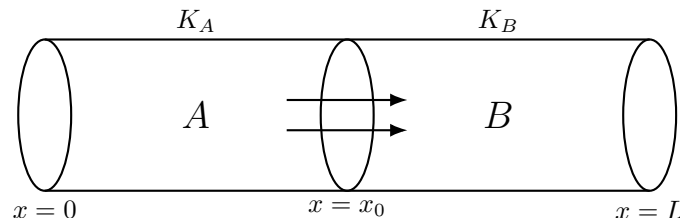
Question: 1.3.2

Two one-dimensional rods of different materials joined at $x = x_0$ are said to be in **perfect thermal contact** if the temperature is continuous at $x = x_0$:

$$u(x_0-, t) = u(x_0+, t)$$

and no heat energy is lost at $x = x_0$ (i.e., the heat energy flowing out of one flows into the other). What mathematical equation represents the latter condition at $x = x_0$? Under what special condition is $\partial u / \partial x$ continuous at $x = x_0$?

Consider two rods labeled A and B . Since we do not assume they consist of identical materials, we must generalize by assuming two different temperature functions u_A , u_B and thermal conductivity constants K_A , K_B .



Some physical insights we must draw: firstly, for there to be no loss of heat energy, the **thermal energy fluxes at the threshold must be equal in**

magnitude, for if they were not, heat energy would either accrue or diminish at $x = x_0$, violating the conservation assumption. There is no addition or subtraction of thermal energy from the outside environment. Secondly, the fluxes must be identical in **direction**, as otherwise we would violate the premise that heat energy propagates from high to cold temperatures. In other words, **the fluxes across A and B are equal** in the limit as $x_0-, x_0+ \rightarrow x_0$.

$$\phi_A(x_0-, t) = \phi_B(x_0+, t)$$

Using the Fourier heat law and canceling the sign, we have

$$K_A(x_0-) \frac{\partial u_A}{\partial x}(x_0-, t) = K_B(x_0+) \frac{\partial u_B}{\partial x}(x_0+, t)$$

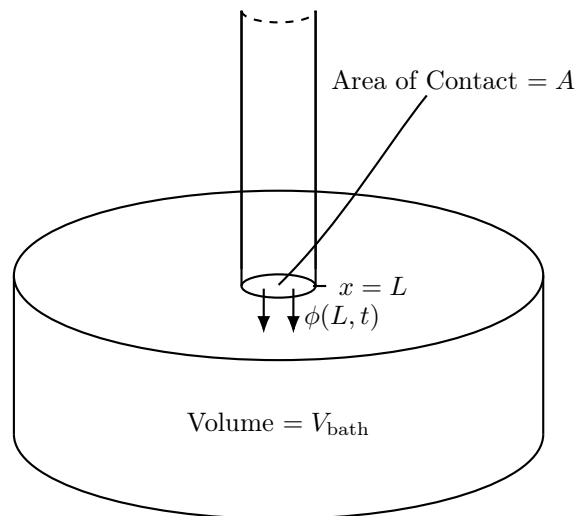
which is our equation for imposing heat conservation at the junction of the rods. To further enforce continuity of the spatial derivative, we must have the partial derivatives equal to each other, which implies that it is only equality of the conductivity constants that leads to continuity of the spatial derivatives.

$$K_A(x_0-) = K_B(x_0+)$$

Question: 1.3.3

Consider a bath containing a fluid of specific heat c_f and mass density ρ_f that surrounds the end $x = L$ of a one-dimensional rod. Suppose that the bath is rapidly-stirred in a manner such that the bath temperature is approximately uniform throughout, equaling the temperature at $x = L$, $u(L, t)$. Assume that the bath is thermally insulated except at its perfect thermal contact with the rod, where the bath may be heated or cooled by the rod. Determine an equation for the temperature in the bath. (This will be a boundary condition at the end $x = L$.) (*Hint: See Exercise 1.3.2.*)

Before we begin, it helps to illustrate the scenario. Consider a bath with volume V_{bath} , and let the area of the contact between the rod and the bath be A . The conceptual setup should look something like this:



With the given quantities, one natural first step might be to calculate the total energy of the bath as

$$e(L, t) = c_f \rho_f V_{\text{bath}} u(L, t)$$

Notice here e is defined in the textbook as an energy density, so we will make a slight alteration to that definition by multiplying through by the bath volume to express in terms of energy alone. Moreover, notice that this is an equation purely in terms of time t . Differentiating with respect to time gives us

$$\frac{\partial e}{\partial t} = c_f \rho_f V_{\text{bath}} \frac{\partial u}{\partial t}(L, t)$$

From exercise 1.3.2, the physical intuition we are meant to derive is the insight that at the boundary $x = L$, **the heat energy fluxes on either infinitesimal flank of the threshold $x = L$ are equal.**

$$\phi(L-, t) = \phi(L+, t)$$

Originally, in deriving the heat equation, we considered a sub-section within a single rod, and identified a spatial gradient of the fluxes between the endpoints of that sub-section. In this case, we only have a *singular* point of contact with an external source or sink (the rod), and thus we only have the flux ϕ to concern ourselves with.

Being perspicacious with units, you will observe that $\partial e / \partial t$ is in units of W. However, the flux ϕ is in units W/m². One natural intuition to consider is the geometry of the rod is relevant, and particularly the area of the cross-sectional slice of the rod that touches the bath. Then

$$\phi(L, t) A = -K_0 \frac{\partial u(L, t)}{\partial x} A$$

Specifically, due to energy conservation, the only contribution or subtraction of the total heat energy in the bath will be due to the exchange of heat energy, or the flux, over the perfect thermal contact. Ergo, our final equation is

$$\frac{\partial e}{\partial t} = c_f \rho_f V_{\text{bath}} \frac{\partial u(L, t)}{\partial t} = -K_0 \frac{\partial u(L, t)}{\partial x} A$$

Conceptually, suppose the temperature of the bath increases with time. Then the initial temperature of the rod is warmer than that of the bath, and $\partial u / \partial x < 0$ corresponds to $\partial u / \partial t > 0$. Vice-versa, if $\partial u / \partial t < 0$ —temperature of the bath drops over time—then the rod is relatively cooler than the bath, and heat energy moves in the opposite direction.

1.4 - Equilibrium Temperature Distribution

Question: 1.4.1

Determine the equilibrium temperature distribution for a one-dimensional rod with constant thermal properties with the following sources and boundary conditions:

- (a) $Q = 0, \quad u(0) = 0, \quad u(L) = T$
- (b) $Q = 0, \quad u(0) = T, \quad u(L) = 0$
- (c) $Q = 0, \quad \frac{\partial u}{\partial x}(0) = 0, \quad u(L) = T$
- (d) $Q = 0, \quad u(0) = T, \quad \frac{\partial u}{\partial x}(L) = \alpha$
- (e) $\frac{Q}{K_0} = 1, \quad u(0) = T_1, \quad u(L) = T_2$
- (f) $\frac{Q}{K_0} = x^2, \quad u(0) = T, \quad \frac{\partial u}{\partial x}(L) = 0$
- (g) $Q = 0, \quad u(0) = T, \quad \frac{\partial u}{\partial x}(L) + u(L) = 0$
- (h) $Q = 0, \quad \frac{\partial u}{\partial x}(0) - [u(0) - T] = 0, \quad \frac{\partial u}{\partial x}(L) = \alpha$

In these you may assume that $u(x, 0) = f(x)$.

- (a) No source, so we satisfy $d^2u/dx^2 = 0$ and derive $u(x) = C_1x + C_2$. The boundary conditions imply

$$\begin{aligned} u(0) &= C_2 = 0 \\ u(L) &= C_1L = T \end{aligned}$$

Leading us to equilibrium temperature distribution $u(x) = \frac{T}{L}x$.

- (b) Flip the boundary conditions from the previous exercise, and find

$$\begin{aligned} u(0) &= C_2 = T \\ u(L) &= C_1L + C_2 = 0 \\ \implies C_1 &= -\frac{T}{L} \end{aligned}$$

The equilibrium temperature distribution is $u(x) = -\frac{T}{L}x + T$.

- (c) We have the same temperature function $u(x) = C_1x + C_2$, but this time with a first derivative boundary condition. Derive

$$\begin{aligned} \frac{\partial u}{\partial x}(0) &= C_1 = 0 \\ u(L) &= C_2 = T \end{aligned}$$

Leaving us with $u(x) = T$. Intuitively, this is correct because to have a zero spatial derivative at one end for a linear temperature distribution immediately implies a horizontal function.

- (d) Once more, $u(x) = C_1x + C_2$. We can intuitively infer that the line will have slope α and intercept T at $x = 0$. Applying the boundary conditions gives us

$$\begin{aligned} u(0) &= C_2 = T \\ \frac{\partial u}{\partial x}(L) &= C_1 = \alpha \end{aligned}$$

Leading us to $u(x) = \alpha x + T$. Our intuition proved correct for this simple case, but it is a lesson in examining how the nature of internal sources or boundary conditions can influence the shape of the equilibrium temperature distribution.

- (e) Now, our source is non-zero. Setting $\partial u / \partial t = 0$, we have

$$K_0 \frac{d^2 u}{dx^2} + K_0 = 0$$

Divide through by K_0 and isolate the second derivative to get

$$\frac{d^2 u}{dx^2} = -1$$

Integrating twice yields the quadratic

$$u(x) = -\frac{x^2}{2} + C_1x + C_2$$

Imposing the boundary conditions, derive

$$\begin{aligned} u(0) &= C_2 = T_1 \\ u(L) &= -\frac{L^2}{2} + C_1L + T_1 = T_2 \\ \implies C_1 &= \frac{2(T_2 - T_1) + L^2}{2L} \end{aligned}$$

Leaving us with

$$u(x) = -\frac{x^2}{2} + \left[\frac{2(T_2 - T_1) + L^2}{2L} \right] x + T_1$$

- (f) From

$$K_0 \frac{d^2 u}{dx^2} + K_0 x^2 = 0$$

Derive

$$u(x) = -\frac{x^4}{12} + C_1x + C_2$$

And apply boundary conditions

$$\begin{aligned} u(0) &= C_2 = T \\ \frac{\partial u}{\partial x}(L) &= -\frac{L^3}{3} + C_1 = 0 \\ \implies C_1 &= \frac{L^3}{3} \end{aligned}$$

Finishing with temperature distribution

$$u(x) = -\frac{x^4}{12} + \frac{L^3}{3}x + T$$

(g) Returning again to the form $u(x) = C_1x + C_2$, apply boundary conditions

$$\begin{aligned} u(0) &= C_2 = T \\ \frac{\partial u}{\partial x}(L) + u(L) &= C_1(1 + L) + T = 0 \\ C_1 &= -\frac{T}{1 + L} \end{aligned}$$

Leading to

$$u(x) = T\left(1 - \frac{x}{1 + L}\right)$$

(h) Lastly, we finish up this set with, once more, temperature distribution form $u(x) = C_1x + C_2$. Applying the boundary conditions gives us coefficients

$$\begin{aligned} \frac{\partial u}{\partial x}(0) - [u(0) - T] &= C_1 - C_2 + T = 0 \\ \frac{\partial u}{\partial x}(L) &= C_1 = \alpha \\ \implies C_2 &= \alpha + T \end{aligned}$$

and concluding with

$$u(x) = \alpha(x + 1) + T$$

Question: 1.4.2

Consider the equilibrium temperature distribution for a uniform one-dimensional rod with sources $Q/K_0 = x$ of thermal energy, subject to the boundary conditions $u(0) = 0$ and $u(L) = 0$.

- Determine the heat energy generated per unit time inside the entire rod.
- Determine the heat energy flowing out of the rod per unit time at $x = 0$ and at $x = L$.
- What relationships should exist between the answers in parts (a) and (b)?

- Conceptually, we have to consider $Q/K_0 = x$ as a **distributed source** of thermal energy, whilst the problem asks about the rate of heat energy generation **for the entire rod**. Recall that Q itself is a measure of heat energy per unit volume per unit time. Therefore, the problem entails summing over all of these distributed sources of heat energy internal to the rod, or

$$Q_{\text{total}} = \int_0^L Q \, dx = K_0 \int_0^L x \, dx = \frac{K_0 L^2}{2}$$

- The objective here is to calculate the fluxes $-K_0 \partial u / \partial x$ at the boundaries. Derive the equilibrium temperature distribution from

$$K_0 \frac{d^2 u}{dx^2} + K_0 x = 0 \quad \implies \quad u(x) = -\frac{x^3}{6} + C_1 x + C_2$$

where applying boundary conditions yields $u(0) = C_2 = 0$ and $u(L) = -L^3/6 + C_1 L = 0$, or $C_1 = L^2/6$. Then our function is

$$u(x) = -\frac{x^3}{6} + \frac{L^2}{6} x$$

which has spatial derivative

$$\frac{\partial u}{\partial x} = -\frac{x^2}{2} + \frac{L^2}{6}$$

The fluxes are then

$$\begin{aligned} \phi(0, t) &= -K_0 \frac{\partial u}{\partial x}(0, t) = -\frac{K_0 L^2}{6} \\ \phi(L, t) &= -K_0 \frac{\partial u}{\partial x}(L, t) = K_0 \frac{L^2}{3} \end{aligned}$$

And since $K_0 > 0$, the directionality of the fluxes is clear. Heat energy propagates leftward at $x = 0$ and rightward at $x = L$. Thus at both ends, **energy leaves the rod**. Now, taking a step back, consider the flux component of the heat equation, which gives us the combined heat energy outflow per unit time at *both* boundaries.

$$-\int_0^L \frac{\partial \phi}{\partial x} \, dx = K_0 \frac{\partial u}{\partial x}(L, t) - K_0 \frac{\partial u}{\partial x}(0, t) = -\frac{K_0 L^2}{2}$$

- (c) Returning to the key physical premise of the problem: that we are at **equilibrium state** for temperature, definitionally we know that $\partial u / \partial t = 0$. Equivalently, $\partial e / \partial t = 0$. **We can then deduce for the entire bar**

$$\frac{\partial}{\partial t} \int_0^L e \, dx = - \int_0^L \frac{\partial \phi}{\partial x} \, dx + Q_{\text{total}} = 0$$

By conservation of heat energy within the rod, the rate of internal heat energy generation must match the rate of outflow at the boundaries. Indeed, we can confirm

$$\frac{\partial}{\partial t} \int_0^L e \, dx = - \int_0^L \frac{\partial \phi}{\partial x} \, dx + Q_{\text{total}} = - \frac{K_0 L^2}{2} + \frac{K_0 L^2}{2} = 0$$

Question: 1.4.3

Determine the equilibrium temperature distribution for a one-dimensional rod composed of two different materials in perfect thermal contact at $x = 1$. For $0 < x < 1$, there is one material ($c\rho = 1$, $K_0 = 1$) with a constant source ($Q = 1$), whereas for the other $1 < x < 2$, there are no sources ($Q = 0$, $c\rho = 2$, $K_0 = 2$) (see Exercise 1.3.2) with $u(0) = 0$ and $u(2) = 0$.

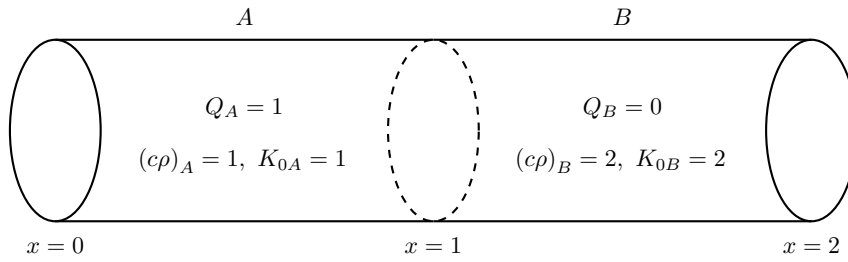
By the assumption of perfect thermal contact, we know that

- (a) the temperatures at the boundary must be equal:

$$u_A(1-, t) = u_B(1+, t)$$

- (b) the fluxes at the boundary are equal:

$$\phi_A(1-, t) = \phi_B(1+, t)$$



First, let us derive the temperature distributions for sections A and B to the extent that we can. In A, we have

$$K_{0A} \frac{d^2 u_A}{dx^2} + 1 = 0$$

Isolating the second derivative and integrating twice, derive

$$u_A(x) = -\frac{x^2}{2} + C_{1A}x + C_{2A}$$

Now, from the boundary condition $u_A(0) = 0$, we immediately deduce $C_{2A} = 0$, so now we are left with

$$u_A(x) = -\frac{x^2}{2} + C_{1A}x$$

Moving onto section B , we have

$$\frac{d^2 u_B}{dx^2} = 0$$

which yields

$$u_B(x) = C_{1B}x + C_{2B}$$

Imposing the second boundary condition, we can further conclude

$$u_B(2) = 2C_{1B} + C_{2B} = 0 \implies C_{2B} = -2C_{1B}, \quad u_B(x) = C_{1B}(x - 2)$$

The last component we need is the fluxes, and thus spatial derivatives for both sections, which are

$$\begin{aligned} K_{0A} \frac{\partial u_A}{\partial x} &= -x + C_{1A} \\ K_{0B} \frac{\partial u_B}{\partial x} &= 2C_{1B} \end{aligned}$$

Finally, we may apply boundary conditions (a) and (b) at $x = 1$.

$$\begin{aligned} u_A(1-, t) &= -\frac{1}{2} + C_{1A} = -C_{1B} = u_B(1+, t) \\ K_{0A} \frac{\partial u_A}{\partial x}(1-, t) &= -1 + C_{1A} = 2C_{1B} = K_{0B} \frac{\partial u_B}{\partial x}(1+, t) \end{aligned}$$

Solve for this using matrix algebra, using which we find

$$C_{1A} = \frac{2}{3}, \quad C_{1B} = -\frac{1}{6}, \quad C_{2B} = \frac{1}{3}$$

Giving us equilibrium temperature distribution

$$u(x) = \begin{cases} -\frac{x^2}{2} + \frac{2}{3}x, & 0 < x < 1 \\ -\frac{x}{6} + \frac{1}{3}, & 1 < x < 2 \end{cases}$$

Observe that $c\rho$ were irrelevant to this problem, as they matter only in the transient, not equilibrium state.

Question: 1.4.4

If both ends of a rod are insulated, derive *from the partial differential equation* that the total thermal energy in the rod is constant.

By the insulation assumption, we must have

$$\frac{\partial u}{\partial x}(0, t) = \frac{\partial u}{\partial x}(L, t) = 0$$

Suppose further that $u(x, 0) = f(x)$, and c and ρ the specific heat and density, respectively. Consider the integral conservation equation

$$\frac{d}{dt} \underbrace{\int_0^L e \, dx}_{\text{total thermal energy}} = \frac{d}{dt} \int_0^L c\rho u(x, t) \, dx = - \int_0^L \frac{\partial \phi}{\partial x} \, dx = K_0 \frac{\partial u}{\partial x}(L, t) - K_0 \frac{\partial u}{\partial x}(0, t) = 0$$

Since the derivative of the total energy density is zero, we may deduce that it is constant.

Question: 1.4.5

Consider a one-dimensional rod $0 \leq x \leq L$ of known length and known constant thermal properties without sources. Suppose that the temperature is an *unknown* constant T at $x = L$. Determine T if we know (in the steady state) both the temperature and the heat flow at $x = 0$.

Let $u(0) = T_1$ and $\partial u / \partial x(0)$ be known quantities. Since the rod is bereft of internal heat sources, the steady state temperature distribution is of form $u(x) = C_1 x + C_2$. From the boundary conditions, $u(0) = C_2 = T_1$ and $\partial u / \partial x(0) = C_1$. Then $u(L)$ is

$$u(L) = \frac{\partial u}{\partial x}(0) L + T_1$$

Question: 1.4.6

The two ends of a uniform rod of length L are insulated. There is a constant source of thermal energy $Q_0 \neq 0$, and the temperature is initially $u(x, 0) = f(x)$.

- Show mathematically that there does not exist any equilibrium temperature distribution. Briefly explain physically.
- Calculate the total thermal energy in the entire rod.

- Since insulation implies there is no heat energy flux at the rod's boundaries, the integral conservation equation reduces to

$$\frac{d}{dt} \int_0^L c\rho u \, dx = \int_0^L Q_0 \, dx = Q_0 L \neq 0$$

which implies that the rate of change of the total thermal energy is never zero. To deduce that we also cannot have $\partial u / \partial t = 0$, suppose otherwise. Then $\partial u / \partial t = 0$ implies

$$\int_0^L c\rho \frac{\partial u}{\partial t} \, dx = 0$$

which immediately contradicts our result. Thus this precludes the existence of an equilibrium temperature distribution.

- (b) Integrating from time 0 to t and using the dummy variable T , derive

$$\int_0^t \left[\frac{d}{dT} \int_0^L c\rho u(x, T) dx \right] dT = \int_0^t \left[\int_0^L Q_0 dx \right] dT$$

$$\int_0^L c\rho u(x, t) dx = \underbrace{\int_0^L c\rho f(x) dx}_{\text{initial thermal energy}} + \underbrace{Q_0 L t}_{\text{internal source}}$$

where we make the substitution $u(x, 0) = f(x)$. Intuitively, the total thermal energy is a sum of the initial thermal energy and the energy contributed by the heat source over time.

Question: 1.4.7

For the following problems, determine an equilibrium temperature distribution (if one exists). For what values of β are there solutions? Explain physically.

- (a) $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + 1$, $u(x, 0) = f(x)$, $\frac{\partial u}{\partial x}(0, t) = 1$, $\frac{\partial u}{\partial x}(L, t) = \beta$
- (b) $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $u(x, 0) = f(x)$, $\frac{\partial u}{\partial x}(0, t) = 1$, $\frac{\partial u}{\partial x}(L, t) = \beta$
- (c) $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + x - \beta$, $u(x, 0) = f(x)$, $\frac{\partial u}{\partial x}(0, t) = 0$, $\frac{\partial u}{\partial x}(L, t) = 0$

- (a) Proceed by letting $\partial u / \partial t = 0$ to see what becomes of this assumption. We have temperature distribution

$$u(x) = -\frac{x^2}{2} + C_1 x + C_2$$

which has spatial derivative $\partial u / \partial x = -x + C_1$. Then $\partial u / \partial x(0, t) = C_1 = 1$ and $\partial u / \partial x(L, t) = 1 - L = \beta$; this is the only permissible value of β for an equilibrium to exist. Unfortunately, we are at a dead end for identifying C_2 , but this merely sets the vertical position of the equilibrium temperature distribution and can be determined if prescribed temperatures at the ends are given.

- (b) Let $\partial u / \partial t = 0$. No source, so our temperature distribution is

$$u(x) = C_1 x + C_2$$

and spatial derivative $\partial u / \partial x = C_1$. Since this is not a function of x and is constant throughout the rod, the boundary conditions at either end must be equal, so we must have $C_1 = \beta = 1$ for an equilibrium to exist. Again, we have a situation in which C_2 cannot be determined, but prescribing temperatures at the boundaries rectifies this.

- (c) Let $\partial u / \partial t = 0$. Incorporating the distributed heat source, we have temperature distribution

$$u(x) = -\frac{x^3}{6} + \frac{\beta x^2}{2} + C_1 x + C_2$$

Then our spatial derivative is

$$\frac{\partial u}{\partial x} = -\frac{x^2}{2} + \beta x + C_1$$

Evaluating at the ends, we get $\partial u / \partial x(0, t) = C_1 = 0$ and $\partial u / \partial x(L, t) = \beta L - L^2/2 = 0$, implying that $\beta = L/2$ for the equilibrium state.

Question: 1.4.8

Express the integral conservation law for the entire rod with constant thermal properties. Assume the heat flow is known to be different constants at both ends. By integrating with respect to time, determine the total thermal energy in the rod. (Hint: Use the initial condition.)

- (a) Assume there are no sources.
- (b) Assume the sources of thermal energy are constant.

In both cases, let the initial condition be $u(x, 0) = f(x)$. Let $\phi(0, t) = \alpha$ and $\phi(L, t) = \beta$.

- (a)

$$\begin{aligned} \frac{d}{dt} \int_0^L c \rho u \, dx &= - \int_0^L \frac{\partial \phi}{\partial x} \, dx \\ &= \phi(0, t) - \phi(L, t) \\ &= \alpha - \beta \end{aligned}$$

Integrate from time 0 to t to get

$$\begin{aligned} \int_0^t \left[\frac{d}{dT} \int_0^L c \rho u(x, T) \, dx \right] dT &= \int_0^t (\alpha - \beta) \, dT \\ \Rightarrow \int_0^L c \rho u(x, t) \, dx &= \int_0^L c \rho f(x) \, dx + (\alpha - \beta) t \end{aligned}$$

- (b)

$$\begin{aligned} \frac{d}{dt} \int_0^L c \rho u \, dx &= - \int_0^L \frac{\partial \phi}{\partial x} \, dx + \int_0^L Q \, dx \\ &= \phi(0, t) - \phi(L, t) + QL \\ &= \alpha - \beta + QL \end{aligned}$$

Integrate from time 0 to t to get

$$\begin{aligned} \int_0^t \left[\frac{d}{dT} \int_0^L c\rho u(x, T) dx \right] dT &= \int_0^t (\alpha - \beta + QL) dT \\ \Rightarrow \int_0^L c\rho u(x, t) dx &= \int_0^L c\rho f(x) dx + (\alpha - \beta + QL)t \end{aligned}$$

Question: 1.4.9

Derive the integral conservation law for the entire rod with constant thermal properties by integrating the heat equation (1.2.10) (assuming no sources). Show the result is equivalent to (1.2.4).

The heat equation is

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

where $k = K_0/c\rho$. Multiply through by $c\rho$ and integrate with respect to x over $a \leq x \leq b$ to get

$$\begin{aligned} \int_a^b c\rho \frac{\partial u}{\partial t} dx &= K_0 \int_a^b \frac{\partial^2 u}{\partial x^2} dx \\ &= K_0 \left[\frac{\partial u}{\partial x}(b, t) - \frac{\partial u}{\partial x}(a, t) \right] \\ &= \phi(a, t) - \phi(b, t) \end{aligned}$$

By Leibniz's rule, we can pull the differential operator out of the integral. We restore the integral conservation law:

$$\frac{d}{dt} \int_a^b c\rho u dx = \phi(a, t) - \phi(b, t)$$

Question: 1.4.10

Suppose $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + 4$, $u(x, 0) = f(x)$, $\frac{\partial u}{\partial x}(0, t) = 5$, $\frac{\partial u}{\partial x}(L, t) = 6$. Calculate the total thermal energy in the one-dimensional rod (as a function of time).

By the integral conservation law,

$$\begin{aligned} \frac{d}{dt} \int_0^L c\rho u dx &= \frac{\partial u}{\partial x}(L, t) - \frac{\partial u}{\partial x}(0, t) + \int_0^L 4 dx \\ &= 6 - 5 + 4L \\ &= 1 + 4L \end{aligned}$$

Integrate from time 0 to t to get

$$\begin{aligned} \int_0^t \left[\frac{d}{dT} \int_0^L c\rho u(x, T) dx \right] dT &= \int_0^t (1 + 4L) dT \\ \Rightarrow \int_0^L c\rho u(x, t) dx &= \int_0^L c\rho f(x) dx + (1 + 4L)t \end{aligned}$$

Question: 1.4.11

Suppose $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + x$, $u(x, 0) = f(x)$, $\frac{\partial u}{\partial x}(0, t) = \beta$, $\frac{\partial u}{\partial x}(L, t) = 7$.

- Calculate the total thermal energy in the one-dimensional rod (as a function of time).
- From part (a), determine the value of β for which an equilibrium exists. For this value of β , determine $\lim_{t \rightarrow \infty} u(x, t)$.

- (a) By the integral conservation law,

$$\begin{aligned} \frac{d}{dt} \int_0^L c\rho u dx &= \frac{\partial u}{\partial x}(L, t) - \frac{\partial u}{\partial x}(0, t) + \int_0^L x dx \\ &= 7 - \beta + \frac{L^2}{2} \end{aligned}$$

Integrating over time from 0 to t , derive

$$\begin{aligned} \int_0^t \left[\frac{d}{dT} \int_0^L c\rho u(x, T) dx \right] dT &= \int_0^t \left(7 - \beta + \frac{L^2}{2} \right) dT \\ \Rightarrow \int_0^L c\rho u(x, t) dx &= \int_0^L c\rho f(x) dx + \left(7 - \beta + \frac{L^2}{2} \right) t \end{aligned}$$

- (b) Let $\partial u / \partial t = 0$. Then

$$u(x) = -\frac{x^3}{6} + C_1 x + C_2$$

with spatial derivative

$$\frac{\partial u}{\partial x} = -\frac{x^2}{2} + C_1$$

Applying the boundary conditions, we have

$$\begin{aligned} \frac{\partial u}{\partial x}(0, t) &= C_1 = \beta \\ \frac{\partial u}{\partial x}(L, t) &= \beta - \frac{L^2}{2} = 7 \\ \beta &= 7 + \frac{L^2}{2} \end{aligned}$$

Then in the steady-state, the temperature distribution is

$$\lim_{t \rightarrow \infty} u(x, t) = u(x) = -\frac{x^3}{6} + \left(7 + \frac{L^2}{2}\right)x + C_2$$

Where C_2 is an arbitrary constant dependent on the initial condition.

Question: 1.4.12

Suppose the concentration $u(x, t)$ of a chemical satisfies Fick's law (1.2.13), and the initial concentration is given as $u(x, 0) = f(x)$. Consider a region $0 < x < L$ in which the flow is specified at both ends $-k \frac{\partial u}{\partial x}(0, t) = \alpha$ and $-k \frac{\partial u}{\partial x}(L, t) = \beta$. Assume α and β are constants.

- Express the conservation law for the entire region.
- Determine the total amount of chemical in the region as a function of time (using the initial condition).
- Under what conditions is there an equilibrium chemical concentration, and what is it?

(a)

$$\frac{d}{dt} \int_0^L u \, dx = - \int_0^L \frac{\partial \phi}{\partial x} \, dx = -k \left[\frac{\partial u}{\partial x}(0, t) - \frac{\partial u}{\partial x}(L, t) \right] = \alpha - \beta$$

- (b) Since we define $u(x, t)$ as the concentration of the chemical per unit volume, we will need to multiply through by the cross-sectional area of the rod A . Integrate over time from 0 to t to get

$$\begin{aligned} \int_0^t \left[\frac{d}{dT} \int_0^L u(x, T) A \, dx \right] dT &= A \int_0^t (\alpha - \beta) dT \\ \Rightarrow \int_0^L u(x, t) A \, dx &= A \left[\int_0^L f(x) \, dx + (\alpha - \beta)t \right] \end{aligned}$$

- (c) Let $\partial u / \partial t = 0$. With zero sources, we have chemical concentration

$$u(x) = C_1 x + C_2$$

Applying the boundary conditions, we have

$$\frac{\partial u}{\partial x}(0, t) = -\frac{\alpha}{k} = C_1 = -\frac{\beta}{k} = \frac{\partial u}{\partial x}(L, t)$$

from which we can conclude that equilibrium exists only when $\alpha = \beta$.

Question: 1.4.13

Do Exercise 1.4.12 if α and β are functions of time.

- (a) Now we assume α and β are functions of time, which changes the conservation equation to

$$\frac{d}{dt} \int_0^L u \, dx = - \int_0^L \frac{\partial \phi}{\partial x} \, dx = -k \left[\frac{\partial u}{\partial x}(0, t) - \frac{\partial u}{\partial x}(L, t) \right] = \alpha(t) - \beta(t)$$

- (b) Since α and β are no longer pure constants, we can only consider their integrals in the abstract as part of the total amount of chemical.

$$\int_0^L u(x, t) A \, dx = A \left[\int_0^L f(x) \, dx + \int_0^t (\alpha(T) - \beta(T)) \, dT \right]$$

- (c) Lastly, by assuming $\partial u / \partial t = 0$ and that $\lim_{t \rightarrow \infty} u(x, t) = u(x)$, it still follows that C_1 is a constant. From the boundary conditions, we still must have

$$- \lim_{t \rightarrow \infty} \frac{\alpha(t)}{k} = C_1 = - \lim_{t \rightarrow \infty} \frac{\beta(t)}{k}$$

In other words, $\alpha(t)$ and $\beta(t)$ must approach constants in the temporal limit for a steady-state to exist.

1.5 - Derivation of the Heat Equation in Two or Three Dimensions

Question: 1.5.1

Let $c(x, y, z, t)$ denote the concentration of a pollutant (the amount per unit volume).

- What is an expression for the total amount of pollutant in the region R ?
- Suppose that the flow $\mathbf{J}$ of the pollutant is proportional to the gradient of the concentration. (Is this reasonable?) Express conservation of the pollutant.
- Derive the partial differential equation governing the diffusion of the pollutant.

(a) $\iiint_R c \, dV$

- (b) Let $\mathbf{J} = A\nabla c$. This is simply Fick's law in three dimensions when $A = -k$.

$$\mathbf{J} = -k\nabla c(x, y, z, t) = \frac{\partial c}{\partial x}\hat{\mathbf{i}} + \frac{\partial c}{\partial y}\hat{\mathbf{j}} + \frac{\partial c}{\partial z}\hat{\mathbf{k}}$$

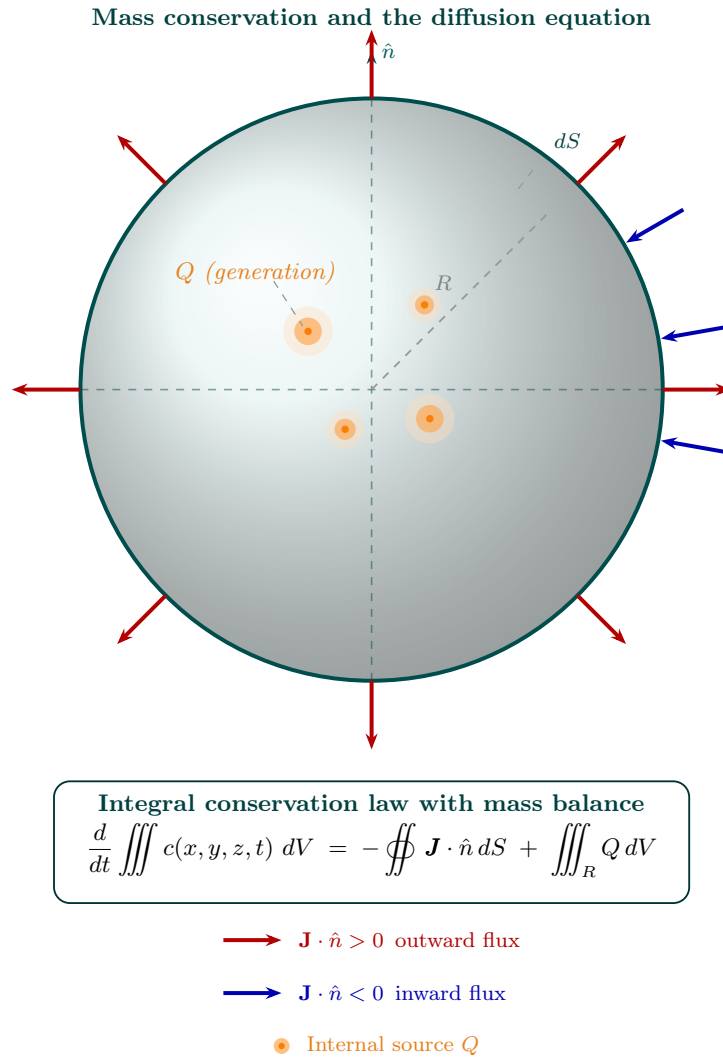
Conservation is when there is no change in mass in the volume, so assuming that the three-dimensional region is invariant with time, we have

$$\frac{d}{dt} \iiint_R c(x, y, z, t) \, dV = 0 \quad \implies \quad \frac{\partial c}{\partial t} = 0$$

- (c) Conceptually, the rate of change of the amount of pollutant is expressed by

$$\begin{array}{ccccc} \text{Rate of change} & & \text{Surface flux} & & \text{Internal pollutant} \\ \text{of amount} & = & \text{of the pollutant} & + & \text{generation} \\ \text{of pollutant} & & \text{per unit time} & & \text{per unit time} \end{array}$$

It is helpful to visualize the physics with a diagram:



Mathematically, we triply integrate over the concentration to derive the total amount of pollutant in the volume, and differentiate to get the rate of change.

$$\text{rate of change of amount of pollutant} = \frac{d}{dt} \iiint_V c(x, y, z, t) dV$$

This is balanced by the sum of the net flux of the pollutant across the volume boundary and the total amount internally generated, which is given by

$$- \oiint_S \mathbf{J} \cdot \hat{n} dS + \iiint_V Q dV$$

where $\hat{n}$ is the unit vector normal to the surface of the volume and Q is the amount of pollutant internally generated per unit volume per unit time. By the divergence theorem, the surface integral can be rewritten as

$$= \iiint_V \nabla \cdot \mathbf{J} dV$$

Combining the terms, we have

$$\frac{d}{dt} \iiint_R c(x, y, z, t) dV = - \iiint_R \nabla \cdot \mathbf{J} dV + \iiint_R Q dV$$

Assembling the integrands together on one side yields

$$\frac{\partial c}{\partial t} + \nabla \cdot \mathbf{J} - Q = 0$$

And using Fick's law in three dimensions as derived in part (b)

$$\mathbf{J} = -k \nabla c$$

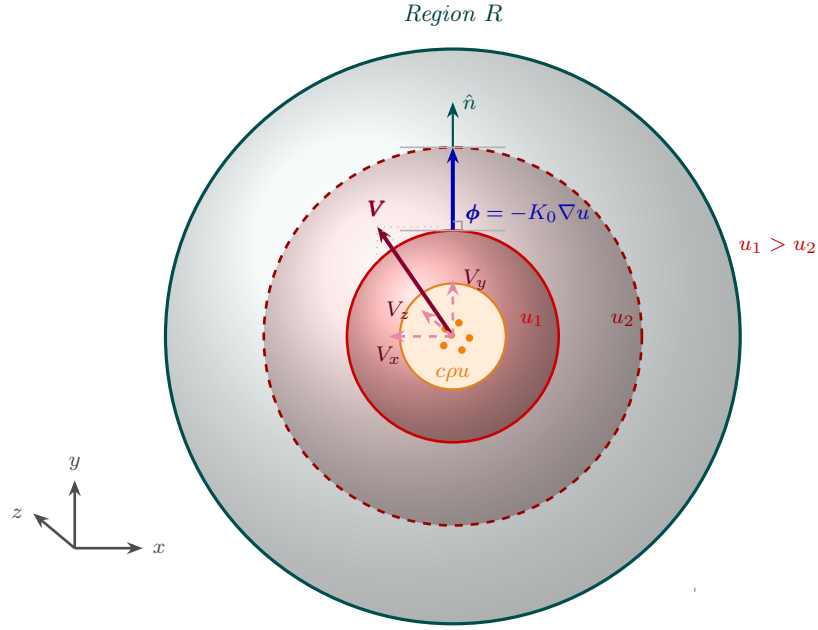
We derive the diffusion equation

$$\frac{\partial c}{\partial t} - k \nabla^2 c - Q = 0$$

Question: 1.5.2

For conduction of thermal energy, the heat flux vector is $\boldsymbol{\phi} = -K_0 \nabla u$. If in addition the molecules move at an average velocity $\mathbf{V}$, a process called **convection**, then briefly explain why $\boldsymbol{\phi} = -K_0 \nabla u + c \rho u \mathbf{V}$. Derive the corresponding equation for heat flow, including both conduction and convection of thermal energy (assuming constant thermal properties with no sources).

Fundamentally, this is a superposition argument. **The total flux is the sum of the energy transfer anti-parallel to the temperature gradient (conduction) and the “movement” of energy by the physical translation of the molecules (convection).**



$$\phi = -K_0 \nabla u + c\rho u \mathbf{V} \quad (\text{conduction} + \text{convection})$$

The addition of the $c\rho u \mathbf{V}$ term arises from the notion of “advecting” energy across space in each of the three spatial dimensions, *in addition* to the spontaneous energy transfer (i.e. heat) in the direction of the negative temperature gradient. We can confirm that $c\rho u \mathbf{V}$ is also in units of flux:

$$\left(\frac{\text{J}}{\text{m}^3} \right) \left(\frac{\text{m}}{\text{s}} \right) = \frac{\text{J}}{\text{m}^2 \cdot \text{s}}$$

Hence total flux is

$$\phi = -K_0 \nabla u + c\rho u \mathbf{V}$$

Substituting into the derivation of the heat equation (with no sources) produces

$$\begin{aligned} c\rho \frac{\partial u}{\partial t} &= -\nabla \cdot (-K_0 \nabla u + c\rho u \mathbf{V}) \\ &= K_0 \nabla^2 u - c\rho (\mathbf{V} \cdot \nabla u + u(\nabla \cdot \mathbf{V})) \end{aligned}$$

But by premise, $\mathbf{V}$ is constant, so that last term vanishes. Dividing through by $c\rho$ and using $k = K_0/c\rho$, we derive

$$\frac{\partial u}{\partial t} = k \nabla^2 u - \mathbf{V} \cdot \nabla u$$

We typically call the first term **diffusion** and the second **advection**.

Question: 1.5.3

Consider the polar coordinates

$$\begin{aligned}x &= r \cos(\theta) \\ y &= r \sin(\theta)\end{aligned}$$

- (a) Since $r^2 = x^2 + y^2$, show that $\frac{\partial r}{\partial x} = \cos(\theta)$, $\frac{\partial r}{\partial y} = \sin(\theta)$, $\frac{\partial \theta}{\partial y} = \frac{\cos(\theta)}{r}$, and $\frac{\partial \theta}{\partial x} = -\frac{\sin(\theta)}{r}$.
- (b) Show that $\hat{\mathbf{r}} = \cos(\theta)\hat{\mathbf{i}} + \sin(\theta)\hat{\mathbf{j}}$ and $\hat{\boldsymbol{\theta}} = -\sin(\theta)\hat{\mathbf{i}} + \cos(\theta)\hat{\mathbf{j}}$.
- (c) Using the chain rule, show that $\nabla = \hat{\mathbf{r}}\frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}}\frac{1}{r}\frac{\partial}{\partial \theta}$ and hence $\nabla u = \frac{\partial u}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial u}{\partial \theta}\hat{\boldsymbol{\theta}}$.
- (d) If $\mathbf{A} = A_r\hat{\mathbf{r}} + A_\theta\hat{\boldsymbol{\theta}}$, show that $\nabla \cdot \mathbf{A} = \frac{1}{r}\frac{\partial}{\partial r}(rA_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(A_\theta)$, since $\partial\hat{\mathbf{r}}/\partial\theta = \hat{\boldsymbol{\theta}}$ and $\partial\hat{\boldsymbol{\theta}}/\partial\theta = -\hat{\mathbf{r}}$ follows from part (b).
- (e) Show that $\nabla^2 u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}$.

(a)

$$2r\frac{\partial r}{\partial x} = 2x\frac{\partial x}{\partial x} = 2r\cos(\theta) \quad \implies \quad \frac{\partial r}{\partial x} = \cos(\theta) \quad (1)$$

$$2r\frac{\partial r}{\partial y} = 2y\frac{\partial y}{\partial y} = 2r\sin(\theta) \quad \implies \quad \frac{\partial r}{\partial y} = \sin(\theta) \quad (2)$$

$$\begin{aligned}\frac{\partial y}{\partial y} &= \frac{\partial r}{\partial y}\sin(\theta) + r\cos(\theta)\frac{\partial \theta}{\partial y} \\ &= \sin^2(\theta) + r\cos(\theta)\frac{\partial \theta}{\partial y}\end{aligned}$$

$$1 = 1 - \cos^2(\theta) + r\cos(\theta)\frac{\partial \theta}{\partial y}$$

$$\implies \cos^2(\theta) = r\cos(\theta)\frac{\partial \theta}{\partial y} \quad \implies \quad \frac{\partial \theta}{\partial y} = \frac{\cos(\theta)}{r} \quad (3)$$

$$\begin{aligned}\frac{\partial x}{\partial x} &= \frac{\partial r}{\partial x}\cos(\theta) - r\sin(\theta)\frac{\partial \theta}{\partial x} \\ &= \cos^2(\theta) - r\sin(\theta)\frac{\partial \theta}{\partial x}\end{aligned}$$

$$1 = 1 - \sin^2(\theta) - r\sin(\theta)\frac{\partial \theta}{\partial x}$$

$$\implies \sin^2(\theta) = -r\sin(\theta)\frac{\partial \theta}{\partial x} \quad \implies \quad \frac{\partial \theta}{\partial x} = -\frac{\sin(\theta)}{r} \quad (4)$$

(b) Let $\mathbf{r} = r\cos(\theta)\hat{\mathbf{i}} + r\sin(\theta)\hat{\mathbf{j}}$. By the Pythagorean theorem, its norm is

$$|\mathbf{r}| = \sqrt{r^2\cos^2(\theta) + r^2\sin^2(\theta)} = r$$

Normalizing, we derive

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|} = \cos(\theta)\hat{\mathbf{i}} + \sin(\theta)\hat{\mathbf{j}}$$

Now, we know that $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ must be spanning basis vectors, which is to say that we must have $\hat{\mathbf{r}} \perp \hat{\boldsymbol{\theta}}$. We need only differentiate with respect to θ to derive this orthogonal basis vector.

$$\hat{\boldsymbol{\theta}} = -\sin(\theta)\hat{\mathbf{i}} + \cos(\theta)\hat{\mathbf{j}}$$

(c) In x and y coordinates, by definition

$$\nabla = \frac{\partial}{\partial x}\hat{\mathbf{i}} + \frac{\partial}{\partial y}\hat{\mathbf{j}}$$

Let $u(r, \theta)$ be a function of r and θ . Consider the total derivatives with respect to x and y :

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x} \\ &= \frac{\partial u}{\partial r} \cos(\theta) - \frac{\partial u}{\partial \theta} \frac{1}{r} \sin(\theta) \\ &= \left(\cos(\theta) \frac{\partial}{\partial r} - \frac{1}{r} \sin(\theta) \frac{\partial}{\partial \theta} \right) u \\ \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y} \\ &= \frac{\partial u}{\partial r} \sin(\theta) + \frac{\partial u}{\partial \theta} \frac{1}{r} \cos(\theta) \\ &= \left(\sin(\theta) \frac{\partial}{\partial r} + \frac{1}{r} \cos(\theta) \frac{\partial}{\partial \theta} \right) u\end{aligned}$$

Combining these expressions for $\partial u/\partial x$ and $\partial u/\partial y$, we have the anticipated result

$$\begin{aligned}\nabla u &= \frac{\partial u}{\partial r} \left[\cos(\theta)\hat{\mathbf{i}} + \sin(\theta)\hat{\mathbf{j}} \right] + \frac{1}{r} \frac{\partial u}{\partial \theta} \left[-\sin(\theta)\hat{\mathbf{i}} + \cos(\theta)\hat{\mathbf{j}} \right] \\ &= \frac{\partial u}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial u}{\partial \theta} \hat{\boldsymbol{\theta}}\end{aligned}$$

(d) The key physical intuition is that the purpose of polar coordinates is to represent *radial symmetry*, and particularly, a frame of reference that *rotates* while retaining the orthogonality of the basis vectors. Put differently, the axes move with rotation, unlike in the Cartesian case, where the positions of $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ are fixed.

Instead of component-wise multiplication as we can afford to do in the Cartesian case, we must completely expand the dot product and calculate

term-by-term.

$$\begin{aligned}
 \nabla \cdot A &= \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} \right) \cdot (A_r \hat{\mathbf{r}} + A_\theta \hat{\boldsymbol{\theta}}) \\
 &= \hat{\mathbf{r}} \frac{\partial}{\partial r} (A_r \hat{\mathbf{r}}) + \hat{\mathbf{r}} \frac{\partial}{\partial r} (A_\theta \hat{\boldsymbol{\theta}}) + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} (A_r \hat{\mathbf{r}}) + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} (A_\theta \hat{\boldsymbol{\theta}}) \\
 &= \hat{\mathbf{r}} \left(\frac{\partial A_r}{\partial r} \hat{\mathbf{r}} + A_r \frac{\partial \hat{\mathbf{r}}}{\partial r} \right) + \hat{\mathbf{r}} \left(\frac{\partial A_\theta}{\partial r} \hat{\boldsymbol{\theta}} + A_\theta \frac{\partial \hat{\boldsymbol{\theta}}}{\partial r} \right) + \hat{\boldsymbol{\theta}} \frac{1}{r} \left(\frac{\partial A_r}{\partial \theta} \hat{\mathbf{r}} + A_r \frac{\partial \hat{\mathbf{r}}}{\partial \theta} \right) \\
 &\quad + \hat{\boldsymbol{\theta}} \frac{1}{r} \left(\frac{\partial A_\theta}{\partial \theta} \hat{\boldsymbol{\theta}} + A_\theta \frac{\partial \hat{\boldsymbol{\theta}}}{\partial \theta} \right) \\
 &= \frac{\partial A_r}{\partial r} + \frac{A_r}{r} + \frac{1}{r} \frac{\partial A_\theta}{\partial \theta} \\
 &= \frac{1}{r} \left(\frac{\partial}{\partial r} (r A_r) + \frac{\partial A_\theta}{\partial \theta} \right)
 \end{aligned}$$

where we use the facts that $\partial \hat{\mathbf{r}} / \partial \theta = \hat{\boldsymbol{\theta}}$, $\partial \hat{\boldsymbol{\theta}} / \partial \theta = -\hat{\mathbf{r}}$, and $\pm \hat{\mathbf{r}} \cdot \pm \hat{\boldsymbol{\theta}} = 0$.

(e) Calculate the Laplacian as

$$\begin{aligned}
 \nabla^2 &= \nabla \cdot \nabla = \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} \right) \cdot \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} \right) \\
 &= \hat{\mathbf{r}} \frac{\partial}{\partial r} \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} \right) + \hat{\mathbf{r}} \frac{\partial}{\partial r} \left(\hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} \right) + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} \left(\hat{\mathbf{r}} \frac{\partial}{\partial r} \right) \\
 &\quad + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} \left(\hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} \right) \\
 &= \hat{\mathbf{r}} \left(\frac{\partial \hat{\mathbf{r}}}{\partial r} \frac{\partial}{\partial r} + \hat{\mathbf{r}} \frac{\partial^2}{\partial r^2} \right) + \hat{\boldsymbol{\theta}} \frac{1}{r} \left(\frac{\partial \hat{\mathbf{r}}}{\partial \theta} \frac{\partial}{\partial r} + \hat{\mathbf{r}} \frac{\partial}{\partial \theta} \frac{\partial}{\partial r} \right) \\
 &\quad + \hat{\boldsymbol{\theta}} \frac{1}{r^2} \left(\frac{\partial \hat{\boldsymbol{\theta}}}{\partial \theta} \frac{\partial}{\partial \theta} + \hat{\boldsymbol{\theta}} \frac{\partial^2}{\partial \theta^2} \right) \\
 &= \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \\
 &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}
 \end{aligned}$$

where in cases where the operator is $\partial / \partial r$, the unit vector is pulled out, and if multiplied by its orthogonal unit vector, the entire term vanishes. Then $\nabla^2 u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$.

Question: 1.5.4

Using Exercise 1.5.3(a) and the chain rule for partial derivatives, derive the special case of Exercise 1.5.3(e) if $u(r)$ only.

Begin with

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

To calculate the second derivatives of u , refer to 1.5.3(c) and apply the chain rule once more. Since here $u(r)$ is not a function of θ , we omit the terms involving it.

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x} \right) \\ &= \frac{\partial^2 u}{\partial r^2} \left(\frac{\partial r}{\partial x} \right)^2 + \frac{\partial u}{\partial r} \frac{\partial^2 r}{\partial x^2} \\ \frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y} \right) \\ &= \frac{\partial^2 u}{\partial r^2} \left(\frac{\partial r}{\partial y} \right)^2 + \frac{\partial u}{\partial r} \frac{\partial^2 r}{\partial y^2}\end{aligned}$$

To get the second derivatives of r , refer to 1.5.3(a) and differentiate once more:

$$\begin{aligned}\frac{\partial}{\partial x} \left(2r \frac{\partial r}{\partial x} \right) &= \frac{\partial}{\partial x} (2x) \\ \Rightarrow 2 \left(\frac{\partial r}{\partial x} \right)^2 + 2r \frac{\partial^2 r}{\partial x^2} &= 2 \quad \Rightarrow \frac{\partial^2 r}{\partial x^2} = \frac{1 - (\partial r / \partial x)^2}{r} = \frac{1 - \cos^2(\theta)}{r} \\ &= \boxed{\frac{\sin^2(\theta)}{r}} \\ \frac{\partial}{\partial y} \left(2r \frac{\partial r}{\partial y} \right) &= \frac{\partial}{\partial y} (2y) \\ \Rightarrow 2 \left(\frac{\partial r}{\partial y} \right)^2 + 2r \frac{\partial^2 r}{\partial y^2} &= 2 \quad \Rightarrow \frac{\partial^2 r}{\partial y^2} = \frac{1 - (\partial r / \partial y)^2}{r} = \frac{1 - \sin^2(\theta)}{r} \\ &= \boxed{\frac{\cos^2(\theta)}{r}}\end{aligned}$$

Now adding the second derivatives of u , derive

$$\begin{aligned}\nabla^2 u &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \\ &= \frac{\partial^2 u}{\partial r^2} \left[\left(\frac{\partial r}{\partial x} \right)^2 + \left(\frac{\partial r}{\partial y} \right)^2 \right] + \frac{\partial u}{\partial r} \left[\frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} \right] \\ &= \frac{\partial^2 u}{\partial r^2} [\cos^2(\theta) + \sin^2(\theta)] + \frac{1}{r} \frac{\partial u}{\partial r} [\sin^2(\theta) + \cos^2(\theta)] \\ &= \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r}\end{aligned}$$

Now, the easiest solution would have been to substitute into the result of 1.5.3(e), in which case we have

$$\nabla^2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r}$$

Since there is no variation with θ , the final $1/r^2 \frac{\partial^2 u}{\partial \theta^2}$ term vanishes.

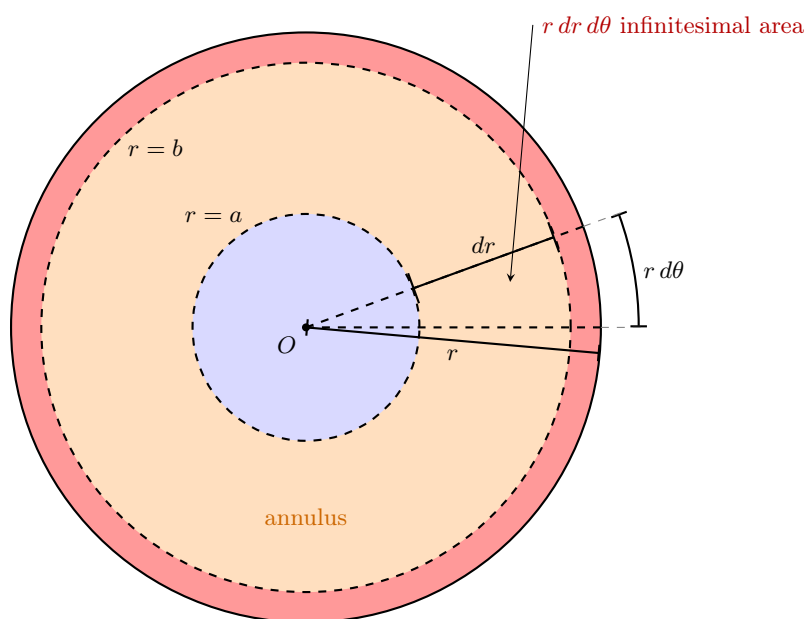
Question: 1.5.5

Assume that the temperature is circularly symmetric: $u = u(r, t)$, where $r^2 = x^2 + y^2$. We will derive the heat equation for this problem. Consider any circular annulus $a \leq r \leq b$.

- Show that the total heat energy is $2\pi \int_a^b c\rho u r dr$.
- Show that the flow of heat energy per unit time out of the annulus at $r = b$ is $-2\pi b K_0 \partial u / \partial r|_{r=b}$. A similar result holds at $r = a$.
- Use parts (a) and (b) to derive the circularly symmetric heat equation without sources:

$$\frac{\partial u}{\partial t} = \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right)$$

- The first observation to make here is since the annulus is a two-dimensional creature, the energy density $c\rho u$ is in units of energy per area. The infinitesimal area is $r dr d\theta$, apparent from the diagram below:



The arc length, $r d\theta$ gives the width of the infinitesimal area, while dr gives its length. Since we consider the whole rotational span of the annulus going from 0 to 2π , the total energy is given by

$$\text{total heat energy} = \int_0^{2\pi} \int_a^b c\rho u r dr d\theta = 2\pi \int_a^b c\rho u r dr$$

- The intuition here is to recall that in the case of the rod, heat flux is in units of thermal energy per unit time per unit *surface area*; for this

two-dimensional annulus, its circumference, a one-dimensional quantity, means flux is now in terms of unit *length*. To represent flow of heat energy per unit time, we need to multiply Fourier's heat law by a length: the circumference, or $2\pi b$.

$$\begin{array}{l} \text{flow of heat energy} \\ \text{per unit time} \end{array} = -2\pi b K_0 \frac{\partial u}{\partial r} \Big|_{r=b}$$

Similarly, we can calculate for $r = a$

$$\begin{array}{l} \text{flow of heat energy} \\ \text{per unit time} \end{array} = -2\pi a K_0 \frac{\partial u}{\partial r} \Big|_{r=a}$$

Heat flows *out* at $r = b$ and *in* at $r = a$.

In general, the rate of flow of heat energy is given by

$$\phi r \Delta\theta = -K_0 r \frac{\partial u}{\partial r} \Delta\theta$$

where $r\Delta\theta$ is the arc length over which flux transpires.

- (c) Assuming no sources, derive the circular heat equation with the integral conservation setup

$$\begin{aligned} \frac{d}{dt} \int_0^{2\pi} \int_a^b c\rho u r dr d\theta &= \phi(a, t) L_a - \phi(b, t) L_b \\ &= -2\pi a K_0 \frac{\partial u}{\partial r} \Big|_a + 2\pi b K_0 \frac{\partial u}{\partial r} \Big|_b \end{aligned}$$

Conceptually, the rate of change of total energy equals the difference in inbound and outbound energy transfer through the annulus. On the right side, we can equivalently write (thanks to the Fundamental Theorem of Calculus)

$$\frac{d}{dt} \int_0^{2\pi} \int_a^b c\rho u r dr d\theta = K_0 \int_0^{2\pi} \int_a^b \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) dr d\theta$$

Moving the differential operator on the left into the integrand (since we do not integrate over time) and setting the integrands equal, we have

$$c\rho \frac{\partial u}{\partial t} r = K_0 \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right)$$

Divide through by $c\rho r$ to conclude

$$\frac{\partial u}{\partial t} = \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right)$$

Question: 1.5.6

Modify Exercise 1.5.5 if the thermal properties depend on r .

The integral conservation law becomes

$$\begin{aligned} \frac{d}{dt} \int_0^{2\pi} \int_a^b c(r) \rho(r) u(r, t) r dr d\theta &= \phi(a, t) L_a - \phi(b, t) L_b \\ &= -2\pi a K_0(a) \left. \frac{\partial u}{\partial r} \right|_a + 2\pi b K_0(b) \left. \frac{\partial u}{\partial r} \right|_b \end{aligned}$$

which then becomes

$$\frac{d}{dt} \int_0^{2\pi} \int_a^b c(r) \rho(r) u(r, t) r dr d\theta = \int_0^{2\pi} \int_a^b \frac{\partial}{\partial r} \left(K_0(r) r \frac{\partial u}{\partial r} \right) dr d\theta$$

Notice here that $K_0(r)$, now a function of r , can no longer be pulled out of the integrand. Moving the derivative inside the integral on the left and setting the integrands equal yields

$$c(r) \rho(r) \frac{\partial u}{\partial r}(r, t) r = \frac{\partial}{\partial r} \left(K_0(r) r \frac{\partial u}{\partial r} \right)$$

Lastly, divide by $c(r) \rho(r) r$ to get

$$\frac{\partial u}{\partial t} = \frac{1}{c(r) \rho(r) r} \frac{\partial}{\partial r} \left(K_0(r) r \frac{\partial u}{\partial r} \right)$$

Question: 1.5.7

Derive the heat equation in two dimensions by using Green's theorem, (1.5.16), the two-dimensional form of the divergence theorem.

In general, for any two-dimensional geometric object R , the rate of change of the total energy is given by

$$\frac{d}{dt} \iint_R c \rho u dS$$

where ρ is an area density and dS is an infinitesimal area. Let $\hat{\phi}$ be the heat flux per time per length. The total flux through the line boundary is

$$-\oint \phi \cdot \hat{n} d\tau$$

Assuming a non-zero source, assembling the integral conservation law gives us

$$\frac{d}{dt} \iint_R c \rho u dS = -\oint \phi \cdot \hat{n} d\tau + \iint_R Q dS$$

It is easiest to convert the line integral to the divergence theorem in two dimensions by way of Green's theorem:

$$\iint_R \nabla \cdot \phi dS = \oint \phi \cdot \hat{n} d\tau$$

which allows us to write and consolidate the integrands to derive

$$\begin{aligned}\frac{d}{dt} \iint_R c\rho u \, dS &= - \iint_R \nabla \cdot \boldsymbol{\phi} \, dS + \iint_R Q \, dS \\ c\rho \frac{\partial u}{\partial t} &= -\nabla \cdot \boldsymbol{\phi} + Q\end{aligned}$$

and finally, substituting Fourier's heat law $\boldsymbol{\phi} = -K_0 \nabla u$ and dividing by $c\rho$ produces the heat equation

$$\frac{\partial u}{\partial t} = k \nabla^2 u + \frac{Q}{c\rho}$$

Question: 1.5.8

If Laplace's equation is satisfied in three dimensions, show that

$$\oiint \nabla u \cdot \hat{\mathbf{n}} \, dS = 0$$

for any closed surface. (*Hint:* Use the divergence theorem.) Give a physical interpretation of this result (in the context of heat flow).

By premise, $\nabla^2 u = 0$. Using the divergence theorem, we can write

$$\oiint \nabla u \cdot \hat{\mathbf{n}} \, dS = \iiint_R \nabla \cdot \nabla u \, dV = \iiint_R \nabla^2 u \, dV = 0$$

Physically, if $\boldsymbol{\phi} \propto \nabla u$, then this says that the net flux is zero. If there are no internal sources, then the *rate of change of total energy is zero*.

Question: 1.5.9

Determine the equilibrium temperature distribution inside a circular annulus ($r_1 \leq r \leq r_2$):

- if the outer radius is at temperature T_2 and the inner at T_1
- if the outer radius is insulated and the inner radius is at temperature T_1

- By premise, $u(r_1, t) = T_1$ and $u(r_2, t) = T_2$. At equilibrium, we assume

$$\frac{\partial u}{\partial t} = 0$$

Then we solve

$$\frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) = \frac{\partial u}{\partial r} + r \frac{\partial^2 u}{\partial r^2} = 0$$

Effectively, this is a homogeneous second-order ordinary differential equation, which we can rewrite as

$$\frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr} = 0$$

An Euler-Cauchy equation, we know to substitute $u = r^n$ to find

$$n(n-1)r^{n-2} + nr^{n-2} = 0 \implies n^2 = 0$$

Which leaves a repeated root in $n = 0$. Thus one null solution is $u_1 = 1$. A second-order equation implies two linearly independent solutions. To find the other, we use reduction of order. Let $u_2 = v(r)u_1 = v(r)$. Then

$$v'' + \frac{1}{r}v' = 0$$

Now, let $w = v'$. Then we rewrite as

$$w' + \frac{1}{r}w = 0$$

Separating variables, solve

$$\int \frac{dw}{w} = - \int \frac{dr}{r} \implies \ln(w) = -\ln(r) \implies w = \frac{1}{r}$$

which implies $u_2 = v(r) = \ln(r)$. Our complete null solution is

$$u(r) = c_1 + c_2 \ln(r)$$

Using this solution, we can write the system of equations

$$\begin{aligned} u(r_1) &= c_1 + c_2 \ln(r_1) = T_1 \\ u(r_2) &= c_1 + c_2 \ln(r_2) = T_2 \end{aligned}$$

Solving for the unknowns gives

$$c_1 = \frac{T_1 \ln(r_2) - T_2 \ln(r_1)}{\ln(r_2/r_1)}, \quad c_2 = \frac{T_2 - T_1}{\ln(r_2/r_1)}$$

- (b) By premise, $\partial u / \partial r|_{r=r_2} = 0$ and $u(r_1, t) = T_1$. From the general solution obtained in the previous part

$$u(r) = c_1 + c_2 \ln(r)$$

we generate the system of equations

$$\begin{aligned} u(r_1) &= c_1 + c_2 \ln(r_1) = T_1 \\ \frac{\partial u}{\partial r} \Big|_{r=r_2} &= \frac{c_2}{r_2} = 0 \end{aligned}$$

which implies $c_1 = T_1$ and $c_2 = 0$. Then $u(r) = T_1$. Interestingly, since heat cannot transfer out of the annulus at $r = r_2$, the temperature must equilibrate to a constant value throughout the entire region.

Question: 1.5.10

Determine the equilibrium temperature distribution inside a circle ($r \leq r_0$) if the boundary is fixed at temperature T_0 .

By premise, $u(r_0) = T_0$. We can derive the general solution as we did before

$$u(r) = c_1 + c_2 \ln(r)$$

But a problem arises when we consider the origin of the circle at $r = 0$ – the temperature approaches infinity, a physical impossibility. Therefore, we require a second “boundary condition” of sorts, namely

$$|u(r)| < \infty \text{ for all } r$$

Since we cannot satisfy this boundary condition with the logarithm’s presence, we necessarily force $c_2 = 0$. That leaves

$$u(r_0) = c_1 = T_0 \implies u(r) = T_0$$

Intuitively, this makes physical sense. For a constant fixed temperature at the boundary, we must have homogeneous temperature throughout the circle at equilibrium.

Question: 1.5.11

Consider

$$\frac{\partial u}{\partial t} = \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right), \quad a < r < b$$

subject to

$$u(r, 0) = f(r), \quad \frac{\partial u}{\partial r}(a, t) = \beta, \quad \text{and} \quad \frac{\partial u}{\partial r}(b, t) = 1$$

Using physical reasoning, for what value(s) of β does an equilibrium temperature distribution exist?

The temperature at equilibrium is given by the general form

$$u(r) = c_1 + c_2 \ln(r)$$

which subjected to the boundary conditions yields

$$\begin{aligned} \frac{\partial u}{\partial r}(a, t) &= \frac{c_2}{a} = \beta \\ \frac{\partial u}{\partial r}(b, t) &= \frac{c_2}{b} = 1 \end{aligned}$$

This immediately implies $c_2 = b$ and $\beta = b/a$. Previously, we derived that the heat flow is proportional to

$$r \frac{\partial u}{\partial r}$$

In our case, we can see that from at $r = a$:

$$\frac{\partial u}{\partial r}(a, t) = \beta = \frac{b}{a} \implies a \frac{\partial u}{\partial r}(a, t) = b$$

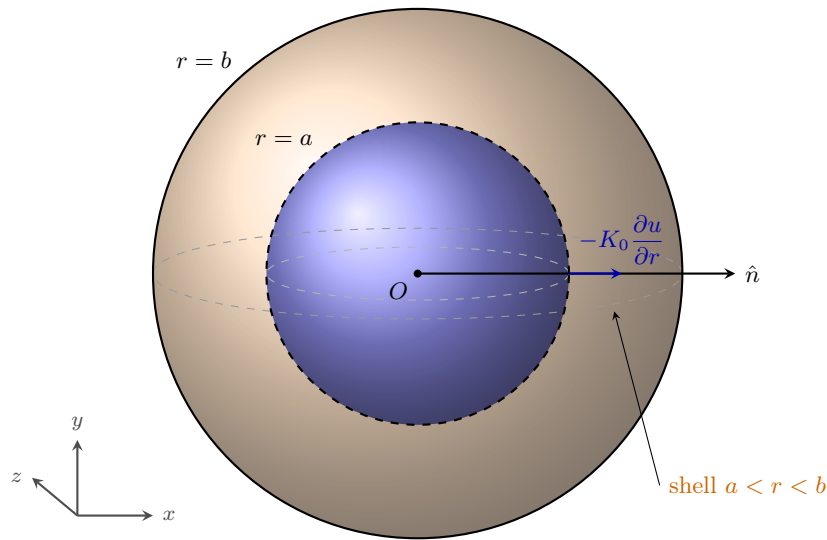
Question: 1.5.12

Assume that the temperature is spherically symmetric, $u = u(r, t)$, where r is the distance from a fixed point ($r^2 = x^2 + y^2 + z^2$). Consider the heat flow (without sources) between any two concentric spheres of radii a and b .

- (a) Show that the total heat energy is $4\pi \int_a^b c\rho u r^2 dr$.
- (b) Show that the flow of heat energy per unit time out of the spherical shell at $r = b$ is $-4\pi b^2 K_0 \partial u / \partial r|_{r=b}$. A similar result holds at $r = a$.
- (c) Use parts (a) and (b) to derive the spherically symmetric heat equation

$$\frac{\partial u}{\partial t} = \frac{k}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right)$$

The diagram below visualizes heat flow through the concentric spheres:



- (a) Let x , y , and z be defined as in (1.5.21):

$$\begin{aligned} x &= r \sin(\phi) \cos(\theta) \\ y &= r \sin(\phi) \sin(\theta) \\ z &= r \cos(\phi) \end{aligned}$$

One potential source of confusion is ϕ is the author's nomenclature for flux *and* the polar angle, but I hope the distinction is clear in the proceeding derivation. Since $c\rho u$ is an energy density, we do a volume integral to get the total heat energy. In spherical coordinates, our set up is

$$\text{total heat energy} = \int_0^{2\pi} \int_0^\pi \int_a^b c\rho u r^2 \sin(\phi) dr d\phi d\theta$$

Integrating over ϕ and θ , we derive

$$4\pi \int_a^b c\rho ur^2 dr$$

Notice how the surface area of a sphere, $4\pi r^2$, emerges in our formula.

- (b) Analogous to the two-dimensional case of the annulus, since flux is given in units of energy per unit time per unit area, we multiply by the entire area through which heat transfer occurs. Since the area of the spherical shell at $r = b$ is $4\pi b^2$, we multiply the Fourier heat conduction law by this area.

$$-4\pi b^2 K_0 \frac{\partial u}{\partial r} \Big|_{r=b}$$

and similarly, the same calculation holds at $r = a$:

$$-4\pi a^2 K_0 \frac{\partial u}{\partial r} \Big|_{r=a}$$

- (c) The integral conservation law is given by

$$\begin{aligned} \frac{d}{dt} \int_0^{2\pi} \int_0^\pi \int_a^b c\rho ur^2 \sin(\phi) dr d\phi d\theta &= \phi(a, t) S_a - \phi(b, t) S_b \\ &= -4\pi a^2 K_0 \frac{\partial u}{\partial r} \Big|_{r=a} + 4\pi b^2 K_0 \frac{\partial u}{\partial r} \Big|_{r=b} \\ &= \int_0^{2\pi} \int_0^\pi \int_a^b \frac{\partial}{\partial r} \left(K_0 r^2 \sin(\phi) \frac{\partial u}{\partial r} \right) dr d\phi d\theta \end{aligned}$$

Setting the integrands equal produces

$$c\rho \frac{\partial u}{\partial t} r^2 \sin(\phi) = \frac{\partial}{\partial r} \left(K_0 r^2 \sin(\phi) \frac{\partial u}{\partial r} \right)$$

On the right-hand side, pull out the K_0 and $\sin(\phi)$ from the derivative, as they are independent of r . Divide by $c\rho r^2 \sin(\phi)$ to conclude

$$\frac{\partial u}{\partial t} = \frac{k}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right)$$

Question: 1.5.13

Determine the *steady-state* temperature distribution between two concentric *spheres* with radii 1 and 4, respectively, if the temperature of the outer sphere is maintained at 80° and the inner sphere at 0° (see Exercise 1.5.12).

Using our result from the previous exercise, set $\partial u / \partial t = 0$ to derive

$$r^2 \frac{\partial^2 u}{\partial r^2} + 2r \frac{\partial u}{\partial r} = 0 \quad \Longleftrightarrow \quad \frac{\partial^2 u}{\partial r^2} + \frac{2}{r} \frac{\partial u}{\partial r} = 0$$

Which equivalently is an second-order ordinary Euler-Cauchy homogeneous differential equation:

$$\frac{d^2 u}{dr^2} + \frac{2}{r} \frac{du}{dr} = 0$$

Let $u = r^n$. Then

$$n(n-1)r^{n-2} + 2nr^{n-2} = 0 \implies n^2 + n = 0 \implies n_1 = 0, n_2 = -1$$

which produces the general solution

$$u(r) = c_1 + c_2 \frac{1}{r}$$

Imposing the boundary conditions, we have the system of equations

$$\begin{aligned} u(1) &= c_1 + c_2 = 0 \\ u(4) &= c_1 + c_2 \frac{1}{4} = 80 \end{aligned}$$

In matrix form,

$$\begin{bmatrix} 1 & 1 \\ 1 & 1/4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 80 \end{bmatrix}$$

Solving for the unknowns, we get

$$c_1 = \frac{320}{3}, \quad c_2 = -\frac{320}{3}$$

and equilibrium temperature distribution

$$u(r) = \frac{320}{3} \left(1 - \frac{1}{r} \right)$$

Question: 1.5.14

Isobars are lines of constant temperature. Show that isobars are perpendicular to any part of the boundary that is insulated.

I believe the author means **isotherms**, which are level curves for temperature. Isobars are for pressure. By definition, the temperature distribution along an isotherm is constant, meaning $u(r) = c$. Additionally, from the construction of a gradient, we know that it is always perpendicular to level curves at any point. Along the normal vector $\hat{n}$, if we have an insulated boundary, then $\nabla u \cdot \hat{n} = 0$. Since the gradient is orthogonal to both the isotherm and the normal vector, they must be parallel, demonstrating that isotherms must be orthogonal to any part of an insulated boundary.

Question: 1.5.15

Derive the heat equation in three dimensions assuming constant thermal properties and no sources.

Let c be the specific heat, ρ the mass density, and $u(x, y, z, t)$ the temperature distribution. The rate of change of the total energy in the region R is

$$\frac{d}{dt} \iiint_R c\rho u \, dV$$

Let ϕ be the heat flux (energy per unit time per unit area). One conceptual simplification is to measure the heat flux along the vector normal to the surface boundary, which we calculate as

$$\phi \cdot \hat{n}$$

which has sign convention $\phi \cdot \hat{n} < 0$ corresponding to inward flux and $\phi \cdot \hat{n} > 0$ for outward flux. Summing over the aggregate of these infinitesimal areas, we have the surface integral

$$- \oiint \phi \cdot \hat{n} \, dS$$

Setting this equal to the rate of change of the total energy, we have

$$\frac{d}{dt} \iiint_R c\rho u \, dV = - \oiint \phi \cdot \hat{n} \, dS$$

Moving the derivative inside the triple integral (as we don't integrate over time) and invoking the divergence theorem to deal with the surface integral, we have

$$\iiint_R c\rho \frac{\partial u}{\partial t} \, dV = - \iiint_R \nabla \cdot \phi \, dV$$

At last, we can set the integrands together to derive

$$c\rho \frac{\partial u}{\partial t} = -\nabla \cdot \phi$$

In this final step, use the Fourier heat conduction law

$$\phi = -K_0 \nabla u$$

where K_0 is the thermal conductivity, substitute for ϕ , and divide by $c\rho$ to get the heat equation in three dimensions

$$\frac{\partial u}{\partial t} = k \nabla^2 u$$

where $k = K_0/c\rho$ is the thermal diffusivity.

Question: 1.5.16

Express the integral conservation law for any three-dimensional object. Assume there are no sources. Also assume the heat flow is specified, $\nabla u \cdot \hat{\mathbf{n}} = \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{z})$, on the entire boundary and does not depend on time. By integrating with respect to time, determine the total thermal energy. (Hint: Use the initial condition.)

As derived in the previous exercise, the integral conservation law with no sources is

$$\iiint_R c\rho \frac{\partial u}{\partial t} dV = - \oint \phi \cdot \hat{\mathbf{n}} dS$$

On the right-hand side, substituting $\phi = -K_0 \nabla u$ gives

$$- \oint \phi \cdot \hat{\mathbf{n}} dS = -K_0 \oint \nabla u \cdot \hat{\mathbf{n}} dS = -K_0 \oint g(x, y, z) dS$$

Integrating all over time, we have

$$\begin{aligned} \int_0^t \left[\iiint_R c\rho \frac{\partial u}{\partial T} dV \right] dT &= -K_0 \int_0^t \left[\oint g(x, y, z) dS \right] dT \\ \Rightarrow \iiint_R c\rho u(x, y, z, t) dV &= \iiint_R c\rho f(x, y, z) dV - K_0 \left[\oint g(x, y, z) dS \right] t \end{aligned}$$

where we use the fact that from the initial condition,

$$\iiint_R c\rho u(x, y, z, 0) dV = \iiint_R c\rho f(x, y, z) dV$$

Question: 1.5.17

Derive the integral conservation law for any three dimensional object (with constant thermal properties) by integrating the heat equation (1.5.11) (assuming no sources). Show that the result is equivalent to (1.5.1).

Integrate over volume on both sides of the heat equation to derive

$$\iiint_R c\rho \frac{\partial u}{\partial t} dV = K_0 \iiint_R \nabla \cdot \nabla u dV$$

where $k = K_0/c\rho$ was split into its constituent terms and $c\rho$ multiplied on both sides. On the right, use the Fourier heat conduction law and substitute:

$$\iiint_R c\rho \frac{\partial u}{\partial t} dV = - \iiint_R \nabla \cdot \phi dV$$

Lastly, use the divergence theorem to restore equation (1.5.1).

$$\iiint_R c\rho \frac{\partial u}{\partial t} dV = - \oint \phi \cdot \hat{\mathbf{n}} dS$$

Orthogonal curvilinear coordinates. A coordinate system (u, v, w) may be introduced and defined by $x = x(u, v, w)$, $y = y(u, v, w)$, and $z = z(u, v, w)$. The radial vector $\mathbf{r} \equiv x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$. Partial derivatives of $\mathbf{r}$ with respect to a coordinate are in the direction of the coordinate. Thus, for example, a vector in the u -direction $\partial\mathbf{r}/\partial u$ can be made a unit vector $\hat{\mathbf{e}}_u$ in the u -direction by dividing by its length $h_u = |\partial\mathbf{r}/\partial u|$ called the **scale factor**: $\hat{\mathbf{e}}_u = \frac{1}{h_u}\partial\mathbf{r}/\partial u$.

Question: 1.5.18

Determine the scale factors for the cylindrical coordinates.

Define the radial vector $\mathbf{r}$ as above. The coordinate transformation is as given in equation (1.5.18). The scale factors and transformed unit vectors are as follows:

$$\begin{aligned}\frac{\partial\mathbf{r}}{\partial r} &= \frac{\partial x}{\partial r}\hat{\mathbf{i}} + \frac{\partial y}{\partial r}\hat{\mathbf{j}} + \frac{\partial z}{\partial r}\hat{\mathbf{k}} \\ &= \cos(\theta)\hat{\mathbf{i}} + \sin(\theta)\hat{\mathbf{j}} \\ h_r &= |\partial\mathbf{r}/\partial r| = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = \boxed{1} \\ \hat{\mathbf{e}}_r &= \frac{1}{h_r}\frac{\partial\mathbf{r}}{\partial r} = \boxed{\frac{\partial\mathbf{r}}{\partial r}} \\ \frac{\partial\mathbf{r}}{\partial\theta} &= \frac{\partial x}{\partial\theta}\hat{\mathbf{i}} + \frac{\partial y}{\partial\theta}\hat{\mathbf{j}} + \frac{\partial z}{\partial\theta}\hat{\mathbf{k}} \\ &= r[-\sin(\theta)\hat{\mathbf{i}} + \cos(\theta)\hat{\mathbf{j}}] \\ h_\theta &= |\partial\mathbf{r}/\partial\theta| = \sqrt{r^2[\cos^2(\theta) + \sin^2(\theta)]} = \boxed{r} \\ \hat{\mathbf{e}}_\theta &= \frac{1}{h_\theta}\frac{\partial\mathbf{r}}{\partial\theta} = \boxed{\frac{1}{r}\frac{\partial\mathbf{r}}{\partial\theta}} \\ \frac{\partial\mathbf{r}}{\partial z} &= \frac{\partial z}{\partial z}\hat{\mathbf{k}} = \hat{\mathbf{k}} \\ h_z &= |\partial\mathbf{r}/\partial z| = \boxed{1} \\ \hat{\mathbf{e}}_z &= \frac{1}{h_z}\frac{\partial\mathbf{r}}{\partial z} = \boxed{\hat{\mathbf{k}}}\end{aligned}$$

Question: 1.5.19

Determine the scale factors for spherical coordinates.

Define the radial vector $\mathbf{r}$ as above. The coordinate transformation is as given in equation (1.5.21). The scale factors and transformed unit vectors are

as follows:

$$\begin{aligned}
 \frac{\partial \mathbf{r}}{\partial \rho} &= \frac{\partial x}{\partial \rho} \hat{\mathbf{i}} + \frac{\partial y}{\partial \rho} \hat{\mathbf{j}} + \frac{\partial z}{\partial \rho} \hat{\mathbf{k}} \\
 &= \sin(\phi) \cos(\theta) \hat{\mathbf{i}} + \sin(\phi) \sin(\theta) \hat{\mathbf{j}} + \cos(\phi) \hat{\mathbf{k}} \\
 h_\rho &= |\partial \mathbf{r} / \partial \rho| = \sqrt{\sin^2(\phi) \cos^2(\theta) + \sin^2(\phi) \sin^2(\theta) + \cos^2(\phi)} = \boxed{1} \\
 \hat{\mathbf{e}}_\rho &= \frac{1}{h_\rho} \frac{\partial \mathbf{r}}{\partial \rho} = \boxed{\frac{\partial \mathbf{r}}{\partial \rho}} \\
 \frac{\partial \mathbf{r}}{\partial \phi} &= \frac{\partial x}{\partial \phi} \hat{\mathbf{i}} + \frac{\partial y}{\partial \phi} \hat{\mathbf{j}} + \frac{\partial z}{\partial \phi} \hat{\mathbf{k}} \\
 &= \rho \cos(\phi) \cos(\theta) \hat{\mathbf{i}} + \rho \cos(\phi) \sin(\theta) \hat{\mathbf{j}} - \rho \sin(\phi) \hat{\mathbf{k}} \\
 h_\phi &= |\partial \mathbf{r} / \partial \phi| = \rho \sqrt{\cos^2(\phi) \cos^2(\theta) + \cos^2(\phi) \sin^2(\theta) + \sin^2(\phi)} = \boxed{\rho} \\
 \hat{\mathbf{e}}_\phi &= \frac{1}{h_\phi} \frac{\partial \mathbf{r}}{\partial \phi} = \boxed{\frac{1}{\rho} \frac{\partial \mathbf{r}}{\partial \phi}} \\
 \frac{\partial \mathbf{r}}{\partial \theta} &= \frac{\partial x}{\partial \theta} \hat{\mathbf{i}} + \frac{\partial y}{\partial \theta} \hat{\mathbf{j}} + \frac{\partial z}{\partial \theta} \hat{\mathbf{k}} \\
 &= -\rho \sin(\phi) \sin(\theta) \hat{\mathbf{i}} + \rho \sin(\phi) \cos(\theta) \hat{\mathbf{j}} \\
 h_\theta &= |\partial \mathbf{r} / \partial \theta| = \rho \sqrt{\sin^2(\phi) \sin^2(\theta) + \sin^2(\phi) \cos^2(\theta)} = \boxed{\rho \sin(\phi)} \\
 \hat{\mathbf{e}}_\theta &= \frac{1}{h_\theta} \frac{\partial \mathbf{r}}{\partial \theta} = \boxed{\frac{1}{\rho \sin(\phi)} \frac{\partial \mathbf{r}}{\partial \theta}}
 \end{aligned}$$

Question: 1.5.20

The gradient of a scalar can be expressed in terms of the new coordinate system $\nabla g = a \partial \mathbf{r} / \partial u + b \partial \mathbf{r} / \partial v + c \partial \mathbf{r} / \partial w$, where you will determine the scalars a , b , c . Using $dg = \nabla g \cdot d\mathbf{r}$, derive that the **gradient** in an orthogonal curvilinear coordinate system is given by

$$\nabla g = \frac{1}{h_u} \frac{\partial g}{\partial u} \hat{\mathbf{e}}_u + \frac{1}{h_v} \frac{\partial g}{\partial v} \hat{\mathbf{e}}_v + \frac{1}{h_w} \frac{\partial g}{\partial w} \hat{\mathbf{e}}_w \quad (1.5.23)$$

An expression for the divergence is more difficult to derive, and we will just state that if a vector $\mathbf{p}$ is expressed in terms of this new coordinate system $\mathbf{p} = p_u \hat{\mathbf{e}}_u + p_v \hat{\mathbf{e}}_v + p_w \hat{\mathbf{e}}_w$, then the **divergence** satisfies

$$\nabla \cdot \mathbf{p} = \frac{1}{h_u h_v h_w} \left[\frac{\partial}{\partial u} (h_v h_w p_u) + \frac{\partial}{\partial v} (h_u h_w p_v) + \frac{\partial}{\partial w} (h_u h_v p_w) \right] \quad (1.5.24)$$

The basis vectors in u , v , and w coordinates are

$$\hat{\mathbf{e}}_u = \frac{1}{h_u} \frac{\partial \mathbf{r}}{\partial u}, \quad \hat{\mathbf{e}}_v = \frac{1}{h_v} \frac{\partial \mathbf{r}}{\partial v}, \quad \hat{\mathbf{e}}_w = \frac{1}{h_w} \frac{\partial \mathbf{r}}{\partial w}$$

And the differentials are

$$\begin{aligned} dg &= \frac{\partial g}{\partial u} du + \frac{\partial g}{\partial v} dv + \frac{\partial g}{\partial w} dw \\ d\mathbf{r} &= \frac{\partial \mathbf{r}}{\partial u} du + \frac{\partial \mathbf{r}}{\partial v} dv + \frac{\partial \mathbf{r}}{\partial w} dw \\ &= h_u du \hat{\mathbf{e}}_{\mathbf{u}} + h_v dv \hat{\mathbf{e}}_{\mathbf{v}} + h_w dw \hat{\mathbf{e}}_{\mathbf{w}} \end{aligned}$$

where the last equality stems from the definition of the basis vectors. Calculating $\nabla g \cdot d\mathbf{r}$ and invoking orthogonality between the basis vectors, we find

$$\nabla g \cdot d\mathbf{r} = h_u^2 a du + h_v^2 b dv + h_w^2 c dw$$

Setting equal to dg , we can derive

$$\frac{\partial g}{\partial u} = h_u^2 a, \quad \frac{\partial g}{\partial v} = h_v^2 b, \quad \frac{\partial g}{\partial w} = h_w^2 c$$

Back to the definition of ∇g . If we isolate a , b , and c and each of the derivatives of $\mathbf{r}$ from the basis vectors, we can conclude

$$\nabla g = \frac{1}{h_u} \frac{\partial g}{\partial u} \hat{\mathbf{e}}_{\mathbf{u}} + \frac{1}{h_v} \frac{\partial g}{\partial v} \hat{\mathbf{e}}_{\mathbf{v}} + \frac{1}{h_w} \frac{\partial g}{\partial w} \hat{\mathbf{e}}_{\mathbf{w}}$$

Question: 1.5.21

Using (1.5.23) and (1.5.24), derive the **Laplacian** in an orthogonal curvilinear coordinate system:

$$\nabla^2 T = \frac{1}{h_u h_v h_w} \left[\frac{\partial}{\partial u} \left(\frac{h_v h_w}{h_u} \frac{\partial T}{\partial u} \right) + \frac{\partial}{\partial v} \left(\frac{h_u h_w}{h_v} \frac{\partial T}{\partial v} \right) + \frac{\partial}{\partial w} \left(\frac{h_u h_v}{h_w} \frac{\partial T}{\partial w} \right) \right] \quad (1.5.25)$$

Replace the components of $\mathbf{p}$ with those of ∇g , switching out g for T , from (1.5.23):

$$p_u \implies \frac{1}{h_u} \frac{\partial T}{\partial u}, \quad p_v \implies \frac{1}{h_v} \frac{\partial T}{\partial v}, \quad p_w \implies \frac{1}{h_w} \frac{\partial T}{\partial w}$$

And then substitute in equation (1.5.24):

$$\nabla \cdot \nabla T = \nabla^2 T = \frac{1}{h_u h_v h_w} \left[\frac{\partial}{\partial u} \left(\frac{h_v h_w}{h_u} \frac{\partial T}{\partial u} \right) + \frac{\partial}{\partial v} \left(\frac{h_u h_w}{h_v} \frac{\partial T}{\partial v} \right) + \frac{\partial}{\partial w} \left(\frac{h_u h_v}{h_w} \frac{\partial T}{\partial w} \right) \right]$$

Question: 1.5.22

Using (1.2.25), derive the Laplacian for cylindrical coordinates.

Let $u = r$, $v = \theta$, and $w = z$. Recall from exercise 18 that the scale factors are

$$h_r = 1, \quad h_\theta = r, \quad h_z = 1$$

Substituting into (1.5.25) yields

$$\begin{aligned}\nabla^2 T &= \frac{1}{h_r h_\theta h_z} \left[\frac{\partial}{\partial r} \left(\frac{h_\theta h_z}{h_r} \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{h_r h_z}{h_\theta} \frac{\partial T}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(\frac{h_r h_\theta}{h_z} \frac{\partial T}{\partial z} \right) \right] \\ &= \frac{1}{r} \left[\frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \frac{\partial T}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(r \frac{\partial T}{\partial z} \right) \right] \\ &= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 T}{\partial \theta^2} + \frac{\partial^2 T}{\partial z^2}\end{aligned}$$

Question: 1.5.23

Using (1.5.25), derive the Laplacian for spherical coordinates.

Let $u = \rho$, $v = \phi$, and $w = \theta$. Recall from exercise 19 that the scale factors are

$$h_\rho = 1, \quad h_\phi = \rho, \quad h_\theta = \rho \sin(\phi)$$

Substituting into (1.5.25) yields

$$\begin{aligned}\nabla^2 T &= \frac{1}{h_\rho h_\phi h_\theta} \left[\frac{\partial}{\partial \rho} \left(\frac{h_\phi h_\theta}{h_\rho} \frac{\partial T}{\partial \rho} \right) + \frac{\partial}{\partial \phi} \left(\frac{h_\rho h_\theta}{h_\phi} \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial \theta} \left(\frac{h_\rho h_\phi}{h_\theta} \frac{\partial T}{\partial \theta} \right) \right] \\ &= \frac{1}{\rho^2 \sin(\phi)} \left[\frac{\partial}{\partial \rho} \left(\rho^2 \sin(\phi) \frac{\partial T}{\partial \rho} \right) + \frac{\partial}{\partial \phi} \left(\sin(\phi) \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial \theta} \left(\frac{1}{\sin(\phi)} \frac{\partial T}{\partial \theta} \right) \right] \\ &= \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial T}{\partial \rho} \right) + \frac{1}{\rho^2 \sin(\phi)} \frac{\partial}{\partial \phi} \left(\sin(\phi) \frac{\partial T}{\partial \phi} \right) + \frac{1}{\rho^2 \sin^2(\phi)} \frac{\partial^2 T}{\partial \theta^2}\end{aligned}$$

Question: 1.5.24

Derive the three-dimensional heat equation assuming constant thermal properties and assuming there are no sources of thermal energy. (*Hint:* Use an arbitrary three-dimensional region, and use the divergence theorem.)

The insight to make here is that the derivation of the heat equation is coordinate-agnostic. By habit we derived it in x , y , and z coordinates, but we could have chosen any basis. Thus refer to exercise 1.5.15.

When evaluating $k \nabla^2 u$, we use (1.5.25) with the appropriate scale factors.

Question: 1.5.25

Suppose a sphere of radius 2 satisfies $\frac{\partial u}{\partial t} = \nabla^2 u + 5$ with $u(x, y, z, 0) = f(x, y, z)$ and on the surface of the sphere it is given that $\nabla u \cdot \hat{\mathbf{n}} = 6$, where $\hat{\mathbf{n}}$ is a unit outward normal vector. Calculate the total thermal energy for this sphere as a function of time. (*Hint:* Use the divergence theorem.)

Using the integral conservation law and the divergence theorem, write

$$\begin{aligned} \int_0^t \left[\frac{d}{dt} \iiint_R u(x, y, z, T) \, dV \right] dT &= \left[\oint \nabla u \cdot \hat{\mathbf{n}} \, dS + \iiint_R 5 \, dV \right] t \\ \implies \iiint_R u(x, y, z, t) \, dV &= \iiint_R f(x, y, z) \, dV + \left[\oint 6 \, dS + \iiint_R 5 \, dV \right] t \\ &= \iiint_R f(x, y, z) \, dV + 6 \cdot 16\pi t + 5 \cdot \frac{32}{3}\pi t \\ &= \iiint_R f(x, y, z) \, dV + \frac{448\pi t}{3} \end{aligned}$$

Using the surface area and volume of a sphere formulae, $S = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$.

2 - Method of Separation of Variables

2.2 - Linearity

Question: 2.2.1

Show that any linear combination of linear operators is a linear operator.

Let c_1 and c_2 be constants and u_1 and u_2 be functions. Define

$$L(c_1u_1 + c_2u_2) = \sum_{i=1}^n d_i L_i(c_1u_1 + c_2u_2)$$

where the L_i 's are each linear operators and the d_i 's are constants. By the definition of linear operator, we can demonstrate

$$\begin{aligned} L(c_1u_1 + c_2u_2) &= \sum_{i=1}^n d_i L_i(c_1u_1 + c_2u_2) \\ &= \sum_{i=1}^n d_i [c_1 L_i(u_1) + c_2 L_i(u_2)] \\ &= c_1 \sum_{i=1}^n d_i L_i(u_1) + c_2 \sum_{i=1}^n d_i L_i(u_2) \end{aligned}$$

Thereby establishing that L is also a linear operator.

Question: 2.2.2

- (a) Show that $L(u) = \frac{\partial}{\partial x} \left[K_0(x) \frac{\partial u}{\partial x} \right]$ is a linear operator.
- (b) Show that usually $L(u) = \frac{\partial}{\partial x} \left[K_0(x, u) \frac{\partial u}{\partial x} \right]$ is not a linear operator.

Let c_1 and c_2 be constants and u_1 and u_2 be functions.

(a)

$$\begin{aligned} L(c_1u_1 + c_2u_2) &= \frac{\partial}{\partial x} \left[K_0(x) \frac{\partial}{\partial x} (c_1u_1 + c_2u_2) \right] \\ &= \frac{\partial}{\partial x} \left[K_0(x) \left(c_1 \frac{\partial u_1}{\partial x} + c_2 \frac{\partial u_2}{\partial x} \right) \right] \\ &= c_1 \frac{\partial}{\partial x} \left[K_0(x) \frac{\partial u_1}{\partial x} \right] + c_2 \frac{\partial}{\partial x} \left[K_0(x) \frac{\partial u_2}{\partial x} \right] \\ &= c_1 L(u_1) + c_2 L(u_2) \end{aligned}$$

(b) As given, one would need to demonstrate

$$\begin{aligned} L(c_1u_1 + c_2u_2) &= \frac{\partial}{\partial x} \left[K_0(x, c_1u_1 + c_2u_2) \frac{\partial}{\partial x} (c_1u_1 + c_2u_2) \right] \\ &= c_1 \frac{\partial}{\partial x} \left[K_0(x, u_1) \frac{\partial u_1}{\partial x} \right] + c_2 \frac{\partial}{\partial x} \left[K_0(x, u_2) \frac{\partial u_2}{\partial x} \right] \end{aligned}$$

And for the sake of argument, suppose

$$K_0(x, c_1 u_1 + c_2 u_2) = c_1 K_0(x, u_1) + c_2 K_0(x, u_2)$$

Then we would have

$$\begin{aligned} L(c_1 u_1 + c_2 u_2) &= \frac{\partial}{\partial x} \left[K_0(x, c_1 u_1 + c_2 u_2) \frac{\partial}{\partial x} (c_1 u_1 + c_2 u_2) \right] \\ &= \frac{\partial}{\partial x} \left[(c_1 K_0(x, u_1) + c_2 K_0(x, u_2)) \left(c_1 \frac{\partial u_1}{\partial x} + c_2 \frac{\partial u_2}{\partial x} \right) \right] \\ &= c_1^2 \frac{\partial}{\partial x} \left[K_0(x, u_1) \frac{\partial u_1}{\partial x} \right] + c_2^2 \frac{\partial}{\partial x} \left[K_0(x, u_2) \frac{\partial u_2}{\partial x} \right] \\ &\quad + c_1 c_2 \frac{\partial}{\partial x} \left[K_0(x, u_1) \frac{\partial u_2}{\partial x} \right] + c_1 c_2 \frac{\partial}{\partial x} \left[K_0(x, u_2) \frac{\partial u_1}{\partial x} \right] \end{aligned}$$

But even here, the cross-terms must be guaranteed to vanish, and the squared constant terms do not adhere to the definition of linearity. Under the given information, there is no way to determine if $K_0(x, u)$ decomposes in the way that it needs to enable this second equality to be true.

Question: 2.2.3

Show that $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} + Q(u, x, t)$ is linear if $Q = \alpha(x, t) u + \beta(x, t)$ and, in addition, homogeneous if $\beta(x, t) = 0$.

Define

$$L(u) = \frac{\partial u}{\partial t} - k \frac{\partial^2 u}{\partial x^2} - \alpha(x, t) u = \beta(x, t)$$

To test for linearity, compute

$$\begin{aligned} L(c_1 u_1 + c_2 u_2) &= \frac{\partial}{\partial t} (c_1 u_1 + c_2 u_2) - k \frac{\partial^2}{\partial x^2} (c_1 u_1 + c_2 u_2) \\ &\quad - \alpha(x, t) (c_1 u_1 + c_2 u_2) \\ &= c_1 \frac{\partial u_1}{\partial t} + c_2 \frac{\partial u_2}{\partial t} - c_1 k \frac{\partial^2 u_1}{\partial x^2} - c_2 k \frac{\partial^2 u_2}{\partial x^2} \\ &\quad - c_1 \alpha(x, t) u_1 - c_2 \alpha(x, t) u_2 \\ &= c_1 L(u_1) + c_2 L(u_2) \end{aligned}$$

By definition of homogeneity, $\beta(x, t) = 0$ produces the homogeneous condition.

Question: 2.2.4

In this exercise we derive superposition principles for nonhomogeneous problems.

- (a) Consider $L(u) = f$. If u_p is a particular solution, $L(u_p) = f$, and if u_1 and u_2 are homogeneous solutions, $L(u_i) = 0$, show that $u = u_p + c_1 u_1 + c_2 u_2$ is another particular solution.
- (b) If $L(u) = f_1 + f_2$, where u_{pi} is a particular solution corresponding to f_i , what is a particular solution for $f_1 + f_2$?

$$(a) \quad L(u_p + c_1 u_1 + c_2 u_2) = L(u_p) + c_1 L(u_1) + c_2 L(u_2) = f + 0 + 0 = f$$

(b) One particular solution is $u = u_{p1} + u_{p2}$. In general, we can have

$$u = c_1 u_1 + c_2 u_2 + u_{p1} + u_{p2}$$

Question: 2.2.5

If L is a linear operator, show that $L\left(\sum_{n=1}^M c_n u_n\right) = \sum_{n=1}^M c_n L(u_n)$. Use this result to show that the principle of superposition may be extended to any finite number of homogeneous solutions.

Compute from the definition of linearity

$$L\left(\sum_{n=1}^M c_n u_n\right) = \sum_{n=1}^M L(c_n u_n) = \sum_{n=1}^M c_n L(u_n)$$

Assuming homogeneity for every solution, we can do (pedantically) induction on M to demonstrate that the result holds for any finite natural number.

$$L\left(\sum_{n=1}^M c_n u_n + c_{M+1} u_{M+1}\right) = \sum_{n=1}^M L(c_n u_n) + L(c_{M+1} u_{M+1}) = \sum_{n=1}^{M+1} c_n L(u_n)$$

2.3 - Heat Equation with Zero Temperatures at Finite Ends

Question: 2.3.1

For the following partial differential equations, what ordinary differential equations are implied by the method of separation of variables?

(a) $\frac{\partial u}{\partial t} = \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right)$

(b) $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} - v_0 \frac{\partial u}{\partial x}$

(c) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial y} = 0$

(d) $\frac{\partial u}{\partial t} = \frac{k}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right)$

(e) $\frac{\partial u}{\partial t} = k \frac{\partial^4 u}{\partial x^4}$

(f) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

(a) Let $u(r, t) = \phi(r) G(t)$. Then

$$\begin{aligned} \frac{\partial u}{\partial t} &= \phi(r) \frac{dG}{dt} \\ \frac{k}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) &= \frac{k}{r} G(t) \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) \end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned} \phi(r) \frac{dG}{dt} &= \frac{k}{r} G(t) \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) \\ \implies \frac{1}{kG(t)} \frac{dG}{dt} &= \frac{1}{r\phi(r)} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = -\lambda \end{aligned}$$

This produces ordinary differential equations

$$\frac{dG}{dt} = -\lambda k G(t), \quad \frac{1}{r} \frac{d}{dr} \left(r \frac{d\phi}{dr} \right) = -\lambda \phi(r)$$

(b) Let $u(x, t) = \phi(x) G(t)$. Then

$$\begin{aligned} \frac{\partial u}{\partial t} &= \phi(x) \frac{dG}{dt} \\ k \frac{\partial^2 u}{\partial x^2} - v_0 \frac{\partial u}{\partial x} &= G(t) \left[k \frac{d^2 \phi}{dx^2} - v_0 \frac{d\phi}{dx} \right] \\ &= kG(t) \left[\frac{d^2 \phi}{dx^2} - \frac{v_0}{k} \frac{d\phi}{dx} \right] \end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned}\phi(x) \frac{dG}{dt} &= kG(t) \left[\frac{d^2\phi}{dx^2} - \frac{v_0}{k} \frac{d\phi}{dx} \right] \\ \implies \frac{1}{kG(t)} \frac{dG}{dt} &= \frac{1}{\phi(x)} \left[\frac{d^2\phi}{dx^2} - \frac{v_0}{k} \frac{d\phi}{dx} \right] = -\lambda\end{aligned}$$

This produces ordinary differential equations

$$\frac{dG}{dt} = -\lambda kG(t), \quad \frac{d^2\phi}{dx^2} - \frac{v_0}{k} \frac{d\phi}{dx} = -\lambda\phi(x)$$

(c) Let $u(x, y) = \phi(x) G(y)$. Then

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial y} = \frac{d^2\phi}{dx^2} G(y) + \phi(x) \frac{dG}{dy} = 0$$

Setting equal to zero and separating variables, we find

$$\frac{1}{\phi(x)} \frac{d^2\phi}{dx^2} = -\frac{1}{G(y)} \frac{dG}{dy} = -\lambda$$

This produces ordinary differential equations

$$\frac{d^2\phi}{dx^2} = -\lambda\phi(x), \quad \frac{dG}{dy} = \lambda G(y)$$

(d) Let $u(r, t) = \phi(r) G(t)$. Then

$$\begin{aligned}\frac{\partial u}{\partial t} &= \phi(r) \frac{dG}{dt} \\ \frac{k}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) &= G(t) \frac{k}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right)\end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned}\phi(r) \frac{dG}{dt} &= G(t) \frac{k}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) \\ \implies \frac{1}{kG(t)} \frac{dG}{dt} &= \frac{1}{r^2\phi(r)} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = -\lambda\end{aligned}$$

This produces ordinary differential equations

$$\frac{dG}{dt} = -\lambda kG(t), \quad \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = -\lambda\phi(r)$$

(e) Let $u(x, t) = \phi(x) G(t)$. Then

$$\begin{aligned}\frac{\partial u}{\partial t} &= \phi(x) \frac{dG}{dt} \\ k \frac{\partial^4 u}{\partial x^4} &= kG(t) \frac{d^4\phi}{dx^4}\end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned}\phi(x) \frac{dG}{dt} &= kG(t) \frac{d^4\phi}{dx^4} \\ \implies \frac{1}{kG(t)} \frac{dG}{dt} &= \frac{1}{\phi(x)} \frac{d^4\phi}{dx^4} = -\lambda\end{aligned}$$

This produces ordinary differential equations

$$\frac{dG}{dt} = -\lambda kG(t), \quad \frac{d^4\phi}{dx^4} = -\lambda\phi(x)$$

(f) Let $u(x, t) = \phi(x) G(t)$. Then

$$\begin{aligned}\frac{\partial^2 u}{\partial t^2} &= \phi(x) \frac{d^2 G}{dt^2} \\ c^2 \frac{\partial^2 u}{\partial x^2} &= c^2 G(t) \frac{d^2 \phi}{dx^2}\end{aligned}$$

Setting equal and separating variables, we find

$$\begin{aligned}\phi(x) \frac{d^2 G}{dt^2} &= c^2 G(t) \frac{d^2 \phi}{dx^2} \\ \implies \frac{1}{c^2 G(t)} \frac{d^2 G}{dt^2} &= \frac{1}{\phi(x)} \frac{d^2 \phi}{dx^2} = -\lambda\end{aligned}$$

This produces ordinary differential equations

$$\frac{d^2 G}{dt^2} = -\lambda c^2 G(t), \quad \frac{d^2 \phi}{dx^2} = -\lambda\phi(x)$$

Question: 2.3.2

Consider the differential equation

$$\frac{d^2\phi}{dx^2} + \lambda\phi = 0$$

Determine the eigenvalues λ (and corresponding eigenfunctions) if ϕ satisfies the following boundary conditions. Analyze three cases ($\lambda > 0$, $\lambda = 0$, $\lambda < 0$). You may assume that the eigenvalues are real.

- (a) $\phi(0) = 0$ and $\phi(\pi) = 0$
- (b) $\phi(0) = 0$ and $\phi(1) = 0$
- (c) $\frac{d\phi}{dx}(0) = 0$ and $\frac{d\phi}{dx}(L) = 0$ (If necessary, see Section 2.4.1.)
- (d) $\phi(0) = 0$ and $\frac{d\phi}{dx}(L) = 0$
- (e) $\frac{d\phi}{dx}(0) = 0$ and $\phi(L) = 0$
- (f) $\phi(a) = 0$ and $\phi(b) = 0$ (You may assume that $\lambda > 0$.)
- (g) $\phi(0) = 0$ and $\frac{d\phi}{dx}(L) + \phi(L) = 0$ (If necessary, see Section 5.8.)

The general forms for each case are

$$\phi = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x) \quad (\lambda > 0)$$

$$\phi = c_1 + c_2x \quad (\lambda = 0)$$

$$\phi = c_1 \cosh(\sqrt{s}x) + c_2 \sinh(\sqrt{s}x) \quad (\lambda < 0, s = -\lambda)$$

- (a) $\lambda > 0$ The boundary conditions produce

$$\phi(0) = c_1 = 0$$

$$\phi(\pi) = c_2 \sin(\sqrt{\lambda}\pi) = 0$$

which forces $\sqrt{\lambda} = n$, implying $\lambda = n^2$ for positive integers. Then $\phi(x) = c_2 \sin(nx)$.

- $\lambda = 0$ The boundary conditions produce

$$\phi(0) = c_1 = 0$$

$$\phi(\pi) = c_2\pi = 0$$

implying $c_2 = 0$, so we only have the trivial solution, confirming that $\lambda = 0$ cannot be an eigenvalue.

- $\lambda < 0$ Let $-\lambda = s$. The boundary conditions produce

$$\phi(0) = c_1 = 0$$

$$\phi(\pi) = c_2 \sinh(\sqrt{s}\pi) = 0$$

which requires either $s = 0$ or $c_2 = 0$. We consider only $s > 0$, so that forces $c_2 = 0$, and the implication that $\lambda < 0$ also cannot be an eigenvalue.

(b) $\lambda > 0$ The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \phi(1) &= c_2 \sin(\sqrt{\lambda}) = 0\end{aligned}$$

which forces $\lambda = (n\pi)^2$ for positive integers. Then $\phi(x) = c_2 \sin(n\pi x)$.

$\lambda = 0$ The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \phi(1) &= c_2 = 0\end{aligned}$$

leaving us only with the trivial solution, implying that $\lambda = 0$ cannot be an eigenvalue.

$\lambda < 0$ Let $-\lambda = s$. The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \phi(1) &= c_2 \sinh(\sqrt{s}) = 0\end{aligned}$$

which requires either $s = 0$ or $c_2 = 0$. We consider only $s > 0$ (equivalent with $\lambda < 0$), so we must have $c_2 = 0$, ergo $\lambda < 0$ also cannot be an eigenvalue.

(c) $\lambda > 0$ The boundary conditions produce

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2 \sqrt{\lambda} = 0 \implies c_2 = 0 \\ \frac{d\phi}{dx}(L) &= -c_1 \sqrt{\lambda} \sin(\sqrt{\lambda}L) = 0\end{aligned}$$

which requires $\lambda = (n\pi/L)^2$ for positive integers. Then $\phi(x) = c_1 \cos(n\pi x/L)$.

$\lambda = 0$ The boundary conditions produce

$$\frac{d\phi}{dx}(0) = \frac{d\phi}{dx}(L) = c_2 = 0$$

Then $\phi(x) = c_1$, thus establishing $\lambda = 0$ as an eigenvalue for this function.

$\lambda < 0$ Let $-\lambda = s$. The boundary conditions produce

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2 \sqrt{s} = 0 \implies c_2 = 0 \\ \frac{d\phi}{dx}(L) &= c_1 \sqrt{s} \sinh(\sqrt{s}L) = 0\end{aligned}$$

which requires $c_1 = 0$ as well, as neither $s = 0$ nor $L = 0$ can be true. Therefore $\lambda < 0$ does not exist as an eigenvalue.

(d) $\lambda > 0$ The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \frac{d\phi}{dx}(L) &= c_2 \sqrt{\lambda} \cos(\sqrt{\lambda}L) = 0\end{aligned}$$

which requires $\lambda = [(2n-1)\pi/2L]^2 = [(n-1/2)\pi/L]^2$ for positive integers. Then

$$\phi(x) = c_2 \sin\left(\frac{(n-\frac{1}{2})\pi x}{L}\right)$$

$\lambda = 0$ The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \frac{d\phi}{dx}(L) &= c_2 = 0\end{aligned}$$

Leaving only the trivial solution, so $\lambda = 0$ is not an eigenvalue.

$\lambda < 0$ Let $s = -\lambda$. The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \frac{d\phi}{dx}(L) &= c_2 \sqrt{s} \cosh(\sqrt{s}L) = 0 \implies c_2 = 0\end{aligned}$$

which confirms that $\lambda < 0$ are ineligible to be eigenvalues of the boundary value problem.

(e) $\lambda > 0$ The boundary conditions produce

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2 \sqrt{\lambda} = 0 \implies c_2 = 0 \\ \phi(L) &= c_1 \cos(\sqrt{\lambda}L) = 0\end{aligned}$$

which forces $\lambda = [(n-1/2)\pi/L]^2$ for positive integers. Then

$$\phi(x) = c_1 \cos\left(\frac{(n-\frac{1}{2})\pi x}{L}\right)$$

$\lambda = 0$ The boundary conditions produce

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2 = 0 \\ \phi(L) &= c_1 = 0\end{aligned}$$

implying that $\lambda = 0$ cannot be an eigenvalue of ϕ .

$\lambda < 0$ Let $s = -\lambda$. The boundary conditions produce

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2 \sqrt{s} = 0 \implies c_2 = 0 \\ \phi(L) &= c_1 \cosh(\sqrt{s}L) = 0 \implies c_1 = 0\end{aligned}$$

once more indicating that the trivial solution is all that is yielded, ensuring that $\lambda < 0$ can never be eigenvalues of the boundary value problem.

(f) $\lambda > 0$ The boundary conditions produce

$$\begin{aligned}\phi(a) &= c_1 \cos(\sqrt{\lambda}a) + c_2 \sin(\sqrt{\lambda}a) = 0 \\ \phi(b) &= c_1 \cos(\sqrt{\lambda}b) + c_2 \sin(\sqrt{\lambda}b) = 0\end{aligned}$$

One insight here is to rewrite these equations as a matrix equation

$$\begin{bmatrix} \cos(\sqrt{\lambda}a) & \sin(\sqrt{\lambda}a) \\ \cos(\sqrt{\lambda}b) & \sin(\sqrt{\lambda}b) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

And calculate the determinant of the left-most matrix using the sine angle difference identity:

$$\det = \cos(\sqrt{\lambda}a) \sin(\sqrt{\lambda}b) - \cos(\sqrt{\lambda}b) \sin(\sqrt{\lambda}a) = \sin(\sqrt{\lambda}(b-a))$$

Now, the question before us that will determine λ : is the determinant zero or non-zero? We know that we cannot have c_1 and c_2 both equal to zero if we want positive eigenvalues, which implies that there exists a non-trivial solution. In turn, we can deduce that *the rows are linearly dependent*, guaranteeing a zero determinant. This forces

$$\lambda = \left(\frac{n\pi}{b-a} \right)^2$$

for positive integers. Since there exist non-zero coefficients c_1 and c_2 , we can guess one solution for both

$$\begin{aligned}c_1 &= -C \sin(\sqrt{\lambda}a) \\ c_2 &= C \cos(\sqrt{\lambda}a)\end{aligned}$$

In the first row the combination of the two terms is clearly zero, and in the second, that is simply equal to the determinant, which we have established is zero. The corresponding eigenfunctions are

$$\begin{aligned}\phi(x) &= -C \sin(\sqrt{\lambda}a) \cos(\sqrt{\lambda}x) + C \cos(\sqrt{\lambda}a) \sin(\sqrt{\lambda}x) \\ &= C \sin(\sqrt{\lambda}(x-a)) \\ &= C \sin\left(\frac{n\pi(x-a)}{b-a}\right)\end{aligned}$$

(g) $\lambda > 0$ The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \frac{d\phi}{dx}(L) + \phi(L) &= c_2 \sqrt{\lambda} \cos(\sqrt{\lambda}L) + c_2 \sin(\sqrt{\lambda}L) = 0\end{aligned}$$

Equivalently, the second boundary condition can be written as

$$\tan(\sqrt{\lambda}L) = -\sqrt{\lambda}$$

which is legal as if $\cos(\sqrt{\lambda}L) = 0$, the corresponding values of λ would not render $\sin(\sqrt{\lambda}L) = 0$, forcing $c_2 = 0$, and producing only the trivial solution. The equation derived is a transcendental equation and must be evaluated with numerical methods. Referring to Figure 5.8.1 in Haberman demonstrates that there are infinitely many solutions. For now, call the solutions $\lambda_1, \lambda_2, \dots$, where $\lambda_1 < \lambda_2 < \dots$. Then $\phi_n(x) = c_2 \sin(\sqrt{\lambda_n}x)$ for positive integers.

$\lambda = 0$ The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \frac{d\phi}{dx}(L) + \phi(L) &= c_2(1 + L) = 0\end{aligned}$$

Forcing $c_2 = 0$, so only the trivial solution exists, implying $\lambda = 0$ cannot be an eigenvalue.

$\lambda < 0$ Let $s = -\lambda$. The boundary conditions produce

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \frac{d\phi}{dx}(L) + \phi(L) &= c_2\sqrt{s} \cosh(\sqrt{s}L) + c_2 \sinh(\sqrt{s}L) = 0\end{aligned}$$

Again, we can rewrite the second boundary condition as

$$\tanh(\sqrt{s}L) = -\sqrt{s}$$

This is once more a transcendental equation, however it is true only at $s = 0$, as when $s > 0$, $\tanh(\sqrt{s}) > 0$ and $-\sqrt{s} < 0$. Therefore we have $c_2 = 0$, and once again only the trivial solution. There exist no eigenvalues $\lambda < 0$ for ϕ .

Question: 2.3.3

Consider the heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

subject to the boundary conditions

$$u(0, t) = 0 \quad \text{and} \quad u(L, t) = 0$$

Solve the initial value problem if the temperature is initially

- (a) $u(x, 0) = 6 \sin\left(\frac{9\pi x}{L}\right)$
- (b) $u(x, 0) = 3 \sin\left(\frac{\pi x}{L}\right) - \sin\left(\frac{3\pi x}{L}\right)$
- (c) $u(x, 0) = 2 \cos\left(\frac{3\pi x}{L}\right)$
- (d) $u(x, 0) = \begin{cases} 1 & 0 < x \leq L/2 \\ 2 & L/2 < x < L \end{cases}$
- (e) $u(x, 0) = f(x)$

Use the general solution

$$u(x, t) = B \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

- (a) Set $B = 6$ and $n = 9$ to get

$$u(x, t) = 6 \sin\left(\frac{9\pi x}{L}\right) e^{-k(9\pi/L)^2 t}$$

- (b) From the principle of superposition, the general form looks like

$$u(x, t) = B_1 \sin\left(\frac{n_1\pi x}{L}\right) e^{-k(n_1\pi/L)^2 t} + B_2 \sin\left(\frac{n_2\pi x}{L}\right) e^{-k(n_2\pi/L)^2 t}$$

Set $B_1 = 3$, $B_2 = -1$, $n_1 = 1$, and $n_2 = 3$ to get

$$u(x, t) = 3 \sin\left(\frac{\pi x}{L}\right) e^{-k(\pi/L)^2 t} - \sin\left(\frac{3\pi x}{L}\right) e^{-k(3\pi/L)^2 t}$$

- (c) The issue here is the initial temperature fails to meet the boundary conditions. In this case, the general solution is

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

From the initial condition, we must set equal

$$u(x, 0) = 2 \cos\left(\frac{3\pi x}{L}\right) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$$

Now, multiply both sides by $\sin(m\pi x/L)$ and integrate from 0 to L :

$$\int_0^L 2 \cos\left(\frac{3\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx = \int_0^L \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx$$

By orthogonality of sines, the right-hand side equals

$$A_m \int_0^L \sin^2\left(\frac{m\pi x}{L}\right) dx$$

And solving for A_m gives us

$$A_m = \frac{\int_0^L 2 \cos\left(\frac{3\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx}{\int_0^L \sin^2\left(\frac{m\pi x}{L}\right) dx} = \frac{2}{L} \int_0^L 2 \cos\left(\frac{3\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx$$

Using the product identity for cosine and sine, we further derive

$$\begin{aligned} A_m &= \frac{2}{L} \int_0^L 2 \cos\left(\frac{3\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \\ &= \frac{2}{L} \left[\int_0^L \sin\left(\frac{(m+3)\pi x}{L}\right) dx + \int_0^L \sin\left(\frac{(m-3)\pi x}{L}\right) dx \right] \\ &= -\frac{2}{L} \left[\frac{L}{(m+3)\pi} \cos\left(\frac{(m+3)\pi x}{L}\right) \Big|_0^L + \frac{L}{(m-3)\pi} \cos\left(\frac{(m-3)\pi x}{L}\right) \Big|_0^L \right] \\ &= \begin{cases} -\frac{2}{L} \left[-\frac{2L}{(m+3)\pi} - \frac{2L}{(m-3)\pi} \right], & m \text{ even} \\ 0, & m \text{ odd} \end{cases} \\ &= \begin{cases} \frac{8m}{(m+3)(m-3)\pi}, & m \text{ even} \\ 0, & m \text{ odd} \end{cases} \end{aligned}$$

Rewriting as

$$A_n = \frac{16n}{(2n+3)(2n-3)\pi}$$

we have temperature distribution

$$u(x, t) = \sum_{n=1}^{\infty} \frac{16n}{(2n+3)(2n-3)\pi} \sin\left(\frac{2n\pi x}{L}\right) e^{-k(2n\pi/L)^2 t}$$

Observe that by ensuring that we sum only over even indices, *every* index throughout the expression must be scaled by a factor of two.

(d) From the general solution

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

Set equal

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$$

And multiply both sides by $\sin(m\pi x/L)$ and integrate from 0 to L to get

$$\begin{aligned} \int_0^L u(x, 0) \sin\left(\frac{m\pi x}{L}\right) dx &= \int_0^{L/2} \sin\left(\frac{m\pi x}{L}\right) dx + \int_{L/2}^L 2 \sin\left(\frac{m\pi x}{L}\right) dx \\ &= \int_0^L \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \\ &= A_m \int_0^L \sin^2\left(\frac{m\pi x}{L}\right) dx \\ &= A_m \frac{L}{2} \end{aligned}$$

Isolating A_m , we find

$$\begin{aligned} A_m &= \frac{2}{L} \left[\int_0^{L/2} \sin\left(\frac{m\pi x}{L}\right) dx + \int_{L/2}^L 2 \sin\left(\frac{m\pi x}{L}\right) dx \right] \\ &= -\frac{2}{L} \frac{L}{m\pi} \left[\cos\left(\frac{m\pi x}{L}\right) \Big|_0^{L/2} + 2 \cos\left(\frac{m\pi x}{L}\right) \Big|_{L/2}^L \right] \\ &= \frac{2}{m\pi} \left[1 + \cos\left(\frac{m\pi}{2}\right) - 2 \cos(m\pi) \right] \end{aligned}$$

Unfortunately, due to the $\cos(m\pi/2)$ term and the modulo 4 arithmetic involved in evaluating it, there is no simpler algebraic expression for this Fourier coefficient. Nevertheless, it does allow us to write the complete temperature distribution:

$$u(x, t) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left[1 + \cos\left(\frac{n\pi}{2}\right) - 2 \cos(n\pi) \right] \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

(e) In the absolute general case for the initial condition, we set equal

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right)$$

Multiply both sides by $\sin(m\pi x/L)$, integrate over 0 to L , and invoke sine orthogonality to get

$$\begin{aligned} \int_0^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx &= \int_0^L \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \\ &= A_m \int_0^L \sin^2\left(\frac{m\pi x}{L}\right) dx \end{aligned}$$

Leading us to

$$A_m = \frac{\int_0^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx}{\int_0^L \sin^2\left(\frac{m\pi x}{L}\right) dx} = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{m\pi x}{L}\right) dx$$

Question: 2.3.4

Consider

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

subject to $u(0, t) = 0$, $u(L, t) = 0$, and $u(x, 0) = f(x)$.

- (a) What is the total heat energy in the rod as a function of time?
- (b) What is the flow of heat energy out of the rod at $x = 0$? at $x = L$?
- (c) What relationship should exist between parts (a) and (b)?

- (a) For the one-dimensional rod, we have the energy per unit length, $c\rho Au(x, t)$. To find the total energy in the rod, we must integrate across the entire length 0 to L . Let B_n be the Fourier coefficient derived in 2.3.3(e). Compute

$$\begin{aligned} \text{Total Energy} &= \int_0^L c\rho Au(x, t) dx \\ &= \int_0^L c\rho A \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} dx \\ &= c\rho A \sum_{n=1}^{\infty} B_n \frac{L}{n\pi} [1 - \cos(n\pi)] e^{-k(n\pi/L)^2 t} \\ &= c\rho A \sum_{j=1}^{\infty} B_{2j-1} \frac{2L}{(2j-1)\pi} e^{-k((2j-1)\pi/L)^2 t} \end{aligned}$$

Where in the final step, we re-index the sum based on whether n is even or odd.

- (b) From the Fourier heat transfer law, we multiply by the area A to get the total flow of energy per time

$$\phi(x) A = -k_0 \frac{\partial u}{\partial x} = -k_0 A \sum_{n=1}^{\infty} B_n \frac{n\pi}{L} \cos\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

At either end

$$\begin{aligned} \phi(0) A &= -k_0 A \sum_{n=1}^{\infty} B_n \frac{n\pi}{L} e^{-k(n\pi/L)^2 t} \\ \phi(L) A &= -k_0 A \sum_{n=1}^{\infty} B_n \frac{n\pi}{L} (-1)^n e^{-k(n\pi/L)^2 t} \end{aligned}$$

where the sign convention is $\phi(x)$ describes the *rightward* flow of heat energy per time per unit area at point x .

- (c) Physically, we can reconstruct the heat equation by balancing the rate of change of total energy with the spatial difference in flux at both ends of

the rod. For the former, if E is the total energy and $k = k_0/c\rho$, then

$$\begin{aligned}\frac{\partial E}{\partial t} &= -c\rho k A \sum_{j=1}^{\infty} B_{2j-1} \left(\frac{(2j-1)\pi}{L} \right)^2 \left(\frac{2L}{(2j-1)\pi} \right) e^{-k((2j-1)\pi/L)^2 t} \\ &= -k_0 A \sum_{j=1}^{\infty} 2B_{2j-1} \frac{(2j-1)\pi}{L} e^{-k((2j-1)\pi/L)^2 t}\end{aligned}$$

and for the latter

$$\begin{aligned}[\phi(0) - \phi(L)] A &= -k_0 A \sum_{n=1}^{\infty} B_n \frac{n\pi}{L} e^{-k(n\pi/L)^2 t} (1 - (-1)^n) \\ &= -k_0 A \sum_{j=1}^{\infty} 2B_{2j-1} \frac{(2j-1)\pi}{L} e^{-k((2j-1)\pi/L)^2 t}\end{aligned}$$

where the last line arises from the elimination of the even modes, leaving only the odd modes. Hence the rate of change of energy balances the spatial difference of heat flux:

$$\frac{\partial E}{\partial t} = [\phi(0) - \phi(L)] A$$

Question: 2.3.5

Evaluate (be careful if $n = m$)

$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) \sin\left(\frac{m\pi x}{L}\right) dx \quad \text{for } n > 0, m > 0$$

Use the trigonometric identity

$$\sin(a) \sin(b) = \frac{1}{2} [\cos(a-b) - \cos(a+b)]$$

From the identity, if $n \neq m$, solve

$$\begin{aligned}\int_0^L \frac{1}{2} \left[\cos\left(\frac{(n-m)\pi x}{L}\right) - \cos\left(\frac{(n+m)\pi x}{L}\right) \right] dx &= \frac{1}{2} \left[\frac{L}{(n-m)\pi} \sin\left(\frac{(n-m)\pi x}{L}\right) - \right. \\ &\quad \left. \frac{L}{(n+m)\pi} \sin\left(\frac{(n+m)\pi x}{L}\right) \right] \Big|_0^L \\ &= 0\end{aligned}$$

which proves the orthogonality of sines when $n \neq m$. In the case when $n = m$, solve using the double angle identity:

$$\begin{aligned}\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx &= \int_0^L \frac{1}{2} \left(1 - \cos\left(\frac{2n\pi x}{L}\right) \right) dx \\ &= \frac{x}{2} \Big|_0^L - \frac{L}{4n\pi} \cos\left(\frac{2n\pi x}{L}\right) \Big|_0^L \\ &= \frac{L}{2}\end{aligned}$$

Explaining the $L/2$ factor that arises when calculating Fourier coefficients.

Question: 2.3.6

Evaluate

$$\int_0^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx \quad \text{for } n \geq 0, m \geq 0$$

Use the trigonometric identity

$$\cos(a) \cos(b) = \frac{1}{2} [\cos(a+b) + \cos(a-b)]$$

(Be careful if $a-b=0$ or $a+b=0$.)Suppose we pick n and m such that $n-m \neq 0$ and $n+m \neq 0$. Then solve

$$\begin{aligned} \int_0^L \frac{1}{2} \left[\cos\left(\frac{(n+m)\pi x}{L}\right) + \cos\left(\frac{(n-m)\pi x}{L}\right) \right] dx &= \frac{1}{2} \left[\frac{L}{(n+m)\pi} \sin\left(\frac{(n+m)\pi x}{L}\right) + \right. \\ &\quad \left. \frac{L}{(n-m)\pi} \sin\left(\frac{(n-m)\pi x}{L}\right) \right] \Big|_0^L \\ &= 0 \end{aligned}$$

which demonstrates the orthogonality of cosines. In the case when $m=n \neq 0$, then

$$\begin{aligned} \int_0^L \cos^2\left(\frac{n\pi x}{L}\right) dx &= \int_0^L \frac{1}{2} \left[1 + \cos\left(\frac{2n\pi x}{L}\right) \right] dx \\ &= \frac{x}{2} \Big|_0^L + \frac{L}{2n\pi} \sin\left(\frac{2n\pi x}{L}\right) \Big|_0^L \\ &= \frac{L}{2} \end{aligned}$$

Again demonstrating that the squared norm of each nonzero cosine mode on $[0, L]$ is $L/2$. Now, in the special case when $m=n=0$, both cosines are 1, and

$$\int_0^L dx = L$$

We are not yet at this point in the textbook, but it is this result that explains the $a_0 = \frac{1}{L} \int_0^L f(x) dx$ term in the Fourier cosine series.

Question: 2.3.7

Consider the following boundary value problem (if necessary, see Section 2.4.1):

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} \text{ with } \frac{\partial u}{\partial x}(0, t) = 0, \frac{\partial u}{\partial x}(L, t) = 0, \text{ and } u(x, 0) = f(x)$$

- (a) Give a one-sentence physical interpretation of this problem.
- (b) Solve by the method of separation of variables. First show that there are no separated solutions which exponentially grow in time. [Hint: The answer is

$$u(x, t) = A_0 + \sum_{n=1}^{\infty} A_n e^{-\lambda_n k t} \cos\left(\frac{n\pi x}{L}\right)]$$

What is λ_n ?

- (c) Show that the initial condition, $u(x, 0) = f(x)$, is satisfied if

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

- (d) Using Exercise 2.3.6, solve for A_0 and A_n ($n \geq 1$).
- (e) What happens to the temperature distribution as $t \rightarrow \infty$? Show that it approaches the steady-state temperature distribution (see Section 1.4).

- (a) The ends of the rod are insulated.
- (b) Let $u(x, t) = \phi(x) G(t)$. Then

$$\begin{aligned} \frac{\partial u}{\partial t} &= \phi(x) \frac{dG}{dt} \\ k \frac{\partial^2 u}{\partial x^2} &= k \frac{d^2 \phi}{dx^2} G(t) \end{aligned}$$

Separating variables, derive

$$\frac{1}{kG(t)} \frac{dG}{dt} = \frac{1}{\phi(x)} \frac{d^2 \phi}{dx^2} = -\lambda$$

From the time equation, we get

$$G(t) = ce^{-\lambda k t}$$

and from the spatial equation, we must solve by assuming different values of λ . First, consider $\lambda < 0$, and let $s = -\lambda$. Then $s > 0$. We solve

$$\frac{d^2 \phi}{dx^2} = -\lambda \phi(x)$$

and find

$$\phi(x) = c_1 \cosh(\sqrt{s}x) + c_2 \sinh(\sqrt{s}x)$$

Now, we must verify that we can find non-trivial solutions subject to the boundary conditions $\partial\phi/\partial x(0, t) = 0$ and $\partial\phi/\partial x(L, t) = 0$. First, the spatial derivative

$$\frac{\partial\phi}{\partial x} = c_1 \sqrt{s} \sinh(\sqrt{s}x) + c_2 \sqrt{s} \cosh(\sqrt{s}x)$$

And evaluating the boundary conditions, we find

$$\frac{\partial\phi}{\partial x}(0) = c_2 \sqrt{s} = 0$$

which implies $c_2 = 0$. For the second,

$$\frac{\partial\phi}{\partial x}(L) = c_1 \sqrt{s} \sinh(\sqrt{s}L) = 0$$

which must imply $c_1 = 0$; for $\sinh(\sqrt{s}L) = 0$, we would require $\sqrt{s}L = 0$, but that cannot be the case. Therefore, only the trivial solution remains. Now, $G(t) = ce^{-\lambda kt}$ grows only when $\lambda < 0$, but since $\lambda < 0$ only admits the trivial solution, we have now established that there cannot be any non-trivial solution that exponentially grows in time.

In the case where $\lambda = 0$, we solve the ordinary differential equation to get

$$\phi(x) = c_1 + c_2 x$$

and imposing the boundary conditions produces

$$\frac{\partial\phi}{\partial x}(0) = \frac{\partial\phi}{\partial x}(L) = c_2 = 0$$

which implies constant $\phi(x)$ and $G(t)$ and thus, constant temperature throughout the rod. We will include this constant term in our solution shortly.

Lastly, when $\lambda > 0$, we have

$$\phi(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

which has spatial derivative

$$\frac{\partial\phi}{\partial x} = -c_1 \sqrt{\lambda} \sin(\sqrt{\lambda}x) + c_2 \sqrt{\lambda} \cos(\sqrt{\lambda}x)$$

Imposing the boundary conditions yield

$$\frac{\partial\phi}{\partial x}(0) = c_2 \sqrt{\lambda} = 0 \implies c_2 = 0, \quad \frac{\partial\phi}{\partial x}(L) = -c_1 \sqrt{\lambda} \sin(\sqrt{\lambda}L) = 0$$

where in the second boundary condition, if we hold $c_1 \neq 0$, a non-trivial solution emerges if we set

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

where n is any positive integer. Since the solution holds for any cosine mode n , appealing to superposition, we can write our complete solution by combining the separated functions into a linear combination:

$$u(x, t) = A_0 + \sum_{n=1}^{\infty} A_n e^{-\lambda_n k t} \cos\left(\frac{n\pi x}{L}\right)$$

where A_0 is our constant solution from the zero eigenvalue case, λ_n our indexed eigenvalue, and A_n arbitrary constants determined by the initial condition $f(x)$.

(c) From part (b), at $t = 0$, we have

$$u(x, 0) = f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

(d) For A_0 , since L is the squared norm of cosine at $n = m = 0$, we simply write

$$A_0 = \frac{\int_0^L f(x) \cos(0) dx}{\int_0^L dx} = \frac{1}{L} \int_0^L f(x) dx$$

As for the remaining coefficients, we have

$$A_n = \frac{\int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx}{\int_0^L \cos^2\left(\frac{n\pi x}{L}\right) dx} = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

(e) In the limit, we can evaluate

$$\lim_{t \rightarrow \infty} u(x, t) = \lim_{t \rightarrow \infty} \left[A_0 + \sum_{n=1}^{\infty} A_n e^{-\lambda_n k t} \cos\left(\frac{n\pi x}{L}\right) \right] = A_0$$

as demonstrated in equation (1.4.21), the steady-state distribution in the case of insulated ends produces the average of the initial temperature distribution:

$$A_0 = \frac{1}{L} \int_0^L f(x) dx$$

Question: 2.3.8

Consider

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} - \alpha u$$

This corresponds to a one-dimensional rod either with heat loss through the lateral sides with outside temperature 0° ($\alpha > 0$, see Exercise 1.2.4) or with insulated lateral sides with a heat sink proportional to the temperature. Suppose that the boundary conditions are

$$u(0, t) = 0 \text{ and } u(L, t) = 0$$

- (a) What are the possible equilibrium temperature distributions if $\alpha > 0$?
- (b) Solve the time-dependent problem $[u(x, 0) = f(x)]$ if $\alpha > 0$. Analyze the temperature for large time ($t \rightarrow \infty$) and compare part (a).

- (a) Let $\partial u / \partial t = 0$. Then isolating the second derivative, we solve

$$\frac{d^2 u}{dx^2} = \frac{\alpha}{k} u$$

Let $u = e^{rx}$. Then the characteristic polynomial $r^2 - \alpha/k = 0$ emerges, producing roots $r = \pm\sqrt{\alpha/k}$. The general solution is

$$u(x) = c_1 e^{\sqrt{\alpha/k}x} + c_2 e^{-\sqrt{\alpha/k}x}$$

The boundary conditions produce the equations

$$u(0) = c_1 + c_2 = 0$$

$$u(L) = c_1 e^{\sqrt{\alpha/k}L} + c_2 e^{-\sqrt{\alpha/k}L} = 0$$

If you write out the matrix equation, it is easy to see that the two columns of the coefficient matrix are linearly independent; the first entries are both 1 while the second entries differ. Therefore only the trivial solution exists. More concretely, from the first equation we deduce $c_1 = -c_2$. Substitute into the second equation to get

$$u(L) = c_1 [e^{\sqrt{\alpha/k}L} - e^{-\sqrt{\alpha/k}L}] = 0$$

The bracketed term is zero only when $\alpha = 0$, so that forces $c_1 = 0$. Immediately from the first equation, it follows that $c_2 = 0$ as well. Therefore $u(x) = 0$ is the only permissible temperature distribution at equilibrium.

- (b) Use the method of separation of variables. Let $u(x, t) = \phi(x) G(t)$. Then after some substitution and division, derive

$$\frac{1}{kG(t)} \frac{dG}{dt} = \frac{1}{\phi(x)} \frac{d^2 \phi}{dx^2} - \frac{\alpha}{k} = -\lambda$$

The time equation is derived similarly as before:

$$G(t) = ce^{-\lambda kt}$$

The spatial equation, on the other hand, looks like

$$\frac{d^2\phi}{dx^2} = \left[-\lambda + \frac{\alpha}{k}\right] \phi(x)$$

There is no difference in technique, we need only handle the additional α/k term. To do so, consider first the case where $\lambda < \alpha/k$. Then

$$\phi(x) = c_1 \cosh\left(\left(\sqrt{\frac{\alpha}{k} - \lambda}\right)x\right) + c_2 \sinh\left(\left(\sqrt{\frac{\alpha}{k} - \lambda}\right)x\right)$$

Imposing the boundary conditions produces

$$\phi(0) = c_1 = 0, \quad \phi(L) = c_2 \sinh\left(\left(\sqrt{\frac{\alpha}{k} - \lambda}\right)L\right) = 0$$

Since the argument of the sinh term is positive, it follows that the term itself cannot be zero, forcing $c_2 = 0$. Therefore only the trivial solution arises.

Next, consider $\lambda = \alpha/k$. Then the right-hand side of the ordinary differential equation vanishes, and integrating twice for the general solution leads to

$$\phi(x) = c_1 + c_2x$$

The boundary conditions once again enable us to conclude that only the trivial solution is viable:

$$\phi(0) = c_1 = 0, \quad \phi(L) = c_2L = 0 \implies c_2 = 0$$

Lastly, examine $\lambda > \alpha/k$. Then we have general solution

$$\phi(x) = c_1 \cos\left(\left(\sqrt{\lambda - \frac{\alpha}{k}}\right)x\right) + c_2 \sin\left(\left(\sqrt{\lambda - \frac{\alpha}{k}}\right)x\right)$$

Applying the boundary conditions yield

$$\phi(0) = c_1 = 0, \quad \phi(L) = c_2 \sin\left(\left(\sqrt{\lambda - \frac{\alpha}{k}}\right)L\right) = 0$$

where in the second boundary condition, a non-trivial solution emerges by holding $c_2 \neq 0$ and setting

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2 + \frac{\alpha}{k}$$

for positive integers n . This yields eigenfunctions

$$\phi(x) = \sin\left(\frac{n\pi x}{L}\right)$$

By superposition and combining the spatial and temporal solutions, we derive the time-dependent temperature distribution

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-[(n\pi/L)^2 + \alpha/k]kt} \\ &= e^{-\alpha t} \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} \end{aligned}$$

where B_n is defined as

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

In the temporal limit, we can see that

$$\lim_{t \rightarrow \infty} u(x, t) = \lim_{t \rightarrow \infty} e^{-\alpha t} \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} = 0$$

which confirms our equilibrium result.

Question: 2.3.9

Redo Exercise 2.3.8 if $\alpha < 0$. [Be especially careful if $-\alpha/k = (n\pi/L)^2$.]

- (a) When $\alpha < 0$, the roots of the characteristic polynomial are $r = \pm i\sqrt{-\alpha/k}$. The general solution now changes to

$$u(x) = c_1 \cos\left(\sqrt{-\frac{\alpha}{k}}x\right) + c_2 \sin\left(\sqrt{-\frac{\alpha}{k}}x\right)$$

The boundary conditions produce the equations

$$\begin{aligned} u(0) &= c_1 = 0 \\ u(L) &= c_2 \sin\left(\sqrt{-\frac{\alpha}{k}}L\right) = 0 \end{aligned}$$

where in the second equation, it is not the general case that the sine term is zero, so we must also have $c_2 = 0$. Thus in the general case, $u(x) = 0$, once again, is the only permissible temperature distribution at equilibrium.

However, consider the case when $-\alpha/k = (n\pi/L)^2$, where n is a positive integer. Then there exists a non-trivial solution, as the sine term goes to zero, and c_2 is left as a free parameter (contingent on the initial temperature). We are left with $u(x) = c_2 \sin(n\pi x/L)$.

- (b) The derivation of the spatial equations proceeds exactly as in the case of positive α . The eigenvalues are also

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2 + \frac{\alpha}{k}$$

However, since $\alpha/k < 0$, we reach an interesting conclusion: $\lambda_n > 0$ is no longer a guarantee.

Let us take a step back. What does $\alpha < 0$ physically represent? Previously, we interpreted the $-\alpha u$ term as a heat sink. Reversing the coefficient sign turns this into a heat *source*. We have three possibilities:

- Net outward diffusion of heat energy exceeds internal generation of heat from the source, contributing to energy loss and decaying temperature over time.
- Net outward diffusion of heat energy perfectly balances internal heat generation, contributing to constant energy and constant temperature distribution over time.
- Net outward diffusion of heat energy is less than internal heat generation, contributing to energy growth and increasing temperature over time.

Consider the bracketed term $(n\pi/L)^2 + \alpha/k$ in the exponential:

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k[(n\pi/L)^2 + \alpha/k]t}$$

A subtle point: we assume that α/k is some fixed quantity. Therefore, its ordering relative to $(n\pi/L)^2$ will depend on n . Conceptually, we must treat each mode on a case-by-case matter. For each respective possibility and each mode, we have

- Net outward diffusion of heat energy $>$ internal heat generation:
 $(n\pi/L)^2 > -\alpha/k, \lim_{t \rightarrow \infty} u_n(x, t) = 0$
- Net outward diffusion of heat energy = internal heat generation:
 $(n\pi/L)^2 = -\alpha/k, \lim_{t \rightarrow \infty} u_n(x, t) = B_n \sin(n\pi x/L)$
- Net outward diffusion of heat energy $<$ internal heat generation:
 $(n\pi/L)^2 < -\alpha/k, \lim_{t \rightarrow \infty} u_n(x, t) = \infty$

where $u_n(x, t)$ corresponds to the n -th mode

$$u_n(x, t) = B_n \sin\left(\frac{n\pi x}{L}\right) e^{-k[(n\pi/L)^2 + \alpha/k]t}$$

Observe that the limit of the case of equality is equivalent to the non-trivial solution derived in part (a): balancing of outward diffusion and internal generation leads to a steady-state temperature distribution.

Lastly, to find the limit of the *entire* temperature distribution $u(x, t)$, choose the smallest mode: $n = 1$. Since we fix α and k , at some point, $(n\pi/L)^2$ will exceed $-\alpha/k$, leading to temporal decay of those modes. What matters is *where* $(\pi/L)^2$ starts off *in relation* to $-\alpha/k$. If $(\pi/L)^2 > -\alpha/k$, then

$$\lim_{t \rightarrow \infty} u(x, t) = 0$$

Otherwise, in the cases when $(\pi/L)^2 = -\alpha/k$ and $(\pi/L)^2 < -\alpha/k$, we have, respectively

$$\lim_{t \rightarrow \infty} u(x, t) = B_1 \sin\left(\frac{\pi x}{L}\right) \quad \lim_{t \rightarrow \infty} u(x, t) = \infty$$

Author's note: I love this problem. It is one of those problems where struggling will confer enormous insight. Who knew that reversing the sign on α would lend itself to such rich physics? My advice: reason through the conceptual physics first, before attempting to rush into mathematical manipulations. Thorough understanding of the physical system will provide the insight that very little manipulation is needed; subtle understanding of the behavior of each independent mode vis-à-vis the complete solution is essential.

Question: 2.3.10

For two- and three-dimensional vectors, the fundamental property of dot products, $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos(\theta)$, implies that

$$|\mathbf{A} \cdot \mathbf{B}| \leq |\mathbf{A}||\mathbf{B}| \quad 2.3.44$$

In this exercise, we generalize this to n -dimensional vectors and functions, in which case (2.3.44) is known as **Schwarz's inequality**. [The names of Cauchy and Buniakovsky are also associated with (2.3.44).]

- (a) Show that $|\mathbf{A} - \gamma \mathbf{B}|^2 \geq 0$ implies (2.3.44), where $\gamma = \mathbf{A} \cdot \mathbf{B} / \mathbf{B} \cdot \mathbf{B}$.
- (b) Express the inequality using both

$$\mathbf{A} \cdot \mathbf{B} = \sum_{n=1}^{\infty} a_n b_n = \sum_{n=1}^{\infty} a_n c_n \frac{b_n}{c_n}$$

- (c) Generalize (2.3.44) to functions. [Hint: Let $\mathbf{A} \cdot \mathbf{B}$ mean the integral $\int_0^L A(x) B(x) dx$.]

- (a) Compute

$$\begin{aligned} |\mathbf{A} - \gamma \mathbf{B}|^2 &= (\mathbf{A} - \gamma \mathbf{B}) \cdot (\mathbf{A} - \gamma \mathbf{B}) \\ &= \mathbf{A} \cdot \mathbf{A} - 2\gamma \mathbf{A} \cdot \mathbf{B} + \gamma^2 \mathbf{B} \cdot \mathbf{B} \\ &= |\mathbf{A}|^2 - \frac{2(\mathbf{A} \cdot \mathbf{B})^2}{|\mathbf{B}|^2} + \frac{(\mathbf{A} \cdot \mathbf{B})^2}{|\mathbf{B}|^2} \\ &\geq 0 \end{aligned}$$

Multiplying through by $|\mathbf{B}|^2$, collecting the $(\mathbf{A} \cdot \mathbf{B})^2$ terms and adding to the other side of the inequality, we get

$$|\mathbf{A}|^2 |\mathbf{B}|^2 \geq (\mathbf{A} \cdot \mathbf{B})^2$$

taking the positive root of both sides, we conclude the Schwarz inequality:

$$|\mathbf{A}||\mathbf{B}| \geq |\mathbf{A} \cdot \mathbf{B}|$$

- (b) From the squared vector definition of the Schwarz inequality, to express in algebraic form, we equivalently derive

$$\begin{aligned}
 (\mathbf{A} \cdot \mathbf{B})^2 &= \left(\sum_{n=1}^{\infty} a_n b_n \right)^2 \\
 &\leq \left(\sum_{n=1}^{\infty} a_n^2 \right) \left(\sum_{n=1}^{\infty} b_n^2 \right) \\
 &= |\mathbf{A}|^2 |\mathbf{B}|^2 \\
 (\mathbf{A} \cdot \mathbf{B})^2 &= \left(\sum_{n=1}^{\infty} a_n c_n \frac{b_n}{c_n} \right)^2 \\
 &\leq \left(\sum_{n=1}^{\infty} a_n^2 c_n^2 \right) \left(\sum_{n=1}^{\infty} \frac{b_n^2}{c_n^2} \right)
 \end{aligned}$$

The second inequality can be interpreted as a weighted inner product.

- (c) Similarly, we proceed from the squared vector definition of the Schwarz inequality, and derive in integral form

$$\left(\int_0^L A(x) B(x) dx \right)^2 \leq \int_0^L A(x)^2 dx \int_0^L B(x)^2 dx$$

Question: 2.3.11

Solve the heat equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$ subject to the following conditions:

$$u(0, t) = 0 \quad u(L, t) = 0 \quad u(x, 0) = f(x)$$

What happens as $t \rightarrow \infty$? [Hints:

- (1) It is known that if $u(x, t) = \phi(x) G(t)$, then $\frac{1}{kG} \frac{dG}{dt} = \frac{1}{\phi} \frac{\partial^2 \phi}{\partial x^2}$.
- (2) Use formula sheet.]

Let $u(x, t) = \phi(x) G(t)$. Then

$$\begin{aligned}
 \frac{\partial u}{\partial t} &= \phi(x) \frac{dG}{dt} \\
 k \frac{\partial^2 u}{\partial x^2} &= k \frac{d^2 \phi}{dx^2} G(t)
 \end{aligned}$$

Collecting similar variables and setting equal to a constant $-\lambda$, derive

$$\frac{1}{kG(t)} \frac{dG}{dt} = \frac{1}{\phi(x)} \frac{d^2 \phi}{dx^2} = -\lambda$$

This produces time and spatial equations and we solve each individually. First the time equation:

$$G(t) = ce^{-\lambda kt}$$

And for the spatial equation, we consider three cases of λ : above, below, or at zero. Try $\lambda > 0$. Then $-\lambda < 0$, and after substituting $u = e^{rx}$, the characteristic polynomial yields complex roots. The ensuing linear combination of Euler equations can be rewritten as a linear combination of cosine and sine:

$$\phi(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions, we find

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \phi(L) &= c_2 \sin(\sqrt{\lambda}L) = 0\end{aligned}$$

where in the second boundary condition, if we assume $c_2 \neq 0$ so as not to be left with a trivial solution, and set $\lambda_n = (n\pi/L)^2$, then we have eigenvalues for the spatial function

$$\phi(x) = c_2 \sin\left(\frac{n\pi x}{L}\right)$$

Now, assume $\lambda = 0$. Then the spatial solution is

$$\phi(x) = c_1 + c_2 x$$

With boundary conditions $\phi(0) = c_1 = 0$ and $\phi(L) = c_2 L = 0$ implying $c_2 = 0$, we are left only with the trivial solution. Lastly, try $\lambda < 0$. Then set $-\lambda = s$, which is positive. With positive roots of the characteristic polynomial, we can express the spatial solution as a linear combination of hyperbolic cosine and sine:

$$\phi(x) = c_1 \cosh(\sqrt{s}x) + c_2 \sinh(\sqrt{s}x)$$

The boundary conditions produce $\phi(0) = c_1$ and $\phi(L) = c_2 \sinh(\sqrt{s}L) = 0$. Since $\sqrt{s}L \neq 0$, it is impossible for the hyperbolic sine to be zero, forcing $c_2 = 0$. Once more, the trivial solution is all that remains.

Therefore we are only permitted eigenvalues $\lambda_n > 0$.

Multiplying $\phi(x)$ and $G(t)$ and appealing to superposition, we have complete solution

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t}$$

where the coefficients A_n are derived from the projection of the initial temperature $f(x)$ onto each of the sine modes spanning the Hilbert space of orthogonal sines:

$$A_n = \frac{\int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx}{\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx} = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Lastly, in the temporal limit, we have

$$\lim_{t \rightarrow \infty} u(x, t) = \lim_{t \rightarrow \infty} \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-k(n\pi/L)^2 t} = 0$$

Physically, as we do not have insulation and the ends of the rod are assumed to be at zero temperature, the rod itself will eventually cool to the uniformly zero equilibrium temperature.

2.4 - Worked Examples with the Heat Equation (Other Boundary Value Problems)

Question: 2.4.1

Solve the heat equation $\partial u / \partial t = k \partial^2 u / \partial x^2$, $0 < x < L$, $t > 0$, subject to

$$\frac{\partial u}{\partial x}(0, t) = 0 \quad t > 0$$

$$\frac{\partial u}{\partial x}(L, t) = 0 \quad t > 0$$

$$(a) \quad u(x, 0) = \begin{cases} 0 & x < L/2 \\ 1 & x > L/2 \end{cases}$$

$$(b) \quad u(x, 0) = 6 + 4 \cos\left(\frac{3\pi x}{L}\right)$$

$$(c) \quad u(x, 0) = -2 \sin\left(\frac{\pi x}{L}\right)$$

$$(d) \quad u(x, 0) = -3 \cos\left(\frac{8\pi x}{L}\right)$$

For all solutions, use the general form

$$u(x, t) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

where

$$u(x, 0) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right)$$

- (a) Multiply the initial temperature by $\cos(m\pi x/L)$ and integrate from 0 to L to find

$$\begin{aligned} \int_0^L u(x, 0) \cos\left(\frac{m\pi x}{L}\right) dx &= \int_0^{L/2} 0 \cdot \cos\left(\frac{m\pi x}{L}\right) dx + \int_{L/2}^L 1 \cdot \cos\left(\frac{m\pi x}{L}\right) dx \\ &= \int_0^L A_0 \cos\left(\frac{m\pi x}{L}\right) dx + \int_0^L \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx \\ &= \begin{cases} A_0 L & m = 0 \\ A_m \int_0^L \cos^2\left(\frac{m\pi x}{L}\right) dx & m \geq 1 \end{cases} \end{aligned}$$

First isolate A_0 by setting $m = 0$, and derive

$$A_0 L = L - \frac{L}{2} = \frac{L}{2} \quad \implies \quad A_0 = \frac{1}{2}$$

Next, isolating A_m , derive

$$\begin{aligned} A_m &= 0 + \frac{\int_{L/2}^L \cos\left(\frac{m\pi x}{L}\right) dx}{\int_0^L \cos^2\left(\frac{m\pi x}{L}\right) dx} \\ &= -\frac{2}{m\pi} \sin\left(\frac{m\pi}{2}\right) \\ &= \begin{cases} 0 & m \text{ even} \\ -\frac{2}{m\pi} \sin\left(\frac{m\pi}{2}\right) & m \text{ odd} \end{cases} \end{aligned}$$

Reindexing, the second case can be rewritten for all positive integers n as

$$A_n = (-1)^n \frac{2}{(2n-1)\pi}$$

Substituting into the general form, we have temperature distribution

$$u(x, t) = \frac{1}{2} + \sum_{n=1}^{\infty} (-1)^n \frac{2}{(2n-1)\pi} \cos\left(\frac{(2n-1)\pi x}{L}\right) e^{-((2n-1)\pi/L)^2 kt}$$

- (b) The boundary conditions are satisfied, so we can conclude $A_0 = 6$, $n = 3$, and $A_3 = 4$.
- (c) In this case, the boundary conditions are not satisfied. Multiply by $\cos(m\pi x/L)$ and integrate from 0 to L to derive

$$\begin{aligned} \int_0^L u(x, 0) \cos\left(\frac{m\pi x}{L}\right) dx &= -2 \int_0^L \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx \\ &= \int_0^L A_0 \cos\left(\frac{m\pi x}{L}\right) dx + \int_0^L \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx \\ &= \begin{cases} A_0 L & m = 0 \\ A_m \int_0^L \cos^2\left(\frac{m\pi x}{L}\right) dx & m \geq 1 \end{cases} \end{aligned}$$

First suppose $m = 0$. Then A_0 is

$$A_0 = -\frac{2}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) dx = -\frac{4}{\pi}$$

and for all other m , we have

$$\begin{aligned}
 A_m &= -\frac{4}{L} \int_0^L \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{m\pi x}{L}\right) dx \\
 &= -\frac{2}{L} \int_0^L \left[\sin\left(\frac{(1+m)\pi x}{L}\right) + \sin\left(\frac{(1-m)\pi x}{L}\right) \right] dx \\
 &= \frac{2}{L} \left[\frac{L}{(1+m)\pi} \cos\left(\frac{(1+m)\pi x}{L}\right) + \frac{L}{(1-m)\pi} \cos\left(\frac{(1-m)\pi x}{L}\right) \right] \Big|_0^L \\
 &= \frac{2}{\pi} \left[\frac{\cos((1+m)\pi) - 1}{1+m} + \frac{\cos((1-m)\pi) - 1}{1-m} \right] \\
 &= \begin{cases} 0 & m \text{ odd} \\ -\frac{4}{\pi} \left[\frac{1}{1+m} + \frac{1}{1-m} \right] & m \text{ even} \end{cases}
 \end{aligned}$$

Note that the $1-m$ term in the denominator shows up in the derivation, but since we eventually determined that the coefficient vanishes for all odd m , we need not worry about a division by zero. Reindexing the second case, we have for positive integers n

$$A_n = -\frac{8}{\pi} \left[\frac{1}{(1+2n)(1-2n)} \right] = \frac{8}{\pi(4n^2 - 1)}$$

Therefore, the complete temperature distribution is

$$u(x, t) = -\frac{4}{\pi} + \sum_{n=1}^{\infty} \frac{8}{\pi(4n^2 - 1)} \cos\left(\frac{2n\pi x}{L}\right) e^{-(2n\pi/L)^2 kt}$$

- (d) Lastly, the boundary conditions are met, so we can simply assign $A_0 = 0$, $n = 8$, and $A_8 = -3$.

Question: 2.4.2

Solve

$$\begin{aligned}
 \frac{\partial u}{\partial t} &= k \frac{\partial^2 u}{\partial x^2} \quad \text{with} \quad \frac{\partial u}{\partial x}(0, t) = 0 \\
 & \quad u(L, t) = 0 \\
 & \quad u(x, 0) = f(x)
 \end{aligned}$$

For this problem you may assume that no solutions of the heat equation exponentially grow in time. You may also guess appropriate orthogonality conditions for the eigenfunctions.

In the case of mixed boundary conditions, proceed normally: let $u(x, t) = \phi(x)G(t)$. From the typical derivation, we have temporal solution $G(t) = ce^{-\lambda kt}$. For the spatial solutions, begin with $\lambda > 0$. Then the characteristic polynomial produces imaginary roots, with general solution

$$\phi(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Imposing the boundary conditions yield

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2\sqrt{\lambda} = 0, \implies c_2 = 0 \\ \phi(L) &= c_1 \cos(\sqrt{\lambda}L) = 0\end{aligned}$$

where we have a non-trivial solution if

$$\lambda_n = \left[\frac{(2n-1)\pi}{2L} \right]^2$$

Next, suppose $\lambda = 0$. Then

$$\phi(x) = c_1 + c_2x$$

The boundary conditions give

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2 = 0 \\ \phi(L) &= c_1 = 0\end{aligned}$$

leaving us only with the trivial solution. Lastly, suppose $\lambda < 0$. Define $s = -\lambda$, so $s > 0$. The characteristic polynomial gives positive roots, so we have a linear combination of hyperbolic cosine and sine for the solution:

$$\phi(x) = c_1 \cosh(\sqrt{s}x) + c_2 \sinh(\sqrt{s}x)$$

and applying the boundary conditions

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2\sqrt{s} = 0 \implies c_2 = 0 \\ \phi(L) &= c_1 \cosh(\sqrt{s}L) = 0\end{aligned}$$

where the second boundary condition forces $c_1 = 0$, as the hyperbolic cosine is never zero. Thus we get the trivial solution once more. So we are left with the eigenfunctions

$$\phi_n(x) = c_1 \cos\left[\frac{(2n-1)\pi x}{2L}\right]$$

And lastly, appealing to superposition and combining with the temporal solution, we have temperature distribution

$$u(x, t) = \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\pi x}{2L}\right] e^{-[(2n-1)\pi/2L]^2 kt}$$

where the Fourier coefficients must be derived from projecting $f(x)$ onto the odd “half-modes” of cosine Hilbert space. We begin with

$$f(x) = \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\pi x}{2L}\right]$$

Now, multiply both sides by $\cos((2m-1)\pi x/2L)$ and integrate from 0 to L to get

$$\begin{aligned}\int_0^L f(x) \cos\left[\frac{(2m-1)\pi x}{2L}\right] dx &= \int_0^L \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\pi x}{2L}\right] \cos\left[\frac{(2m-1)\pi x}{2L}\right] dx \\ &= A_m \int_0^L \cos^2\left[\frac{(2m-1)\pi x}{2L}\right] dx \\ &= A_m \frac{L}{2} \\ \Rightarrow A_m &= \frac{2}{L} \int_0^L f(x) \cos\left[\frac{(2m-1)\pi x}{2L}\right] dx\end{aligned}$$

where the orthogonality condition still holds between differing values at the half-modes, enabling the truth of the second equality.

Question: 2.4.3

Solve the eigenvalue problem

$$\frac{d^2\phi}{dx^2} = -\lambda\phi$$

subject to

$$\phi(0) = \phi(2\pi) \quad \text{and} \quad \frac{d\phi}{dx}(0) = \frac{d\phi}{dx}(2\pi)$$

$\lambda > 0$: We have general solution

$$\phi(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

with boundary conditions

$$\begin{aligned}\phi(0) = c_1 &= c_1 \cos(2\sqrt{\lambda}\pi) + c_2 \sin(2\sqrt{\lambda}\pi) &= \phi(2\pi) \\ \frac{d\phi}{dx}(0) = c_2\sqrt{\lambda} &= -c_1\sqrt{\lambda} \sin(2\sqrt{\lambda}\pi) + c_2\sqrt{\lambda} \cos(2\sqrt{\lambda}\pi) &= \frac{d\phi}{dx}(2\pi)\end{aligned}$$

This system of equations is solved if we observe that $\lambda_n = n^2$, for $n = 0, 1, \dots$ is an eigenvalue for a corresponding non-trivial solution, namely

$$\phi_n(x) = c_1 \cos(nx) + c_2 \sin(nx)$$

More concretely, we can solve the matrix equation

$$\begin{bmatrix} 1 - \cos(2\sqrt{\lambda}\pi) & -\sin(2\sqrt{\lambda}\pi) \\ \sqrt{\lambda} \sin(2\sqrt{\lambda}\pi) & \sqrt{\lambda} [1 - \cos(2\sqrt{\lambda}\pi)] \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

In order for the existence of a non-trivial solution, we must have a zero determinant, as that guarantees a non-trivial linear combination of the columns to

yield the zero vector. Compute from the coefficient matrix

$$\begin{aligned}\det &= \sqrt{\lambda} \left[1 - \cos(2\sqrt{\lambda}\pi) \right]^2 + \sqrt{\lambda} \sin^2(2\sqrt{\lambda}\pi) \\ &= \sqrt{\lambda} \left[1 - 2\cos(2\sqrt{\lambda}\pi) + \sin^2(2\sqrt{\lambda}\pi) + \cos^2(2\sqrt{\lambda}\pi) \right] \\ &= \sqrt{\lambda} \left[2 - 2\cos(2\sqrt{\lambda}\pi) \right]\end{aligned}$$

where the equality is zero if and only if $\lambda_n = n^2$ for $n = 0, 1, \dots$. Moreover, note that the entire coefficient matrix goes to rank zero, meaning that c_1 and c_2 are unconstrained. This leaves us with the aforementioned eigenfunctions ϕ_n .

$\lambda = 0$: We have general solution

$$\phi(x) = c_1 + c_2 x$$

with boundary conditions

$$\begin{aligned}\phi(0) &= c_1 = c_1 + 2c_2\pi = \phi(2\pi) \\ \frac{d\phi}{dx}(0) &= c_2 = c_2 = \frac{d\phi}{dx}(2\pi)\end{aligned}$$

The boundary conditions force $c_2 = 0$, and c_1 is a free-standing constant. Thus for eigenvalue $\lambda_0 = 0$, we have corresponding eigenfunction

$$\phi_0 = c_1$$

$\lambda < 0$: Let $s = -\lambda$ so $s > 0$. Then we have general solution

$$\phi(x) = c_1 \cosh(\sqrt{s}x) + c_2 \sinh(\sqrt{s}x)$$

with boundary conditions

$$\begin{aligned}\phi(0) &= c_1 = c_1 \cosh(2\sqrt{s}\pi) + c_2 \sinh(2\sqrt{s}\pi) = \phi(2\pi) \\ \frac{d\phi}{dx}(0) &= c_2\sqrt{s} = c_1\sqrt{s} \sinh(2\sqrt{s}\pi) + c_2\sqrt{s} \cosh(2\sqrt{s}\pi) = \frac{d\phi}{dx}(2\pi)\end{aligned}$$

that correspond to the matrix equation

$$\begin{bmatrix} 1 - \cosh(2\sqrt{s}\pi) & \sinh(2\sqrt{s}\pi) \\ -\sqrt{s} \sinh(2\sqrt{s}\pi) & \sqrt{s}[1 - \cosh(2\sqrt{s}\pi)] \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Computing the determinant of the coefficient matrix produces

$$\begin{aligned}\det &= \sqrt{s} [1 - \cosh(2\sqrt{s}\pi)]^2 + \sqrt{s} \sinh^2(2\sqrt{s}\pi) \\ &= \sqrt{s} [1 - 2\cosh(2\sqrt{s}\pi) + \sinh^2(2\sqrt{s}\pi) + \cosh^2(2\sqrt{s}\pi)] \\ &= \sqrt{s} [-2\cosh(2\sqrt{s}\pi) + 2\cosh^2(2\sqrt{s}\pi)] \\ &= 2\sqrt{s} \cosh(2\sqrt{s}\pi) [\cosh(2\sqrt{s}\pi) - 1]\end{aligned}$$

which vanishes only when $2\sqrt{s}\pi = 0$, but since we specified positive s , we can only have the trivial solution: $c_1 = c_2 = 0$.

Question: 2.4.4

Explicitly show that there are no negative eigenvalues for

$$\frac{d^2\phi}{dx^2} = -\lambda\phi \quad \text{subject to} \quad \frac{d\phi}{dx}(0) = 0 \quad \text{and} \quad \frac{d\phi}{dx}(L) = 0$$

Let $\lambda < 0$. Then let $s = -\lambda$, so $s > 0$. We have general solution

$$\phi(x) = c_1 \cosh(\sqrt{s}x) + c_2 \sinh(\sqrt{s}x)$$

with boundary conditions

$$\frac{d\phi}{dx}(0) = c_2\sqrt{s} = 0 = c_1\sqrt{s} \sinh(\sqrt{s}L) + c_2\sqrt{s} \cosh(\sqrt{s}L) = \frac{d\phi}{dx}(L)$$

writing as the matrix equation

$$\begin{bmatrix} 0 & \sqrt{s} \\ \sqrt{s} \sinh(\sqrt{s}L) & \sqrt{s} \cosh(\sqrt{s}L) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

From the coefficient matrix, calculate the determinant, which if zero, guarantees a non-trivial pair (c_1, c_2) :

$$\det = -s \sinh(\sqrt{s}L) = 0$$

which holds true only when $\sinh(\sqrt{s}L) = 0$, but this is impossible as that holds only at $\sqrt{s}L = 0$, and we assumed $s > 0$. Therefore no negative eigenvalues exist for this eigenfunction and boundary conditions.

Question: 2.4.5

This problem presents an alternative derivation of the heat equation for a thin wire. The equation for a circular wire of finite thickness is the two-dimensional heat equation (in polar coordinates). Show that this reduces to (2.4.25) if the temperature does not depend on r and if the wire is very thin.

The difficult part is in conceptualizing the problem. From section 1.5, recall that the polar heat equation can describe a disk or an annulus, depending on whether the lower bound for r is zero or not. In this scenario, imagine the annulus case, where the radial bounds a and b differ by some very small amount δ :

$$b - a = \delta \ll 1$$

It is this assumption that allows us to conclude that there is no radial variation in the temperature; put differently, r is constant. From (1.5.19), the two-dimensional heat equation in polar coordinates is

$$\frac{\partial u}{\partial t} = k \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \right)$$

Since temperature is independent of r , we immediately see that the first two terms vanish. Now we are left with

$$\frac{\partial u}{\partial t} = \frac{k}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

Recall that the arc length $s = r\theta$. The task will be to convert from (r, θ) -coordinates to s -coordinates. First compute

$$\frac{\partial r}{\partial s} = 0, \quad \frac{\partial \theta}{\partial s} = \frac{1}{r}, \quad \frac{\partial^2 \theta}{\partial s^2} = -\frac{1}{r^2} \frac{\partial r}{\partial s} = 0$$

where the first and last equalities arise from the result that r is constant. By the chain rule, derive (once again appealing to the premise of all radial derivatives vanishing)

$$\begin{aligned} \frac{\partial u}{\partial s} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial s} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial s} = \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial s} \\ &= \left(\frac{1}{r} \frac{\partial}{\partial \theta} \right) u \\ \frac{\partial^2 u}{\partial s^2} &= \frac{\partial}{\partial s} \left(\frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial s} \right) \\ &= \frac{\partial^2 u}{\partial \theta^2} \left(\frac{\partial \theta}{\partial s} \right)^2 \\ &= \left(\frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} \right) u \end{aligned}$$

Substituting back to the heat equation, make the final step to get

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial s^2}$$

Question: 2.4.6

Determine the equilibrium temperature distribution for the thin circular ring of Section 2.4.2:

- (a) directly from the equilibrium problem (see Section 1.4)
- (b) by comparing the limit as $t \rightarrow \infty$ of the time-dependent problem

- (a) Set $\partial u / \partial t = 0$, leaving us with

$$\frac{\partial^2 u}{\partial x^2} = 0$$

Integrating twice gives us general solution

$$u(x) = c_1 + c_2 x$$

Imposing the boundary conditions produces

$$\begin{aligned} u(-L) &= c_1 - c_2 L = c_1 + c_2 L = u(L) \\ \frac{du}{dx}(-L) &= c_2 = c_2 = \frac{du}{dx}(L) \end{aligned}$$

which forces $c_2 = 0$, leaving only

$$u(x) = c_1$$

At equilibrium, the temperature in the ring is constant, with c_1 determined by the initial temperature (due to the conservation of thermal energy), namely

$$c_1 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

(b) From equation (2.4.38)

$$u(x, t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt} + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

Taking the temporal limit, it is simple to derive

$$\lim_{t \rightarrow \infty} u(x, t) = a_0$$

from which it follows that $a_0 = c_1$ from the previous part.

Question: 2.4.7

Solve the heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

[Hints: It is known that if $u(x, t) = \phi(x) G(t)$, then $\frac{1}{kG} \frac{dG}{dt} = \frac{1}{\phi} \frac{d^2 \phi}{dx^2}$. Appropriately assume $\lambda > 0$. Assume the eigenfunctions $\phi_n(x)$ satisfy the following integral condition (orthogonality):

$$\int_0^L \phi_n(x) \phi_m(x) dx = \begin{cases} 0 & n \neq m \\ L/2 & n = m \end{cases}$$

subject to the following conditions:

- (a) $u(0, t) = 0, u(L, t) = 0, u(x, 0) = f(x)$
- (b) $u(0, t) = 0, \frac{\partial u}{\partial x}(L, t) = 0, u(x, 0) = f(x)$
- (c) $\frac{\partial u}{\partial x}(0, t) = 0, u(L, t) = 0, u(x, 0) = f(x)$
- (d) $\frac{\partial u}{\partial x}(0, t) = 0, \frac{\partial u}{\partial x}(L, t) = 0, u(x, 0) = f(x)$ and modify orthogonality condition [using Table 2.4.1.]

In all cases, the temporal solution is

$$G(t) = ce^{-\lambda kt}$$

and the spatial solution has general form

$$\phi(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

(a) Impose the boundary conditions

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \phi(L) &= c_2 \sin(\sqrt{\lambda}L) = 0\end{aligned}$$

where a non-trivial solution corresponds to eigenvalues

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, \dots$$

Leaving us eigenfunctions

$$\phi_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

By superposition, we have temperature distribution

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-(n\pi/L)^2 kt}$$

where A_n is

$$A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

(b) Impose the boundary conditions

$$\begin{aligned}\phi(0) &= c_1 = 0 \\ \frac{d\phi}{dx}(L) &= c_2 \sqrt{\lambda} \cos(\sqrt{\lambda}L) = 0\end{aligned}$$

where a non-trivial solution corresponds to

$$\lambda_n = \left[\frac{(2n-1)\pi}{2L}\right]^2, \quad n = 1, 2, \dots$$

Leaving us eigenfunctions

$$\phi_n(x) = \sin\left[\frac{(2n-1)\pi x}{2L}\right]$$

By superposition, we have temperature distribution

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin\left[\frac{(2n-1)\pi x}{2L}\right] e^{-[(2n-1)\pi/2L]^2 kt}$$

where A_n is derived by projecting onto the Hilbert sine space of “half-modes” to get

$$\begin{aligned}\int_0^L f(x) \sin\left[\frac{(2n-1)\pi x}{2L}\right] dx &= \int_0^L \sum_{n=1}^{\infty} A_n \sin\left[\frac{(2n-1)\pi x}{2L}\right] \sin\left[\frac{(2m-1)\pi x}{2L}\right] dx \\ &= A_m \int_0^L \sin^2\left[\frac{(2m-1)\pi x}{2L}\right] dx \\ &= A_m \frac{L}{2} \\ \implies A_m &= \frac{2}{L} \int_0^L f(x) \sin\left[\frac{(2m-1)\pi x}{2L}\right] dx\end{aligned}$$

(c) Impose the boundary conditions

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2\sqrt{\lambda} = 0 \implies c_2 = 0 \\ \phi(L) &= c_1 \cos(\sqrt{\lambda}L) = 0\end{aligned}$$

where a non-trivial solution corresponds to eigenvalues

$$\lambda_n = \left[\frac{(2n-1)\pi}{2L} \right]^2, \quad n = 1, 2, \dots$$

Leaving us eigenfunctions

$$\phi_n(x) = \cos\left[\frac{(2n-1)\pi x}{2L} \right]$$

By superposition, we have temperature distribution

$$u(x, t) = \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\pi x}{2L} \right] e^{-[(2n-1)\pi/2L]^2 kt}$$

where A_n is derived by projecting onto the Hilbert cosine space of “half-modes” to get

$$\begin{aligned}\int_0^L f(x) \cos\left[\frac{(2n-1)\pi x}{2L} \right] dx &= \int_0^L \sum_{n=1}^{\infty} A_n \cos\left[\frac{(2n-1)\pi x}{2L} \right] \cos\left[\frac{(2m-1)\pi x}{2L} \right] dx \\ &= A_m \int_0^L \cos^2\left(\frac{(2m-1)\pi x}{2L} \right) dx \\ &= A_m \frac{L}{2} \\ \implies A_m &= \frac{2}{L} \int_0^L f(x) \cos\left[\frac{(2m-1)\pi x}{2L} \right] dx\end{aligned}$$

(d) Impose the boundary conditions

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2\sqrt{\lambda} = 0 \implies c_2 = 0 \\ \frac{d\phi}{dx}(L) &= -c_1\sqrt{\lambda} \sin(\sqrt{\lambda}L) = 0\end{aligned}$$

where a non-trivial solution corresponds to eigenvalues

$$\lambda_n = \left(\frac{n\pi}{L} \right)^2, \quad n = 0, 1, \dots$$

Leaving us eigenfunctions

$$\phi_n(x) = \cos\left(\frac{n\pi x}{L} \right)$$

By superposition, we have temperature distribution

$$u(x, t) = \sum_{n=0}^{\infty} A_n \cos\left(\frac{n\pi x}{L} \right) e^{-(n\pi/L)^2 kt}$$

where the orthogonality conditions for A_0 and A_n , $n \neq 0$, slightly differ:

$$A_n = \begin{cases} \frac{1}{L} \int_0^L f(x) \, dx & n = 0 \\ \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) \, dx & n \neq 0 \end{cases}$$

2.5 - Laplace's Equation: Solutions and Qualitative Properties

Question: 2.5.1

Solve Laplace's equation inside a rectangle $0 \leq x \leq L$, $0 \leq y \leq H$, with the following boundary conditions [*Hint*: Separate variables. If there are two homogeneous boundary conditions in y , let $u(x, y) = h(x) \phi(y)$, and if there are two homogeneous boundary conditions in x , let $u(x, y) = \phi(x) h(y)$.]:

- (a) $\frac{\partial u}{\partial x}(0, y) = 0$, $\frac{\partial u}{\partial x}(L, y) = 0$, $u(x, 0) = 0$, $u(x, H) = f(x)$
- (b) $\frac{\partial u}{\partial x}(0, y) = g(y)$, $\frac{\partial u}{\partial x}(L, y) = 0$, $u(x, 0) = 0$, $u(x, H) = 0$
- (c) $\frac{\partial u}{\partial x}(0, y) = 0$, $u(L, y) = g(y)$, $u(x, 0) = 0$, $u(x, H) = 0$
- (d) $u(0, y) = g(y)$, $u(L, y) = 0$, $\frac{\partial u}{\partial y}(x, 0) = 0$, $u(x, H) = 0$
- (e) $u(0, y) = 0$, $u(L, y) = 0$, $u(x, 0) - \frac{\partial u}{\partial y}(x, 0) = 0$, $u(x, H) = f(x)$
- (f) $u(0, y) = f(y)$, $u(L, y) = 0$, $\frac{\partial u}{\partial y}(x, 0) = 0$, $\frac{\partial u}{\partial y}(x, H) = 0$
- (g) $\frac{\partial u}{\partial x}(0, y) = 0$, $\frac{\partial u}{\partial x}(L, y) = 0$, $u(x, 0) = \begin{cases} 0 & x > L/2 \\ 1 & x < L/2 \end{cases}$, $\frac{\partial u}{\partial y}(x, H) = 0$
- (h) $u(0, y) = 0$, $u(L, y) = g(y)$, $u(x, 0) = 0$, $u(x, H) = 0$

For this specific class of problem – Sturm-Liouville – over the given intervals, we will omit consideration of negative eigenvalues, and simply investigate the cases of $\lambda \geq 0$.

- (a) Since the boundary conditions are homogeneous in x , we have

$$u(x, y) = \phi(x) h(y)$$

Then we write the Laplacian as

$$\nabla^2 u = h(y) \frac{d^2 \phi}{dx^2} + \phi(x) \frac{d^2 h}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{\phi(x)} \frac{d^2 \phi}{dx^2} = -\frac{1}{h(y)} \frac{d^2 h}{dy^2} = -\lambda$$

where we choose the negative sign to ensure that we have oscillation in x (to satisfy the two homogeneous boundary conditions) and exponentials in y . Now, to solve the eigenvalue problem, first consider the positive case:

$\lambda > 0$: The general spatial solution is

$$\phi(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Imposing the boundary conditions, derive

$$\begin{aligned}\frac{d\phi}{dx}(0) &= c_2\sqrt{\lambda} = 0 \implies c_2 = 0 \\ \frac{d\phi}{dx}(L) &= -c_1\sqrt{\lambda}\sin(\sqrt{\lambda}L) = 0\end{aligned}$$

where in the second boundary condition, a non-trivial solution corresponds to eigenvalues

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

with eigenfunctions

$$\phi(x) = \cos\left(\frac{n\pi x}{L}\right)$$

$\lambda = 0$: The general spatial solution in this case is

$$\phi(x) = c_1 + c_2x$$

The boundary conditions are simply

$$\frac{d\phi}{dx}(0) = \frac{d\phi}{dx}(L) = c_2 = 0$$

Leaving us with $\phi(x) = c_1$.

Now we move onto the y -equations. For the positive eigenvalues λ_n we derived, we have general solution

$\lambda > 0$:

$$h(y) = a_1 \cosh\left(\frac{n\pi y}{L}\right) + a_2 \sinh\left(\frac{n\pi y}{L}\right)$$

where the boundary conditions produce

$$\begin{aligned}h(0) &= a_1 = 0 \\ h(H) &= a_2 \sinh\left(\frac{n\pi H}{L}\right)\end{aligned}$$

which yields corresponding eigenfunction

$$h(y) = \sinh\left(\frac{n\pi y}{L}\right)$$

Lastly, we examine the zero eigenvalue case.

$\lambda = 0$:

$$h(y) = a_1 + a_2y$$

with boundary conditions

$$\begin{aligned}h(0) &= a_1 = 0 \\ h(H) &= a_2H\end{aligned}$$

leaving us with eigenfunction

$$h(y) = a_2y$$

We summarize our equations in the table below. For the following problems, I will simply list the eigenfunctions derived from the boundary conditions in such tables for the sake of brevity, unless special considerations warrant a more detailed derivation.

	$\lambda = 0$	$\lambda = \left(\frac{n\pi}{L}\right)^2 > 0$
$\phi(x)$	c_1	$\cos\left(\frac{n\pi x}{L}\right)$
$h(y)$	$a_2 y$	$\sinh\left(\frac{n\pi y}{L}\right)$

Combining solutions and using superposition, the full temperature distribution is

$$u(x, y) = A_0 y + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi y}{L}\right)$$

To identify the coefficients we rely on the non-homogeneous boundary condition: $u(x, H) = f(x)$. At that point, we have

$$f(x) = u(x, H) = A_0 H + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi H}{L}\right)$$

First we consider the case when $n > 0$, i.e. for the positive eigenvalues. Here is the key point: **we must project our boundary condition $f(x)$ onto each of the directions of the Hilbert space spanned by the orthogonal eigenbasis corresponding to the non-zero eigenvalues. The coefficients A_m correspond to the “coordinates” of $f(x)$ as projected on that specific direction in the Hilbert space.** By orthogonality, multiply all terms by $\cos(m\pi x/L)$ and integrate from 0 to L to get

$$\begin{aligned} \int_0^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx &= \int_0^L A_0 H \cos\left(\frac{m\pi x}{L}\right) dx \\ &+ \int_0^L A_m \cos^2\left(\frac{m\pi x}{L}\right) \sinh\left(\frac{m\pi H}{L}\right) dx \end{aligned}$$

This in turn allows us to compute A_m :

$$A_m = \frac{2}{L \sinh(m\pi H/L)} \int_0^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx$$

Lastly, to compute A_0 , we analogously project the boundary condition onto the eigenfunctions corresponding to the zero eigenvalue. Truly, we deal with only a singular eigenfunction: the constant, which we can reduce to 1. Another way to conceptualize this is accounting for all of the basis vectors of the Hilbert space in x

$$\left\{ \cos\left(\frac{n\pi x}{L}\right) \mid n \in \mathbb{N} \cup \{0\} \right\}$$

where in the $n > 0$ case we considered the subspace spanned by

$$\{\cos(n\pi x/L) \mid n > 0\}$$

and in the $n = 0$ case we account for the basis vector $\{1\}$. With that out of the way, calculate

$$\int_0^L f(x) dx = \int_0^L A_0 H dx + \int_0^L \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi H}{L}\right) dx$$

Observe that most conveniently, the entirety of the second term vanishes due to the orthogonality condition in the cosine function against the constant eigenfunction 1. This leaves us with

$$A_0 = \frac{1}{HL} \int_0^L f(x) dx$$

(b) With boundary conditions homogeneous in y , we have

$$u(x, y) = h(x) \phi(y)$$

The Laplacian is then

$$\nabla^2 u = \phi(y) \frac{d^2 h}{dx^2} + h(x) \frac{d^2 \phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)} \frac{d^2 h}{dx^2} = -\frac{1}{\phi(y)} \frac{d^2 \phi}{dy^2} = \lambda$$

where the positive sign is chosen to ensure oscillation in y and exponentials in x . The eigenfunctions and corresponding eigenvalues are

	$\lambda = \left(\frac{n\pi}{H}\right)^2 > 0$
$\phi(y)$	$\sin\left(\frac{n\pi y}{H}\right)$
$h(x)$	$\sinh\left(\frac{n\pi}{H}(x - L)\right)$

By superposition, we have temperature distribution

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi y}{H}\right) \sinh\left(\frac{n\pi}{H}(x - L)\right)$$

To derive the coefficients A_n , first differentiate with respect to x and examine the point $(0, y)$, where the non-homogeneous condition is realized

$$\frac{\partial u}{\partial x}(0, y) = \sum_{n=1}^{\infty} A_n \left(\frac{n\pi}{H}\right) \sin\left(\frac{n\pi y}{H}\right) \cosh\left(-\frac{n\pi L}{H}\right) = g(y)$$

Then multiplying the right equality by $\sin(m\pi y/H)$, integrating from 0 to H , and isolating the coefficient produces

$$A_m = \frac{2}{m\pi \cosh(m\pi L/H)} \int_0^H g(y) \sin\left(\frac{m\pi y}{H}\right) dy$$

where we appeal to the evenness of the hyperbolic cosine function to change the argument sign to positive.

(c) Once again we are homogeneous in y , so we define

$$u(x, y) = h(x) \phi(y)$$

with corresponding Laplacian

$$\nabla^2 u = \phi(y) \frac{d^2 h}{dx^2} + h(x) \frac{d^2 \phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)} \frac{d^2 h}{dx^2} = -\frac{1}{\phi(y)} \frac{d^2 \phi}{dy^2} = \lambda$$

where the separation constant sign is chosen to ensure oscillation in y . The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = \left(\frac{n\pi}{H}\right)^2 > 0$
$\phi(y)$	$\sin\left(\frac{n\pi y}{H}\right)$
$h(x)$	$\cosh\left(\frac{n\pi x}{H}\right)$

By superposition, we have complete temperature distribution

$$u(x, y) = \sum_{n=1}^{\infty} A_n \cosh\left(\frac{n\pi x}{H}\right) \sin\left(\frac{n\pi y}{H}\right)$$

where A_n is determined by evaluating at (L, y) :

$$u(L, y) = \sum_{n=1}^{\infty} A_n \cosh\left(\frac{n\pi L}{H}\right) \sin\left(\frac{n\pi y}{H}\right) = g(y)$$

and then multiplying by $\sin(m\pi y/H)$ and integrating from 0 to H to derive

$$A_m = \frac{2}{H \cosh(m\pi L/H)} \int_0^H g(y) \sin\left(\frac{m\pi y}{H}\right) dy$$

(d) Homogeneity of the boundary conditions in y produces

$$u(x, y) = h(x) \phi(y)$$

The Laplacian is

$$\nabla^2 u = \phi(y) \frac{d^2 h}{dx^2} + h(x) \frac{d^2 \phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)} \frac{d^2 h}{dx^2} = -\frac{1}{\phi(y)} \frac{d^2 \phi}{dy^2} = \lambda$$

where the sign of the separation constant ensures oscillation in y and exponentials in x . The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = \left[\frac{(2n-1)\pi}{2H} \right]^2 > 0$
$\phi(y)$	$\cos \left[\frac{(2n-1)\pi y}{2H} \right]$
$h(x)$	$\sinh \left[\frac{(2n-1)\pi(x-L)}{2H} \right]$

By superposition, we have temperature distribution

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sinh \left[\frac{(2n-1)\pi(x-L)}{2H} \right] \cos \left[\frac{(2n-1)\pi y}{2H} \right]$$

where the Fourier coefficient is derived as:

$$A_m = -\frac{2}{H \sinh[(2m-1)\pi L/2H]} \int_0^H g(y) \cos \left[\frac{(2m-1)\pi y}{2H} \right] dy$$

(e) We switch to homogeneity of the boundary conditions in x , so we define

$$u(x, y) = \phi(x) h(y)$$

which has Laplacian

$$\nabla^2 u = h(y) \frac{d^2 \phi}{dx^2} + \phi(x) \frac{d^2 h}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{\phi(x)} \frac{d^2 \phi}{dx^2} = -\frac{1}{h(y)} \frac{d^2 h}{dy^2} = -\lambda$$

which guarantees oscillation in x and exponentials in y . The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = \left(\frac{n\pi}{L} \right)^2$
$\phi(x)$	$\sin \left(\frac{n\pi x}{L} \right)$
$h(y)$	$\left(\frac{n\pi}{L} \right) \cosh \left(\frac{n\pi y}{L} \right) + \sinh \left(\frac{n\pi y}{L} \right)$

By superposition, we have temperature distribution

$$u(x, y) = \sum_{n=1}^{\infty} A_n h_n(y) \sin \left(\frac{n\pi x}{L} \right)$$

where

$$h_n(y) = \left(\frac{n\pi}{L} \right) \cosh \left(\frac{n\pi y}{L} \right) + \sinh \left(\frac{n\pi y}{L} \right)$$

and the Fourier coefficient is derived as

$$A_m = \frac{2}{L h_m(H)} \int_0^L f(x) \sin \left(\frac{m\pi x}{L} \right) dx$$

(f) Homogeneity of the boundary conditions in y yields

$$u(x, y) = h(x) \phi(y)$$

with corresponding Laplacian

$$\nabla^2 u = \phi(y) \frac{d^2 h}{dx^2} + h(x) \frac{d^2 \phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)} \frac{d^2 h}{dx^2} = -\frac{1}{\phi(y)} \frac{d^2 \phi}{dy^2} = \lambda$$

which ensures oscillation in y and exponentials in x . The eigenfunctions and corresponding eigenvalues are

	$\lambda_0 = 0$	$\lambda_n = \left(\frac{n\pi}{H}\right)^2 > 0$
$\phi(y)$	c	$\cos\left(\frac{n\pi y}{H}\right)$
$h(x)$	$1 - \frac{x}{L}$	$\sinh\left[\frac{n\pi}{H}(x - L)\right]$

By superposition, we have solution

$$u(x, y) = A_0 \left(1 - \frac{x}{L}\right) + \sum_{n=1}^{\infty} A_n \sinh\left[\frac{n\pi}{H}(x - L)\right] \cos\left(\frac{n\pi y}{H}\right)$$

where the Fourier coefficients are given by

$$A_m = \begin{cases} \frac{1}{H} \int_0^H f(y) dy, & m = 0 \\ -\frac{2}{H \sinh(m\pi L/H)} \int_0^H f(y) \cos\left(\frac{m\pi y}{H}\right) dy, & m > 0 \end{cases}$$

(g) Our boundary conditions are homogeneous in x , which gives us

$$u(x, y) = \phi(x) h(y)$$

which has corresponding Laplacian

$$\nabla^2 u = h(y) \frac{d^2 \phi}{dx^2} + \phi(x) \frac{d^2 h}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, find

$$\frac{1}{\phi(x)} \frac{d^2 \phi}{dx^2} = -\frac{1}{h(y)} \frac{d^2 h}{dy^2} = -\lambda$$

which guarantees oscillation in x and exponentials in y . The eigenfunctions and corresponding eigenvalues are

	$\lambda_0 = 0$	$\lambda_n = \left(\frac{n\pi}{L}\right)^2 > 0$
$\phi(x)$	c_1	$\cos\left(\frac{n\pi x}{L}\right)$
$h(y)$	a_1	$\cosh\left[\frac{n\pi}{L}(y - H)\right]$

which combine and with superposition produce

$$u(x, y) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \cosh\left[\frac{n\pi}{L}(y - H)\right]$$

where

$$A_0 = \frac{1}{2}, \quad A_m = \frac{2}{m\pi \cosh(m\pi H/L)} \sin\left(\frac{m\pi}{2}\right)$$

Note that the derivation proceeds analogously to the problem posed in (a), where the presence of non-trivial eigenfunctions corresponding to $\lambda_0 = 0$ indicates two different projections of the boundary condition onto the eigenbases associated with the zero and non-zero eigenvalues.

- (h) We end this first set of exercises with homogeneity of the boundary conditions in y , giving us form

$$u(x, y) = h(x) \phi(y)$$

The corresponding Laplacian is

$$\nabla^2 u = \phi(y) \frac{d^2 h}{dx^2} + h(x) \frac{d^2 \phi}{dy^2} = 0$$

Separating variables and setting equal to the separation constant, derive

$$\frac{1}{h(x)} \frac{d^2 h}{dx^2} = -\frac{1}{\phi(y)} \frac{d^2 \phi}{dy^2} = \lambda$$

which ensures oscillation in y and exponentials in x . The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = \left(\frac{n\pi}{H}\right)^2 > 0$
$\phi(y)$	$\sin\left(\frac{n\pi y}{H}\right)$
$h(x)$	$\sinh\left(\frac{n\pi x}{H}\right)$

By superposition, we have temperature distribution

$$u(x, y) = \sum_{n=1}^{\infty} A_n \sinh\left(\frac{n\pi x}{H}\right) \sin\left(\frac{n\pi y}{H}\right)$$

with Fourier coefficient

$$A_m = \frac{2}{H \sinh(m\pi L/H)} \int_0^H g(y) \sin\left(\frac{m\pi y}{H}\right) dy$$

Question: 2.5.2

Consider $u(x, y)$ satisfying Laplace's equation inside a rectangle ($0 < x < L$, $0 < y < H$) subject to the boundary conditions

$$\begin{aligned} \frac{\partial u}{\partial x}(0, y) &= 0, & \frac{\partial u}{\partial y}(x, 0) &= 0 \\ \frac{\partial u}{\partial x}(L, y) &= 0, & \frac{\partial u}{\partial y}(x, H) &= f(x) \end{aligned}$$

- Without* solving this problem, briefly explain the physical condition under which there is a solution to this problem.
- Solve this problem by the method of separation of variables. Show that the method works only under the condition of part (a). [*Hint:* You may use (2.5.16) without derivation.]
- The solution [part (b)] has an arbitrary constant. Determine it by consideration of the time-dependent heat equation (1.5.11) subject to the initial condition

$$u(x, y, 0) = g(x, y)$$

- Conceptually, net heat flow through the boundary must be zero in order for the steady-state to exist, namely the solvability condition

$$\oint \nabla u \cdot \hat{\mathbf{n}} ds = \int_0^L \frac{\partial u}{\partial y}(x, H) dx = \int_0^L f(x) dx = 0$$

- The boundary conditions are homogeneous in x , so we separate variables as

$$u(x, y) = \phi(x) h(y)$$

The corresponding Laplacian at equilibrium is

$$\nabla^2 u = h(y) \frac{d^2 \phi}{dx^2} + \phi(x) \frac{d^2 h}{dy^2} = 0$$

And separating variables and setting equal to the separation constant produces

$$\frac{1}{\phi(x)} \frac{d^2 \phi}{dx^2} = -\frac{1}{h(y)} \frac{d^2 h}{dy^2} = -\lambda$$

which guarantees oscillation in x and exponentials in y . The eigenfunctions and corresponding eigenvalues are

	$\lambda_0 = 0$	$\lambda_n = \left(\frac{n\pi}{L}\right)^2 > 0$
$\phi(x)$	1	$\cos\left(\frac{n\pi x}{L}\right)$
$h(y)$	1	$\cosh\left(\frac{n\pi y}{L}\right)$

By superposition, we have temperature distribution

$$u(x, y) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \cosh\left(\frac{n\pi y}{L}\right)$$

where the Fourier coefficients for the eigenbases with non-zero eigenvalues are given by

$$A_m = \frac{2}{m\pi \sinh(m\pi H/L)} \int_0^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx$$

However, in the case of multiplying through by the constant eigenfunction $\phi(x) = 1$ and integrating, we find

$$\begin{aligned} \int_0^L \frac{\partial u}{\partial y}(x, H) dx &= \sum_{n=1}^{\infty} A_n \left(\frac{n\pi}{L}\right) \sinh\left(\frac{n\pi H}{L}\right) \int_0^L \cos\left(\frac{n\pi x}{L}\right) dx \\ &= \int_0^L f(x) dx \\ &= 0 \end{aligned}$$

Which is consistent with the solvability condition in part (a). The other insight here is that A_0 is *undetermined from the boundary conditions*, and in equilibrium, *any* value of A_0 is consistent with all four boundary conditions.

- (c) Where we do determine A_0 is from the initial condition. First, recall that one plan of approach for the time-dependent case is to compute the rate of change of the total energy, namely

$$\begin{aligned} \frac{\partial E}{\partial t} &= \iint c\rho \frac{\partial u}{\partial t} dA = K_0 \iint \nabla \cdot \nabla u dA \\ &= \oint \nabla u \cdot \hat{\mathbf{n}} ds \\ &= \int_0^L f(x) dx \\ &= 0 \end{aligned}$$

where we use the divergence theorem and the solvability condition in the second and third lines, respectively. We have now established that **energy is conserved**.

What next? This implies

$$\iint u(x, y, 0) \, dA = \iint u(x, y) \, dA = \iint g(x, y) \, dA$$

which conceptually translates to initial energy equals to energy at equilibrium. We may simply integrate

$$\begin{aligned} \int_0^H \int_0^L \left[A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) \cosh\left(\frac{n\pi y}{L}\right) \right] dx \, dy &= A_0 HL \\ &= \iint g(x, y) \, dA \end{aligned}$$

implying

$$A_0 = \frac{1}{HL} \iint g(x, y) \, dA$$

Observe that physically, the transient modes carry net-zero energy, which is the interpretation of the entire summation vanishing by the area integral.

Question: 2.5.3

Solve Laplace's equation *outside* a circular disk ($r \geq a$) subject to the boundary condition [*Hint*: In polar coordinates,

$$\nabla^2 u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

it is known that if $u(r, \theta) = \phi(\theta) G(r)$, then $\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2 \phi}{d\theta^2}$.]

- (a) $u(a, \theta) = \ln(2) + 4 \cos(3\theta)$
- (b) $u(a, \theta) = f(\theta)$

You may assume that $u(r, \theta)$ remains finite as $r \rightarrow \infty$.

By separable variables, let

$$u(r, \theta) = \phi(\theta) G(r)$$

Which produces the equation

$$\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2 \phi}{d\theta^2} = \lambda$$

First assume $\lambda > 0$ and examine the angular equation. The periodicity conditions

$$\begin{aligned} \phi(-\pi) &= \phi(\pi) \\ \frac{d\phi}{d\theta}(-\pi) &= \frac{d\phi}{d\theta}(\pi) \end{aligned}$$

must be satisfied. Since we have oscillation in θ , derive general form

$$\phi(\theta) = c_1 \cos(\sqrt{\lambda}\theta) + c_2 \sin(\sqrt{\lambda}\theta)$$

Enforcing the first boundary condition produces

$$\phi(-\pi) = c_1 \cos(\sqrt{\lambda}\pi) - c_2 \sin(\sqrt{\lambda}\pi) = c_1 \cos(\sqrt{\lambda}\pi) + c_2 \sin(\sqrt{\lambda}\pi) = \phi(\pi)$$

implying

$$2c_2 \sin(\sqrt{\lambda}\pi) = 0$$

The second:

$$\begin{aligned} \frac{d\phi}{d\theta}(-\pi) &= c_1 \sqrt{\lambda} \sin(\sqrt{\lambda}\pi) + c_2 \sqrt{\lambda} \cos(\sqrt{\lambda}\pi) \\ \frac{d\phi}{d\theta}(\pi) &= -c_1 \sqrt{\lambda} \sin(\sqrt{\lambda}\pi) + c_2 \sqrt{\lambda} \cos(\sqrt{\lambda}\pi) \end{aligned}$$

implying

$$2c_1 \sqrt{\lambda} \sin(\sqrt{\lambda}\pi) = 0$$

Combined, the boundary conditions suggest that either $c_1 = c_2 = 0$ or $\lambda = n^2$. Since we desire a non-trivial solution, this is what produces our eigenvalues.

From the radial equation, we seek to solve

$$\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = n^2$$

where, when rearranging and evaluating terms, we have the equidimensional equation

$$r^2 \frac{d^2 G}{dr^2} + r \frac{dG}{dr} - n^2 G = 0$$

Which is solved by supposing $G(r) = r^p$, solving for p , and deriving $p = \pm n$. Therefore the general radial solution is

$$G(r) = a_1 r^n + a_2 r^{-n}$$

Now, here is where some physical intuition comes in. We are considering the area *beyond* $r = a$. Originally, we had to guarantee that $|G(0)| < \infty$; that is, we bound $G(r)$ at the origin to be finite. In this case, since we consider the distribution of heat far from the origin, it is reasonable to suggest that the boundary condition in this case be

$$|G(\infty)| < \infty$$

Imposing this boundary condition implies

$$G(\infty) = a_1 r^n + a_2 r^{-n} < \infty \implies a_1 = 0$$

Repeat the process for $\lambda = 0$. For the angular equation, derive

$$\phi(\theta) = c_1 + c_2 \theta$$

where the periodicity conditions suggest $c_2 = 0$, leaving us with the constant eigenfunction. In the radial equation, we are left to solve

$$\frac{d}{dr} \left(r \frac{dG}{dr} \right) = 0$$

which has solution

$$G(r) = b_1 + b_2 \ln(r)$$

Similarly, we apply the limiting condition $|G(\infty)| < \infty$, forcing $b_2 = 0$, and leaving us once again with the constant eigenfunction. The complete set of eigenfunctions and corresponding eigenvalues is thus

	$\lambda_0 = 0$	$\lambda_n = n^2 > 0$
$\phi(\theta)$	1	$\sin(n\theta), \cos(n\theta)$
$G(r)$	1	r^{-n}

Therefore, combining the radial and angular solutions and appealing to superposition gives us general temperature distribution

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n r^{-n} \cos(n\theta) + \sum_{n=1}^{\infty} B_n r^{-n} \sin(n\theta)$$

(a)

$$\begin{aligned} u(a, \theta) &= \sum_{n=0}^{\infty} A_n a^{-n} \cos(n\theta) + \sum_{n=1}^{\infty} B_n a^{-n} \sin(n\theta) \\ &= \ln(2) + 4 \cos(3\theta) \end{aligned}$$

implies

$$A_0 = \ln(2), \quad A_3 = 4a^3, \quad A_{n \neq 0,3} = 0, \quad B_n = 0$$

(b)

$$u(a, \theta) = \sum_{n=0}^{\infty} A_n a^{-n} \cos(n\theta) + \sum_{n=1}^{\infty} B_n a^{-n} \sin(n\theta) = f(\theta)$$

To calculate the Fourier coefficients, go case-by-case. For A_0 , multiply through by $\phi(\theta) = 1$ and integrate from $-\pi$ to π to get

$$A_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) d\theta$$

For A_n and B_n , use orthogonality and integrate over the angular bounds to get

$$\begin{aligned} A_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \cos(n\theta) d\theta \\ B_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(\theta) \sin(n\theta) d\theta \end{aligned}$$

Question: 2.5.4

For Laplace's equation inside a circular disk ($r \leq a$), using (2.5.45) and (2.5.47), show that

$$u(r, \theta) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\bar{\theta}) \left[-\frac{1}{2} + \sum_{n=0}^{\infty} \left(\frac{r}{a} \right)^n \cos(n(\theta - \bar{\theta})) \right] d\bar{\theta}$$

Using $\cos(z) = \operatorname{Re}[e^{iz}]$, sum the resulting geometric series to obtain Poisson's integral formula.

Using the formulae for the Fourier coefficients, substitute into the solution to the polar Laplacian. Note that we define a dummy variable of integration $\bar{\theta}$ for the coefficient formulae:

$$\begin{aligned} u(r, \theta) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\bar{\theta}) d\bar{\theta} + \sum_{n=1}^{\infty} \frac{1}{\pi} \left(\frac{r}{a} \right)^n \cos(n\theta) \int_{-\pi}^{\pi} f(\bar{\theta}) \cos(n\bar{\theta}) d\bar{\theta} \\ &\quad + \sum_{n=1}^{\infty} \frac{1}{\pi} \left(\frac{r}{a} \right)^n \sin(n\theta) \int_{-\pi}^{\pi} f(\bar{\theta}) \sin(n\bar{\theta}) d\bar{\theta} \end{aligned}$$

Now, the advantage of defining this dummy variable is to be able to move the θ functions inside of the integrals, and exploiting the angle sum and difference trigonometric identities to simplify. Do this and combine terms to derive

$$u(r, \theta) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\bar{\theta}) \left[\frac{1}{2} + \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n (\cos(n\theta) \cos(n\bar{\theta}) + \sin(n\theta) \sin(n\bar{\theta})) \right] d\bar{\theta}$$

The morass of trigonometric terms can be combined into $\cos(n(\theta - \bar{\theta}))$, giving us

$$u(r, \theta) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\bar{\theta}) \left[\frac{1}{2} + \sum_{n=1}^{\infty} \left(\frac{r}{a} \right)^n \cos(n(\theta - \bar{\theta})) \right] d\bar{\theta}$$

And lastly, adding one to the second of the bracketed terms and subtracting from the first allows us to readjust the summation index to commence from $n = 0$, as desired:

$$u(r, \theta) = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\bar{\theta}) \left[-\frac{1}{2} + \sum_{n=0}^{\infty} \left(\frac{r}{a} \right)^n \cos(n(\theta - \bar{\theta})) \right] d\bar{\theta}$$

To derive Poisson's integral formula, first isolate the summation and rewrite as

$$\sum_{n=0}^{\infty} \left(\frac{r}{a} \right)^n \cos(n(\theta - \bar{\theta})) = \operatorname{Re} \left[\sum_{n=0}^{\infty} \left(\frac{r e^{in(\theta - \bar{\theta})}}{a} \right)^n \right]$$

Since the modulus $|r/a| < 1$, we apply the geometric series formula to derive

$$\sum_{n=0}^{\infty} \left(\frac{r e^{in(\theta - \bar{\theta})}}{a} \right)^n = \frac{1}{1 - \frac{r}{a} e^{i(\theta - \bar{\theta})}}$$

Now, the complex conjugate of the denominator is

$$1 - \frac{r}{a} e^{-i(\theta - \bar{\theta})}$$

Multiplying the numerator and denominator with this complex conjugate and explicitly writing out the trigonometric functions, calculate the *real* part as

$$\frac{a^2 - ar \cos(\theta - \bar{\theta})}{a^2 + r^2 - 2ar \cos(\theta - \bar{\theta})}$$

We still have the $-1/2$ term to deal with. Summing together, find

$$-\frac{1}{2} + \frac{a^2 - ar \cos(\theta - \bar{\theta})}{a^2 + r^2 - 2ar \cos(\theta - \bar{\theta})} = \frac{a^2 - r^2}{2[a^2 + r^2 - 2ar \cos(\theta - \bar{\theta})]}$$

For the finale, look back to $u(r, \theta)$. Replace the bracketed term with our result to derive Poisson's integral formula:

$$u(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\bar{\theta}) \frac{a^2 - r^2}{a^2 + r^2 - 2ar \cos(\theta - \bar{\theta})} d\bar{\theta}$$

Question: 2.5.5

Solve Laplace's equation inside the quarter-circle of radius 1 ($0 \leq \theta \leq \pi/2$, $0 \leq r \leq 1$) subject to the boundary conditions [*Hint*: In polar coordinates,

$$\nabla^2 u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

it is known that if $u(r, \theta) = \phi(\theta) G(r)$, then $\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2 \phi}{d\theta^2}$.]

- (a) $\frac{\partial u}{\partial \theta}(r, 0) = 0$, $u(r, \frac{\pi}{2}) = 0$, $u(1, \theta) = f(\theta)$
- (b) $\frac{\partial u}{\partial \theta}(r, 0) = 0$, $\frac{\partial u}{\partial \theta}(r, \frac{\pi}{2}) = 0$, $u(1, \theta) = f(\theta)$
- (c) $u(r, 0) = 0$, $u(r, \frac{\pi}{2}) = 0$, $\frac{\partial u}{\partial r}(1, \theta) = f(\theta)$
- (d) $\frac{\partial u}{\partial \theta}(r, 0) = 0$, $\frac{\partial u}{\partial \theta}(r, \frac{\pi}{2}) = 0$, $\frac{\partial u}{\partial r}(1, \theta) = g(\theta)$

Show that the solution [part (d)] exists only if $\int_0^{\pi/2} g(\theta) d\theta = 0$. Explain this condition physically.

- (a) Since the boundary conditions in θ are homogeneous, we set separation constant

$$\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2 \phi}{d\theta^2} = \lambda$$

$\lambda > 0$:

Our angular equation is

$$\phi(\theta) = c_1 \cos(\sqrt{\lambda}\theta) + c_2 \sin(\sqrt{\lambda}\theta)$$

which is subject to the boundary conditions

$$\begin{aligned}\frac{d\phi}{d\theta}(0) &= c_2\sqrt{\lambda} = 0 \quad \implies \quad c_2 = 0 \\ \phi\left(\frac{\pi}{2}\right) &= c_1 \cos\left(\frac{\sqrt{\lambda}\pi}{2}\right) = 0\end{aligned}$$

We derive eigenvalues

$$\lambda_n = (2n - 1)^2, \quad n = 1, 2, \dots$$

which gives angular eigenfunction

$$\phi(\theta) = \cos[(2n - 1)\theta]$$

For the radial equation, we end up with the equidimensional equation

$$r^2 \frac{d^2 G}{dr^2} + r \frac{dG}{dr} - (2n - 1)^2 G = 0$$

Substitute $G = r^p$ and solve the characteristic polynomial

$$p(p - 1) + p - (2n - 1)^2 = 0 \quad \implies \quad p = \pm(2n - 1)$$

Leaving us with radial equation

$$G(r) = a_1 r^p + a_2 r^{-p}$$

Now, we must impose the bounding condition that $|G(0)| < \infty$. This forces $a_2 = 0$, as otherwise G would diverge to infinity. Then our radial eigenfunction is

$$G(r) = r^{2n-1}$$

$\lambda = 0$:

We have angular equation

$$\phi(\theta) = c_1 + c_2 \theta$$

with boundary conditions

$$\begin{aligned}\frac{d\phi}{d\theta}(0) &= c_2 = 0 \\ \phi\left(\frac{\pi}{2}\right) &= c_1 = 0\end{aligned}$$

meaning no mode can exist for $n = 0$. We summarize our eigenfunctions and corresponding eigenvalues as

	$\lambda_n = (2n - 1)^2 > 0$
$\phi(\theta)$	$\cos[(2n - 1)\theta]$
$G(r)$	r^{2n-1}

By superposition, we have temperature distribution

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^{2n-1} \cos[(2n-1)\theta]$$

where A_n is given by first setting

$$u(1, \theta) = \sum_{n=1}^{\infty} A_n \cos[(2n-1)\theta] = f(\theta)$$

and then appealing to orthogonality of the eigenfunctions to derive

$$A_m = \frac{\int_0^{\pi/2} f(\theta) \cos[(2m-1)\theta] d\theta}{\int_0^{\pi/2} \cos^2[(2m-1)\theta] d\theta} = \frac{4}{\pi} \int_0^{\pi/2} f(\theta) \cos[(2m-1)\theta] d\theta$$

(b) With boundary conditions homogeneous in θ , write

$$\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2 \phi}{d\theta^2} = \lambda$$

which produces eigenfunctions with corresponding eigenvalues

	$\lambda_0 = 0$	$\lambda_n = (2n)^2 > 0$
$\phi(\theta)$	1	$\cos(2n\theta)$
$G(r)$	1	r^{2n}

By superposition, the temperature distribution is

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n r^{2n} \cos(2n\theta)$$

with Fourier coefficients

$$A_0 = \frac{2}{\pi} \int_0^{\pi/2} f(\theta) d\theta$$

$$A_m = \frac{4}{\pi} \int_0^{\pi/2} f(\theta) \cos(2m\theta) d\theta$$

(c) Given that the boundary conditions in θ are homogeneous, we set separation constant

$$\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2 \phi}{d\theta^2} = \lambda$$

which produce eigenfunctions and corresponding eigenvalues

	$\lambda_n = (2n)^2 > 0$
$\phi(\theta)$	$\sin(2n\theta)$
$G(r)$	r^{2n}

By superposition, we have temperature distribution

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^{2n} \sin(2n\theta)$$

with Fourier coefficients

$$A_m = \frac{2}{m\pi} \int_0^{\pi/2} f(\theta) \sin(2m\theta) d\theta$$

- (d) Homogeneity in θ from the boundary conditions enables us to set separation constant

$$\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2 \phi}{d\theta^2} = \lambda$$

and since the boundary conditions in θ are exactly as those of part (b), we have temperature distribution

$$u(r, \theta) = \sum_{n=0}^{\infty} A_n r^{2n} \cos(2n\theta)$$

For A_n , $n > 0$, we appeal to cosine orthogonality to derive

$$A_m = \frac{2}{m\pi} \int_0^{\pi/2} g(\theta) \cos(2m\theta) d\theta$$

In the case when we multiply through by the constant eigenfunction and integrate, however, we find

$$\int_0^{\pi/2} \frac{\partial u}{\partial r}(1, \theta) d\theta = \int_0^{\pi/2} \sum_{n=1}^{\infty} 2A_n n \cos(2n\theta) d\theta = \int_0^{\pi/2} g(\theta) d\theta = 0$$

Which forces our solvability condition, or that net flux is zero. Note that A_0 is undetermined (as in the case of exercise 2.5.2); it will remain so unless we have an initial condition to determine it.

Question: 2.5.6

Solve Laplace's equation inside a semicircle of radius a ($0 < r < a$, $0 < \theta < \pi$) subject to the boundary conditions [*Hint: In polar coordinates,*

$$\nabla^2 u = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

it is known that if $u(r, \theta) = \phi(\theta) G(r)$, then $\frac{r}{G} \frac{d}{dr} \left(r \frac{dG}{dr} \right) = -\frac{1}{\phi} \frac{d^2 \phi}{d\theta^2}$.]

- (a) $u = 0$ on the diameter and $u(a, \theta) = g(\theta)$
- (b) the diameter is insulated and $u(a, \theta) = g(\theta)$

- (a) The two boundary conditions in θ are

$$u(r, 0) = u(r, \pi) = 0$$

The eigenfunctions and corresponding eigenvalues are

	$\lambda_n = n^2 > 0$
$\phi(\theta)$	$\sin(n\theta)$
$G(r)$	r^n

By superposition, the temperature distribution is

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n r^n \sin(n\theta)$$

where the Fourier coefficients are

$$A_m a^m = \frac{2}{\pi} \int_0^{\pi} g(\theta) \sin(m\theta) d\theta$$

- (b) Now, the two boundary conditions in θ are

$$\frac{\partial u}{\partial \theta}(r, 0) = \frac{\partial u}{\partial \theta}(r, \pi) = 0$$

The eigenfunctions and eigenvalues switch to

	$\lambda_0 = 0$	$\lambda_n = n^2 > 0$
$\phi(\theta)$	1	$\cos(n\theta)$
$G(r)$	1	r^n

And the temperature distribution superposes to become

$$u(r, \theta) = A_0 + \sum_{n=1}^{\infty} A_n r^n \cos(n\theta)$$

with Fourier coefficients

$$A_0 = \frac{1}{\pi} \int_0^{\pi} g(\theta) d\theta$$

$$A_n a^n = \frac{2}{\pi} \int_0^{\pi} g(\theta) \cos(n\theta) d\theta$$

Question: 2.5.11

Do Exercise 1.5.8.

See solution manual for Chapter 1.

3 - Fourier Series

3.2 - Statement of Convergence Theorem

Question